

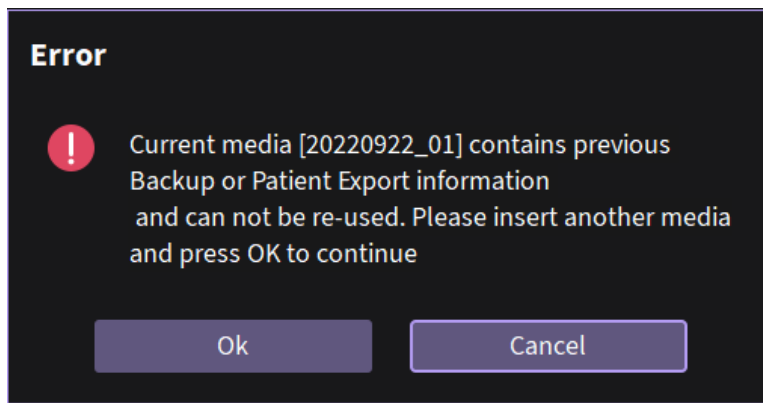


Figure 5-87 The Replace current media window

Insert a new media and select **OK**.

NOTE

To reuse a Backup CD/DVD when performing a new archive backup, the media has to be re-formatted first.

10. During backup, progress windows are displayed showing the current operation being performed.

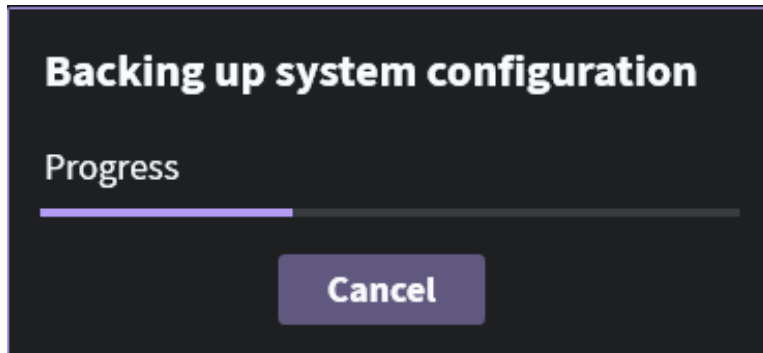


Figure 5-88 The Backup progress window

11. At the end of the process, the **Backup completed** window is displayed.

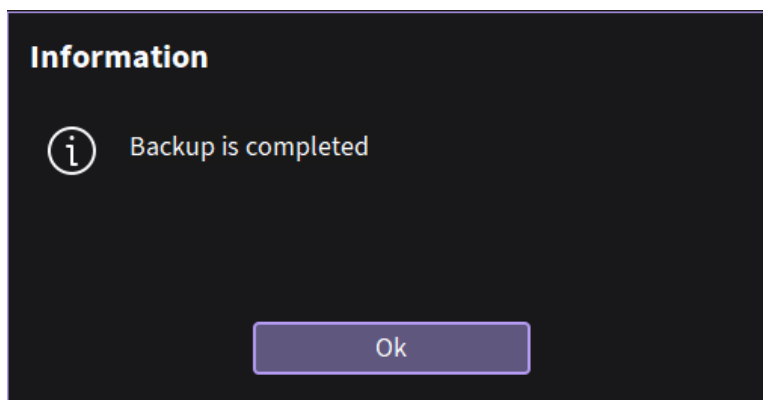


Figure 5-89 The Backup completed window

Select **OK**.

The Backup result is displayed on the **Backup** sheet.

12. Make sure to physically label the media. An identification of the system should also be noted on the media and a backup log should be kept.

File the media in a safe place.

Restore procedure

1. Press **Utility > Config** from the Touch panel.
2. Select the category **Admin**.
3. Select the **Restore** sheet.

Figure 5-90 The Restore sheet

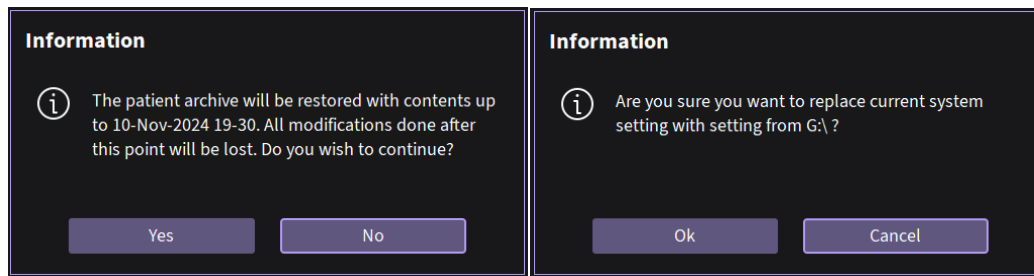
4. In the **Restore** sheet, select as needed:
 - **Patient archive** to restore the patient archive.
 - **System configuration** to restore all system settings and user presets.
OR
 - One or several **system configuration items** to restore parts of the system settings and user presets (see [Figure 5-90 The Restore sheet](#) on page 201).
5. If restore is done from a backup on a removable media, insert the media in the drive. Make sure that **Restore from Source Device** is selected.
6. Select the appropriate **Source device**.



CAUTION

The Restore procedure will **OVERWRITE** the existing data on the local harddrive. Make sure to insert the correct media and select the correct source device.

7. Select **Restore now**.
Depending on the selection of items to be restored, one or two restore confirmation windows are displayed:



A: Displayed if the patient archive is to be restored.

B: Displayed if any of the system settings are to be restored.

Figure 5-91 The Restore confirmation window

8. Ensure that the correct source is selected and select **OK**.
The selected items are copied to the systems.
9. If connectivity configuration settings are restored, the following information window is displayed.

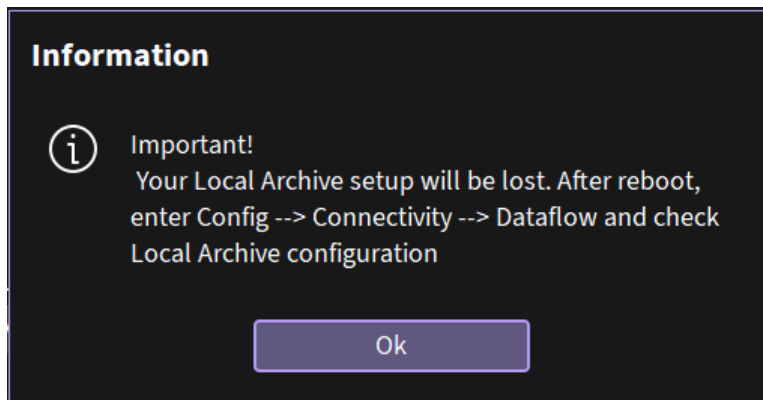


Figure 5-92 Information window

10. Select **OK**.
The **System shutdown** window is displayed.

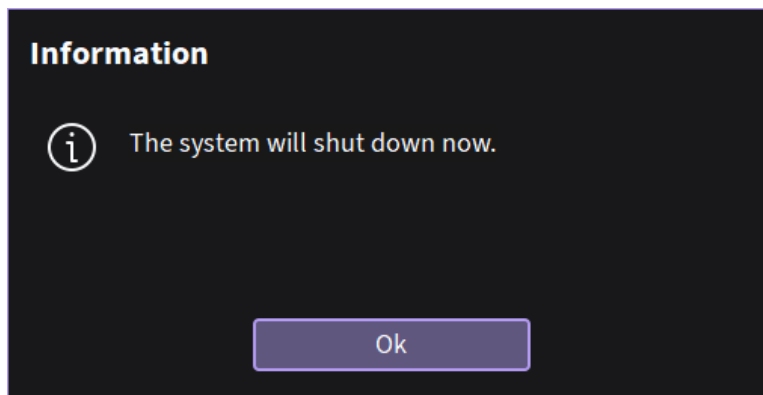


Figure 5-93 The System shutdown window

11. Select **OK** to shut down the system.
12. Restart the system.

5.3.7 Service tab

5.3.7.1 Open Service Screen

1. Select **Config** and log on as **ADM**.
2. Select the **Service** tab to view the **Service** screen.

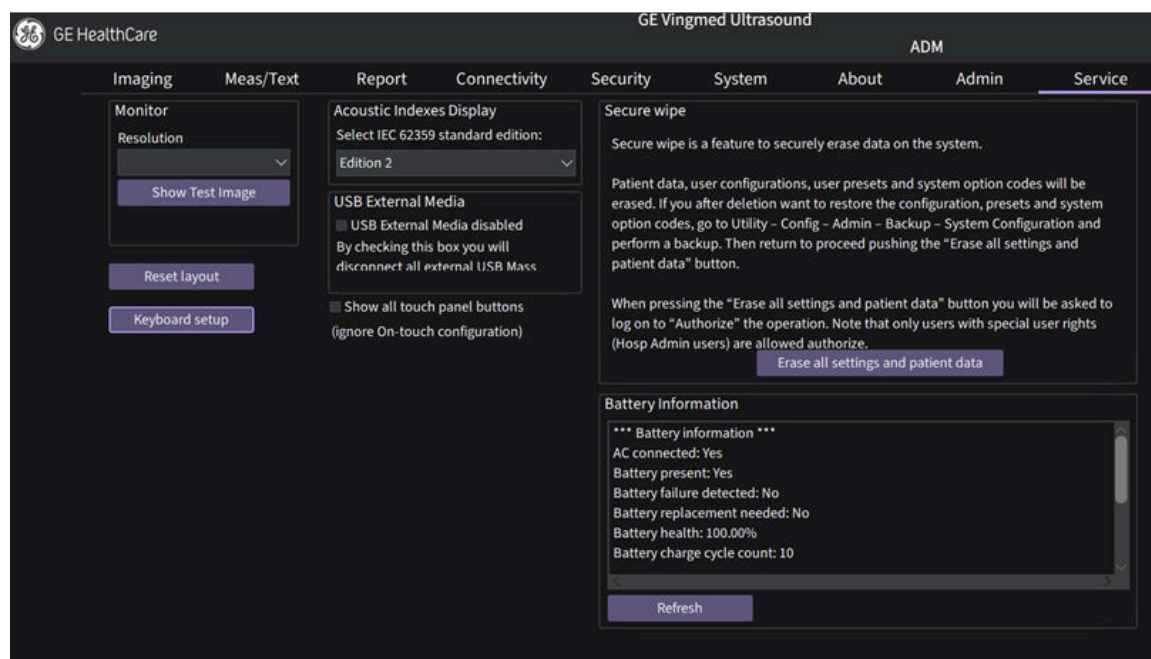


Figure 5-94 Service Screen

The following choices are available on the **Service** tab:

- Monitor
 - Resolution
 - Show Test Image
- Reset layout
- Keyboard setup
- Acoustic Indexes Display
- USB External Media
 - USB External Media disabled
- Show all touch panel buttons
- Secure wipe: Erase all settings and patient data
- Battery information
 - Refresh

5.3.7.2 Set Monitor Resolution

Select monitor resolution from the **Resolution** pull-down menu.

5.3.7.3 Select Show Test Image

1. Select **Show Test Image**.
2. Use the **Arrow** keys on the alphanumeric keyboard to browse the test images.
3. Press **Esc** to exit from the test images.

5.3.7.4 Accoustic Indexes Display

Select the IEC 62359 standard edition from the Accoustic Indexes Display pulldown menu.

5.3.7.4.1 The real-time display of acoustic output indices

The Vivid Explorer/Vivid Advanced/Vivid Focus systems have real-time display features according to IEC 62359 Ed.2. They display both a thermal (TI) and a mechanical (MI) index in all operating modes according to the required standard. These two indices are intended to estimate the potential for thermal and mechanical bioeffects induced by ultrasound. Both TI and MI are displayed with increments of 0.1. The displayed (estimated) TI and MI are nominal values.

5.3.7.5 Disable USB External Media

1. Click the **USB External Media disabled** check-box (so it is checked) to disable the use of external USB Mass Storage devices.
This will disconnect all external USB Mass Storage devices.
2. Click on the checked **USB External Media disabled** check-box to enable the use of USB External Media.

5.3.7.6 Keyboard Setup

NOTE

You don't need to perform this procedure if the alphanumeric keyboard is a US keyboard, since the default setting is set to US English keyboards.

5.3.7.6.1 Alphanumeric Keyboard Configuration

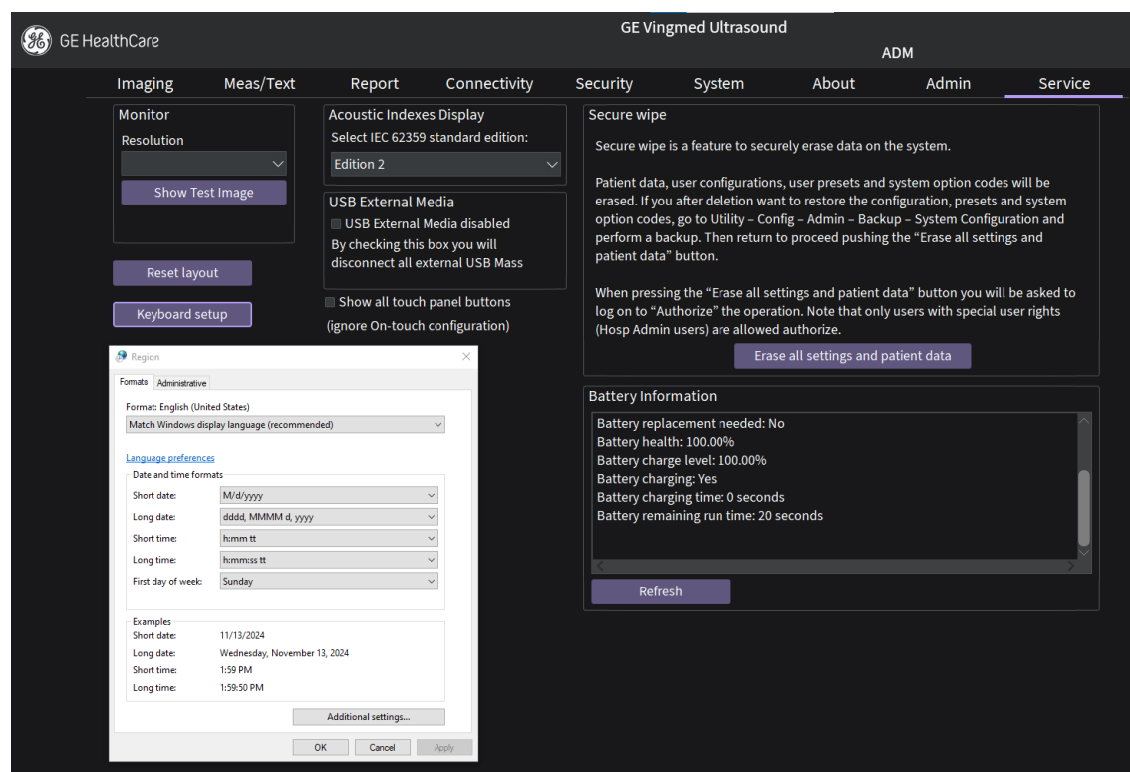


Figure 5-95 Select Keyboard Setup

1. Select **Keyboard Setup** to get access to **Keyboard Properties**.
2. Select the **Formats** tab.

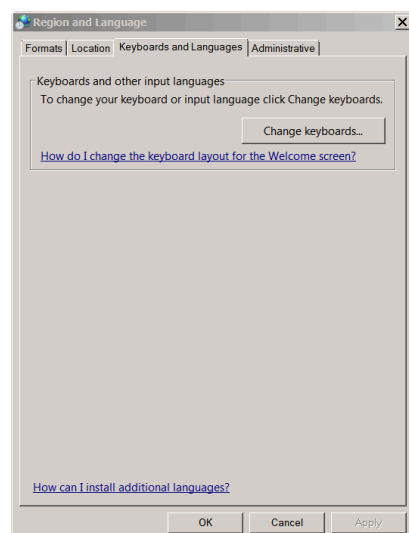


Figure 5-96 Keyboard Language dialog

3. Select **Change Keyboards...**

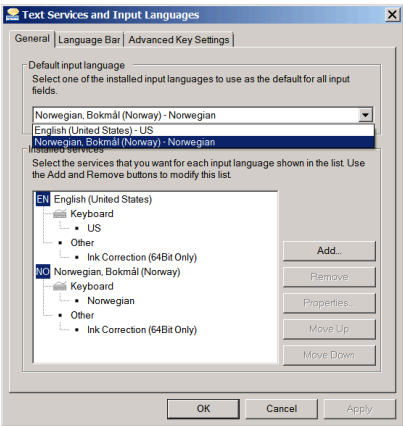


Figure 5-97 Select input language

- 4. Select the installed keyboard and then select **OK**.

5.3.7.7 Secure Wipe

Secure wipe is a function allowing for permanent deletion of data on the disk, thus eliminating the possibility of recovering sensitive data. This provides an extra layer of protection for customer data.

Configure User Rights

To perform the secure wipe, the user must have both the **HospAdmin** and **SecureWipe** privileges. The privileges could be configured on the **Security > Users** tab as shown below.

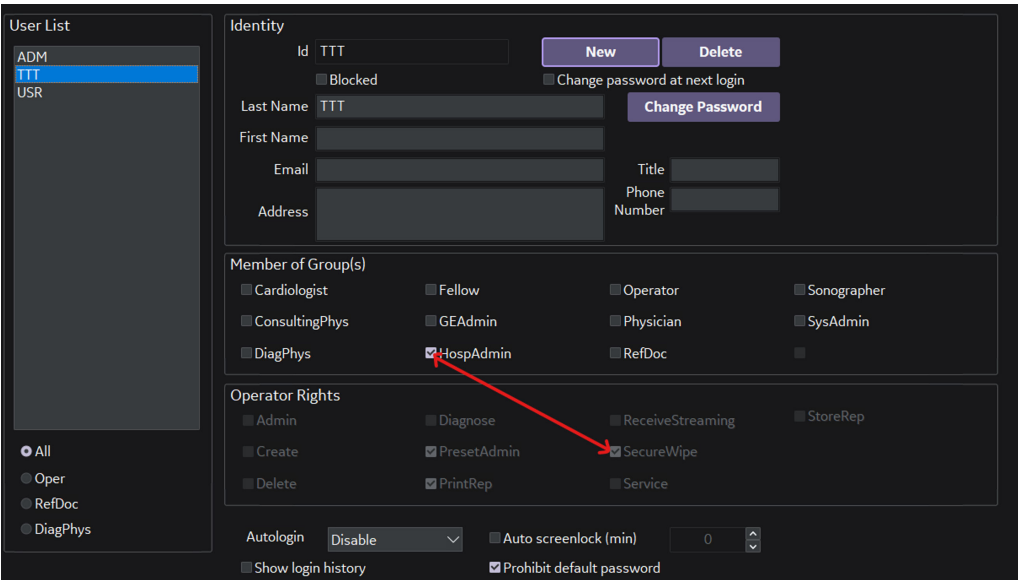


Figure 5-98 Configuring user privileges

Perform Secure Wipe

Once the privileges are correctly configured, the user can perform the secure wipe from the **Service** tab (see [Figure 5-99](#) on page 207below).

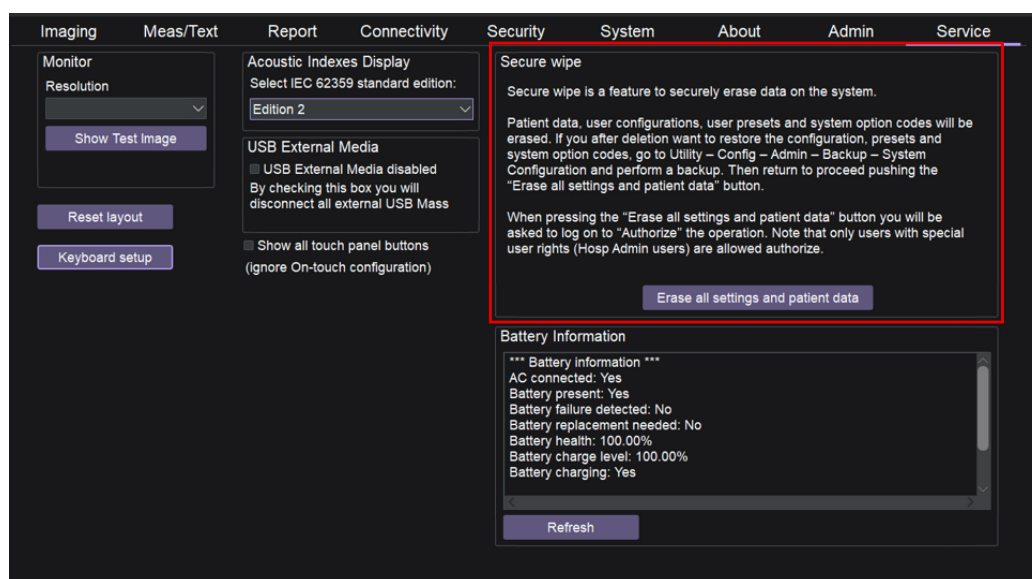


Figure 5-99 Performing secure wipe

Press **Erase all settings and patient data** to start the secure wipe process.

Authentication is required to start the secure wipe to prevent accidental deletion of all the patient data.

Figure 5-100 Authorization request for secure wipe launch

Enter your credentials and press **Authorize** to launch the process.



CAUTION

Secure Wipe will erase all data on the E: drive including all images and patient data stored locally in the system. There is no way to retrieve or restore this data when Secure Wipe has been completed.

A warning dialog pops up to give the user a final chance to change their mind before starting the secure wipe process. Press **Yes** to proceed.

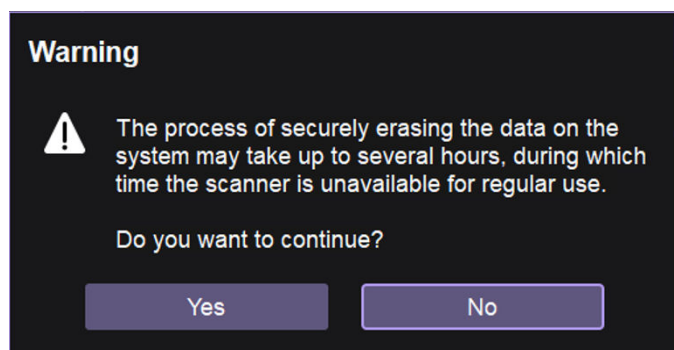


Figure 5-101 Secure wipe warning popup

5.3.7.8 Battery information

For details about the battery installed on the system, navigate to **Config > Utility > Service**.

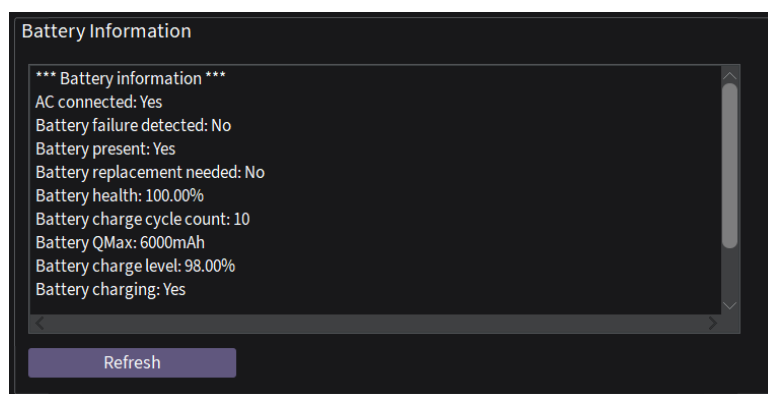


Figure 5-102 Battery information

5.4 Calibration

5.4.1 Monitor and Touch Screen Alignment

To ensure the correct alignment of the Main Monitor and the Touch Screen, perform the following procedure:

1. Insert a service dongle, restart the application with **Ctrl+Alt+R**, and select **Exit to Windows**.
2. Open the folder C:\700BE_Installs.

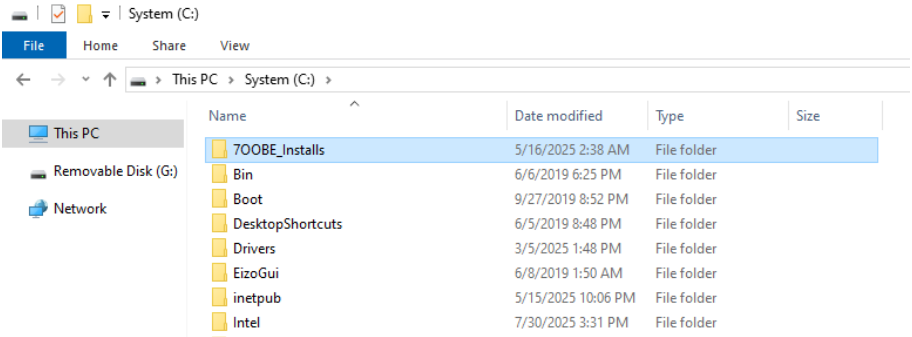


Figure 5-103 Open folder C:\700BE_Installs

3. Run the file DisplaySettings.exe.

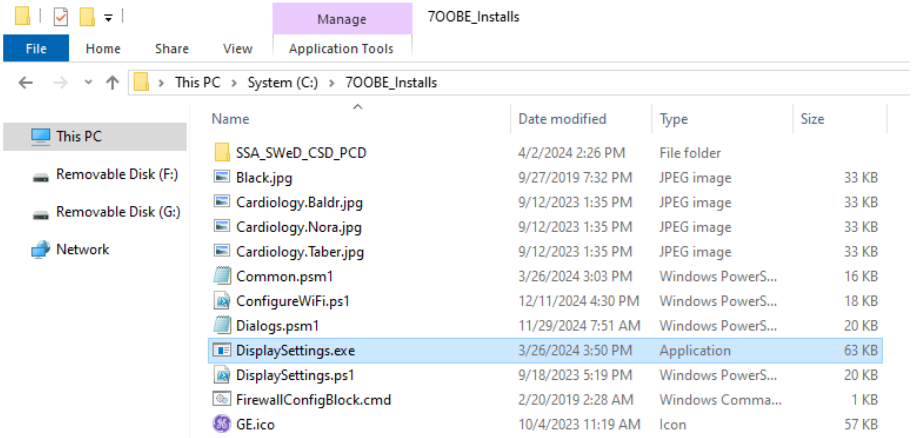


Figure 5-104 Run the file DisplaySettings.exe.

Align the main monitor and the touch panel correctly.

Display

Rearrange your displays

Select a display below to change the settings for it. Press and hold (or select) a display, then drag to rearrange it.

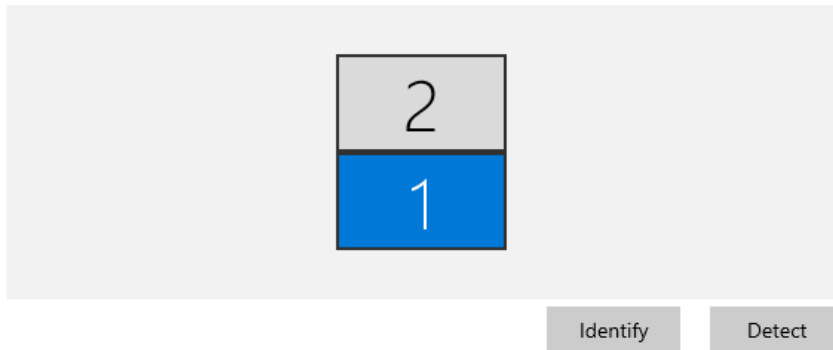


Figure 5-105 Arrange your displays: LCD main monitor (upper), Touch Screen (lower)

4. Verify that the cursor moves correctly from the Main Monitor to the Touch Screen by moving the trackball cursor vertically.

5.4.2 Touch Screen Calibration

While in Windows, calibrate the touch panel by performing the following steps:

1. Run the application **Tablet PC Settings**, available from the desktop.
2. Select **2. Displayport**.

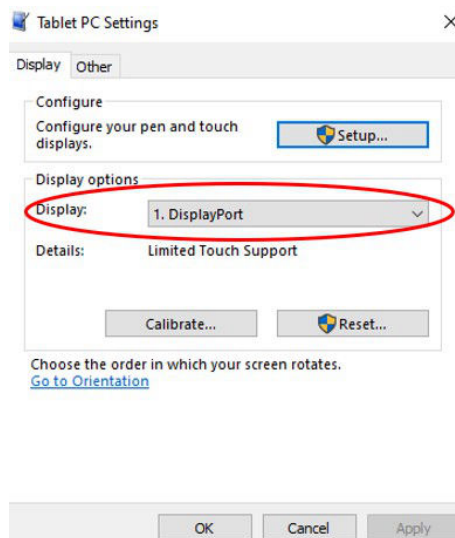


Figure 5-106 Select: 1. Displayport

3. Press the **Reset...** button and select **Ok** when prompted.

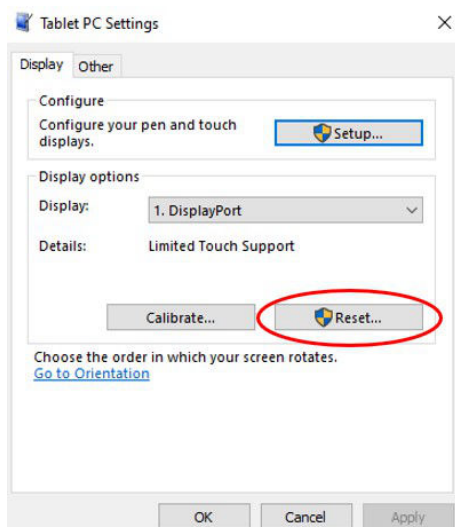


Figure 5-107 Select Reset...

4. Select: **2. Displayport** again.

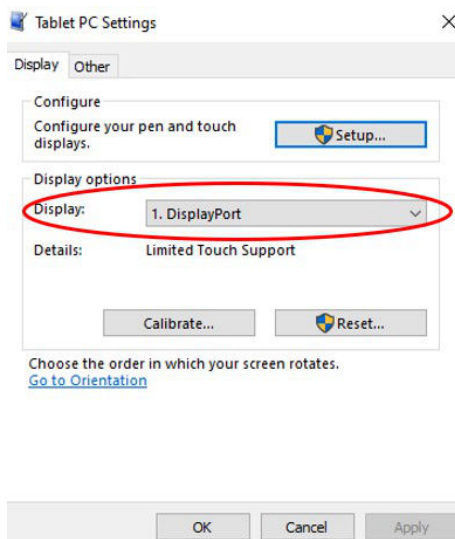


Figure 5-108 Select: 1. Displayport again

5. Press the **Calibrate...** button.

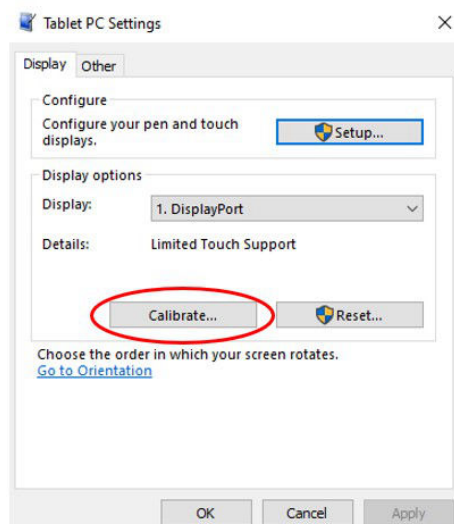


Figure 5-109 Press Calibrate....

6. Touch the crosses that come up on Touch Screen.
7. Select **Yes** on the Touch Screen to save the calibration data.
8. Select **OK** to close the Table PC Settings dialogue box and restart the Application SW.

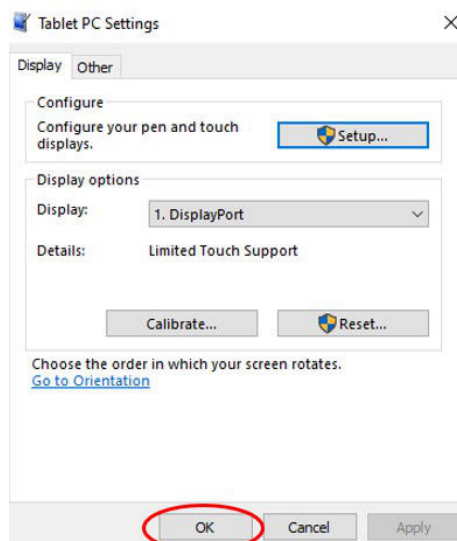


Figure 5-110 Select OK to close the Tablet PC Settings dialogue box.

5.5 InSite

5.5.1 Introduction

InSite is your direct link with a GE HealthCare Online Service Engineer or Applications Support Engineer, or a Request for Service via the InSite link at the bottom of the display screen.

5.5.2 Request for Service (RFS)

Create a Request

Users can create and submit an error report to GE Healthcare digital service.

NOTE

Submission will only be successful if the agent is already connected.

1. Select the Contact GE Icon on the status bar at the bottom of the screen.



2. The InSite service menu is displayed. You will see these options:

- **Service Desktop**
- **Request for Service**
- **Connect to GE Healthcare**
- **Connect to Clinical Lifeline**
- **Cancel**

3. Select **Request for Service**

4. The **Contact GE** screen is displayed

5. Select **Create RFS** from the top right of the screen.
6. Fill out the required information and press **Submit**

Table 5-1 Request for Service - Required Information

Element	Description
System ID	
Problem Type	What type of issue are you experiencing? Valid values are: • Service • Application This is a required field
Problem Area	In which area are you experiencing an issue? Valid values are: • Hardware Electrical • Hardware Mechanical • Network • Image Quality • Peripherals This is a required field
Problem Description	Description of the issue in more detail. This is a required field
First Name	First name of submitter. This is a required field
Last Name	Last name of submitter. This is a required field
Phone Number	Phone number of submitter. This is a required field
Email	Email of submitter.

Selecting **RFS History** on the **Contact GE** screen will display all previously submitted requests and include the information described in table [Table 10-31 Request for Service - Required Information](#) on page 307.

Chapter 6

Functional Checks and Checkout Procedures

In this chapter

Functional Checks.	216
Ultrasound Image/Equipment Quality Check (IQC and EQC).	218

6.1 Functional Checks

Overview

Functional checks are used to verify that the Vivid Explorer/Vivid Advanced/Vivid Focus works as intended.

The functional checks may also be used during troubleshooting.

Functional Checks in this Chapter

1. [Basic Control Functional Checks](#) on page 216

Preparations

6.1.1 Basic Control Functional Checks

After setting up the ultrasound system, perform these functional checks before returning the ultrasound system to the customer.

1. Power up the ultrasound system.
2. Connect a probe to one of the probe connectors and select it.
3. Test scanning in Doppler mode and in 2D mode.
4. Perform power shut down before returning the system to the end user.

6.1.2 Mechanical Functions Checks

6.1.2.1 Castors and Brakes Functional Checks

6.1.2.2 Operator Panel Movement

To check up and down of the system, hold the front handle with both hands, compress the release button and pull the platform up or push it down. Release the button when the system height is set at desired position.

Refer to [Control Console Positioning](#) on page 84 for details.

6.1.2.3 Main monitor positions and lock functional checks

Check the the lock and unlock function for the monitor position works as described in the section below ([Unlock and move the main display](#) on page 216 and [Lock the main display](#) on page 217).

It is possible to lock and unlock the monitor position as described.

6.1.2.3.1 Unlock and move the main display

1. Turn the release knob clockwise to unlock the main display ([Figure 6-2 Main display locking mechanism](#) on page 217).

The main display can be moved freely in all directions.
2. Grab the bottom of the main display to adjust the position of the display ([Figure 6-1 Positioning the main display](#) on page 217).



Figure 6-1 Positioning the main display

NOTE

When flipping up the display from the flip down state, you can grab the upper corner of the display.

6.1.2.3.2 Lock the main display

1. Turn the release knob counter-clockwise to raise the lock.
2. Move the main display sideways to lock the arm's joints.
3. Push the display down into parked position.



1. Unlocked main display
2. Locked main display
3. Release knob

Figure 6-2 Main display locking mechanism

6.2 Ultrasound Image/Equipment Quality Check (IQC and EQC)

Based on the region/area of the world that the customer is based in, the customers Quality Assurance Program may require alignment with local standards. Thus, all testing per IEC 60601 and/or IEC 62353 may or may not be required for EQC Care and Maintenance.

For example, United States National Standards for Health Care Facilities (NFPA 99) only requires touch current testing in normal operating configurations (this standard differs from IEC 62353 requirements for recurrent test and test after repair of medical electrical equipment).

Contact your GE HealthCare Service Representative to perform Equipment and Image Quality Checks.

6.2.1 Image Quality Check (IQC) preset for service

Image Quality Check (IQC) is intended to facilitate Image Quality checks during Quality Assurance Evaluation. Quality Assurance tests are used to determine whether a scanner is providing the same level of performance year after year.

By using the same setting year after year, this ensures that the data collection consistent, independently of who performs the test.

This preset only includes fundamental settings for 2D mode. Processing modes like SRI, Harmonics, etc., are turned off.

To do IQC, follow the steps below:

1. Press **Probe** button on the control panel.
2. Select the proper probe, and then select **IQC** in the Applications column.

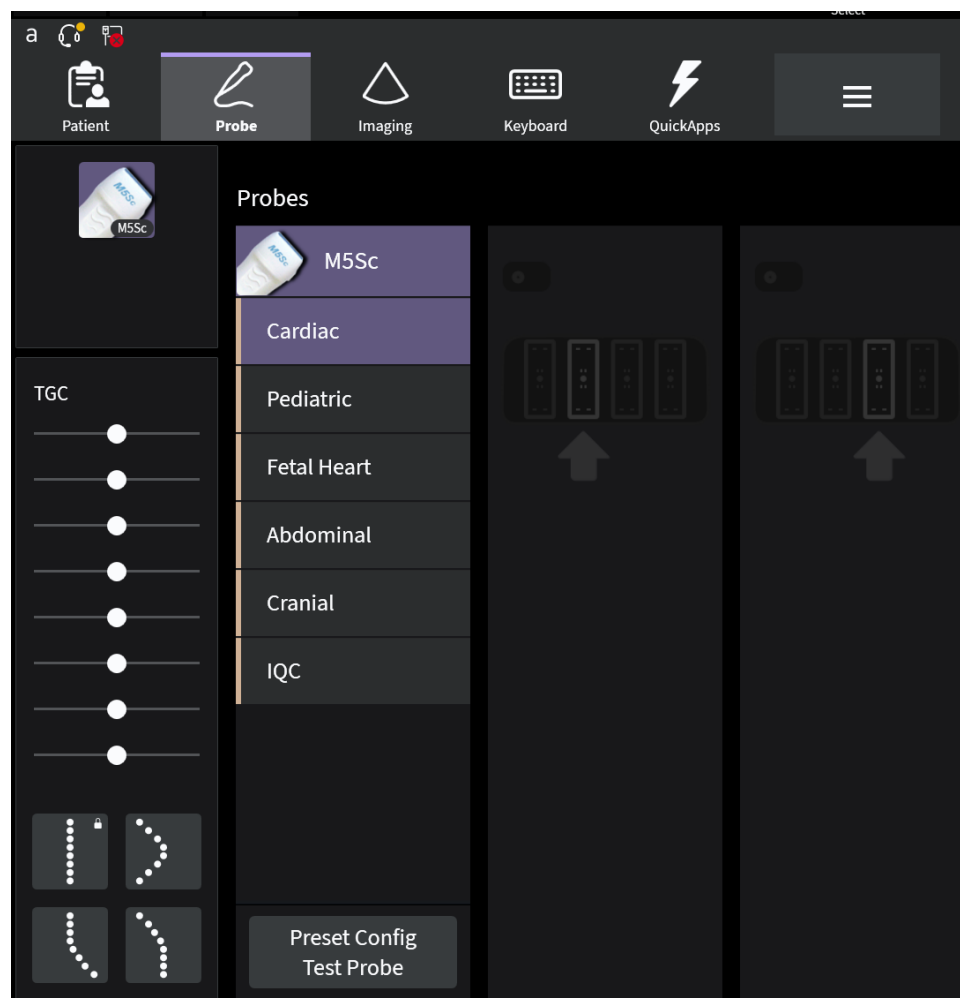


Figure 6-3 Image Quality Check

Chapter 7

Software

In this chapter

General Information.	220
eDelivery - Software update.	222
Setup after software loading.	240
Backup and Restore.	241

7.1 General Information

7.1.1 Customer Provided Prerequisites

- Server solution or formatted and labelled media for Images storage
- Formatted and labelled media for Patient Archive and Presets (User Defined Settings)
- Security Certificates, per customer site IT policy
- Password for the user ADM (see below)

7.1.1.1 Password for the user ADM

The default password for the user ADM is **ulsadm**.

The customer must type in the password when needed if the password for the user ADM has been changed.

7.1.2 Tools Provided with the System at Delivery or After an Software Installation

- System software
- Application software

If present:

- patch(es) for the ultrasound system

Verify that the software is current. Updated software may be available.

Check for any available FMIs.

NOTE

If installing a new software revision, ensure that you have the previous software version available, in case the software load fails, and you have to reload the 'old' software.

7.1.3 Space Management - Moving All Images



CAUTION

In order to complete a successful restoration of the Patient Database if needed, the images must be moved away from the ultrasound system before doing backup of the Patient Database. Depending on the location set-up, either move the images to a remote server or to removable media. As the images are moved, the database will point to the new location. If this procedure is not completed correctly, the images and database information could be lost.



CAUTION

If the drives are encrypted, a software update will have a catastrophic effect if the customer does not have the recovery key. All data will be lost.

Speak with the personnel at the site to determine which patient images need to be backed up prior to starting.

- Move the images to a remote server or to removable media.
For instructions, see "Disk management" in the User Manual.

If the drives are unencrypted, software can be reloaded without a recovery key.

7.1.4 Backing up the Patient Archive and System Configurations

**CAUTION**

An error, or a power loss may occur during the software loading.

Always backup the Patient Images, Patient Archive and the System Configurations before starting a software loading!

In order to complete a successful restore of the Patient Database, if needed, the images must be moved away from the Vivid Explorer/Vivid Advanced/Vivid Focus before doing backup of Patient Database.

Depending on the location set-up, either move the images to a remote server or to removable media. If the backup procedure is not completed correctly, the images and database information could be lost.

Backup the Patient Archive and System Configurations.

7.2 eDelivery - Software update

7.2.1 Introduction

As part of the product lifecycle management, GE HealthCare regularly analyzes and integrates software updates from our third party vendors into our products. These are typically released as part of regular updates or software releases.

The user can update to the latest software in two ways:

- Through the GE HealthCare service platform on the ultrasound system. This requires RSvP connectivity. See [Software Update via Insite Remote Service Platform \(RSvP\)](#) on page 222.
- Download the latest software from an end-user portal to a local storage location and install it on the ultrasound system. See [Software update via End-User Portal](#) on page 229.

NOTE

Software includes application software, 3rd part application software, and OS update patch.

7.2.2 Software Update via Insite Remote Service Platform (RSvP)

Software updates may be made available for download and installation through the Insite Remote Service Platform (RSvP).

All users can download new software, but only users with administrative (ADM) rights can install new software.



CAUTION

Please back up presets and database before installation of the software (see [Data Backup and restore](#) on page 197).

Remote software download should not change user presets or affect customer database; however, it is always best practice to ensure patient data and presets are backed up before proceeding with any software installation.



CAUTION

If the drives are encrypted, software update will have a catastrophic effect if the customer does not have the recovery key. All patient data will be lost. If the drives are not encrypted, software can be reloaded without a recovery key.

NOTE

Please allow for several hours to complete a software download (the download time may vary due to network connection speed). Please allow approximately one hour for complete installation.

NOTE

Software update through the Insite Remote Service Platform (RSvP) may not be available in all markets.

7.2.3 Software download and installation

1. Log on as ADM.
2. Ensure the Agent Configuration is correctly configured.

NOTE

Refer to Service Manual or contact your Service Representative for Agent Configuration information.

The icon  indicates the Agent Configuration is not configured.

The icon  indicates the Agent Configuration is correctly configured.

3. Press the **Download** button ( or ) at the bottom of the display screen.

NOTE

The system automatically queries if new SW is available. There is an icon on the bottom left corner of the title bar to illustrate the download connection status.

The icon  indicates the last query was negative. You may press it to initiate another query.

The icon  indicates that new software is currently available for download.

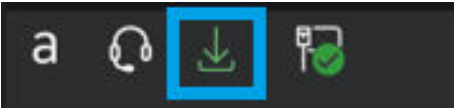


Figure 7-1 Press Download Button

4. Available software updates are displayed in the list. If you want to refresh the query for available updates, press **Refresh**.

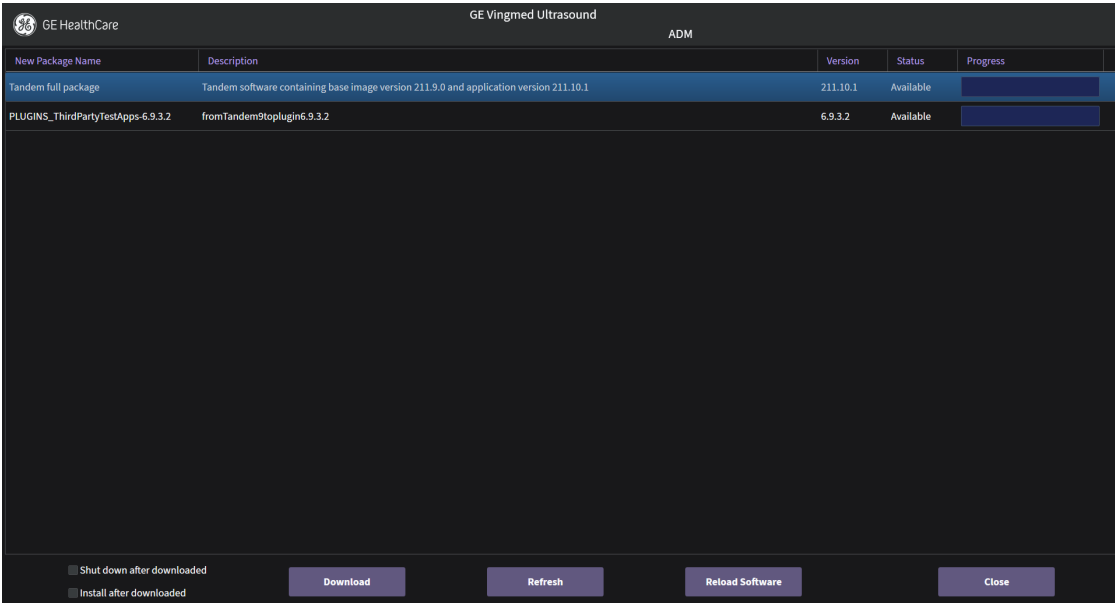


Figure 7-2 Software List

5. A window will pop up and ask for confirmation of refresh. Press **Yes**.

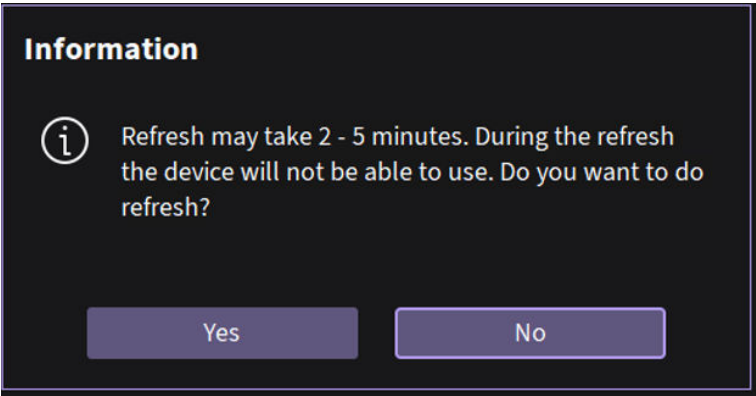


Figure 7-3 Refresh Confirmation

6. A dialogue informs that the refreshed list of available updates is being queried.

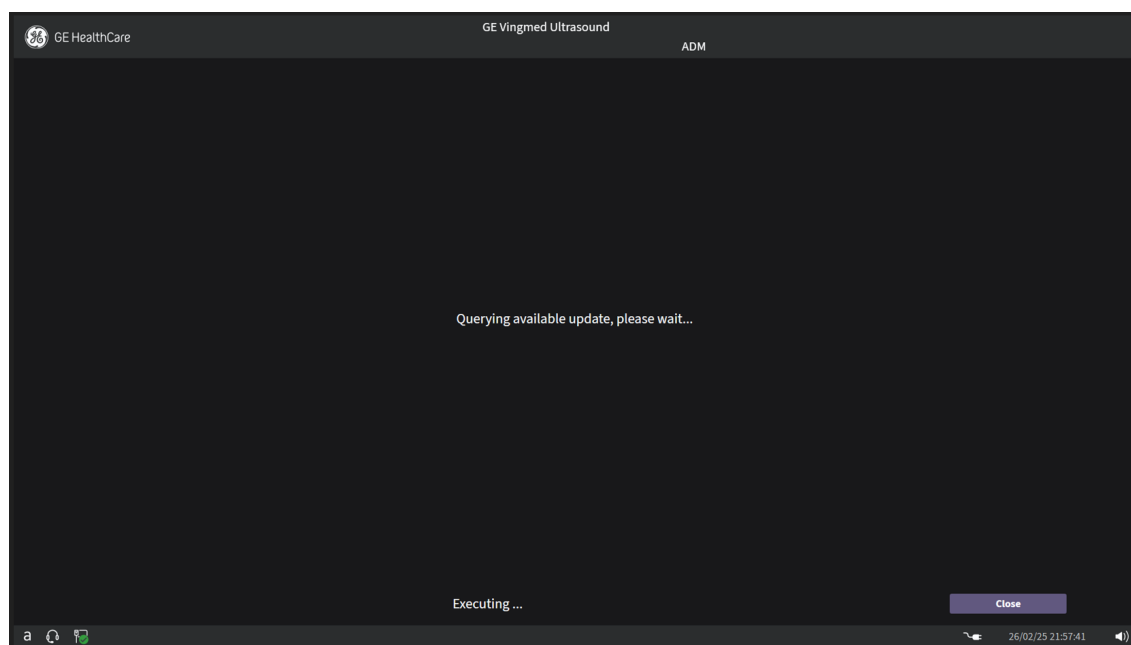


Figure 7-4 Receiving List

7. Select the desired software and press **Download** to download the software, or press **Close** to exit the window.

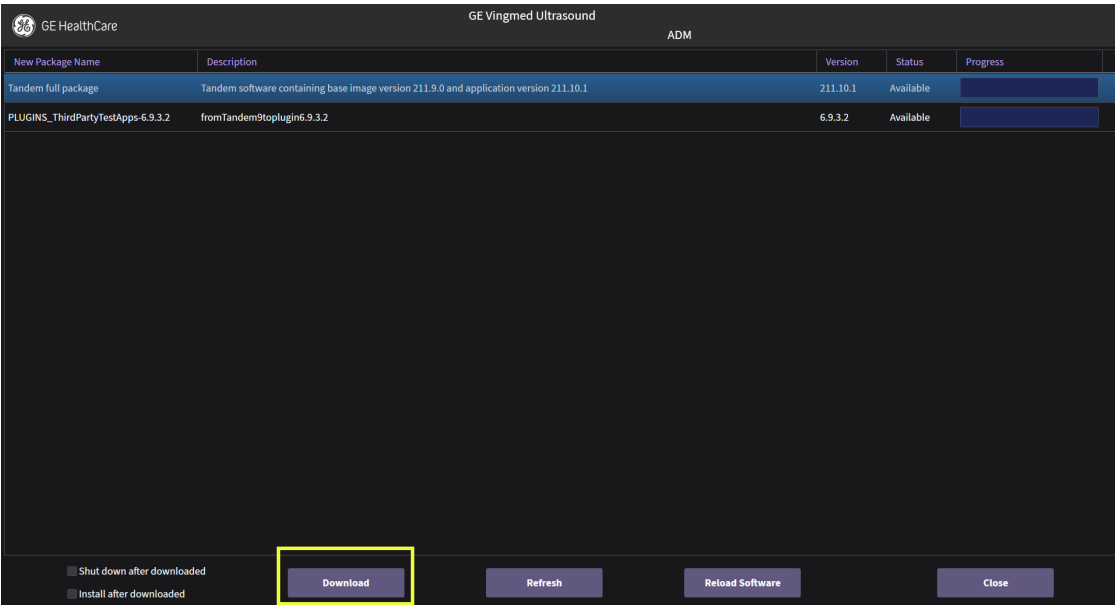


Figure 7-5 Start the Download

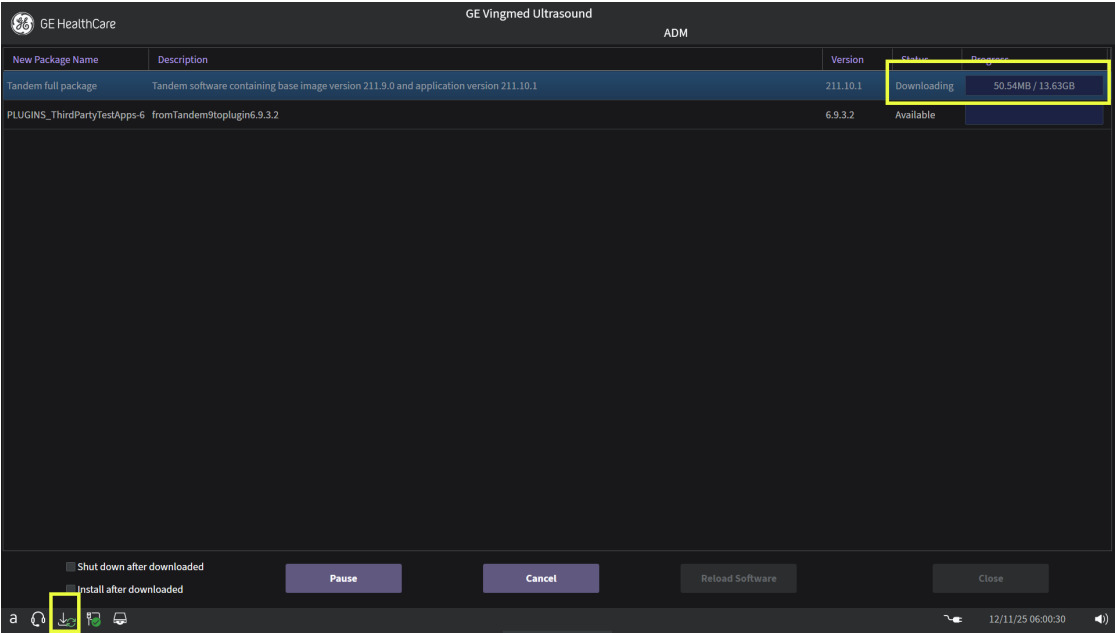


Figure 7-6 Software Download in Progress

NOTE
If the system doesn't have enough space for download and installation, please contact your GE HealthCare service representative. It is suggested that service engineer clear the disk space first.

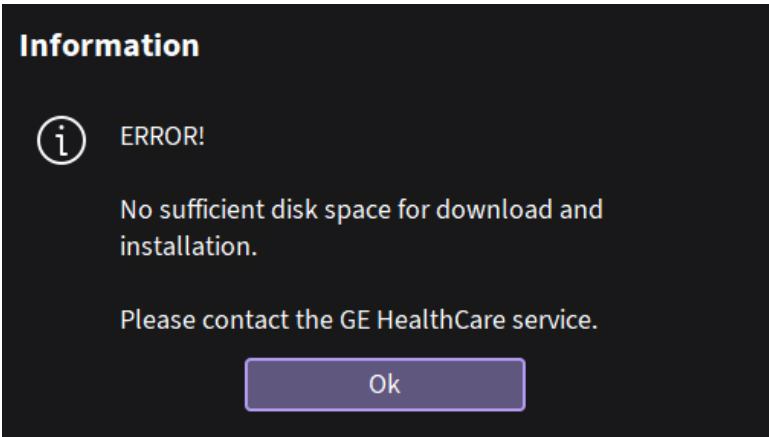


Figure 7-7 No Space Error

8. During the software download process, the Status will be displayed as **Downloading**.

Status	Progress
Downloading	200.00MB / 13.18GB

Figure 7-8 Status displays: 'Downloading'

You can press **Pause** to suspend the download or press **Cancel** to exit the download.

- If the download process is suspended, press the **Resume** button to continue the download process from the point where it is stopped.
- If there is an error during the downloading process, press the **Retry** button to restart the download process from the beginning.

By selecting **Pause** (lower left on the screen) it is possible to pause the download.

If **Pause** has been pressed, it will be indicated by that the Status field is displaying **Paused**.

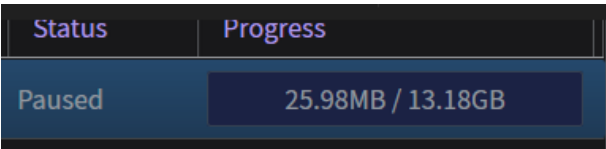


Figure 7-9 Status displays: 'Paused'

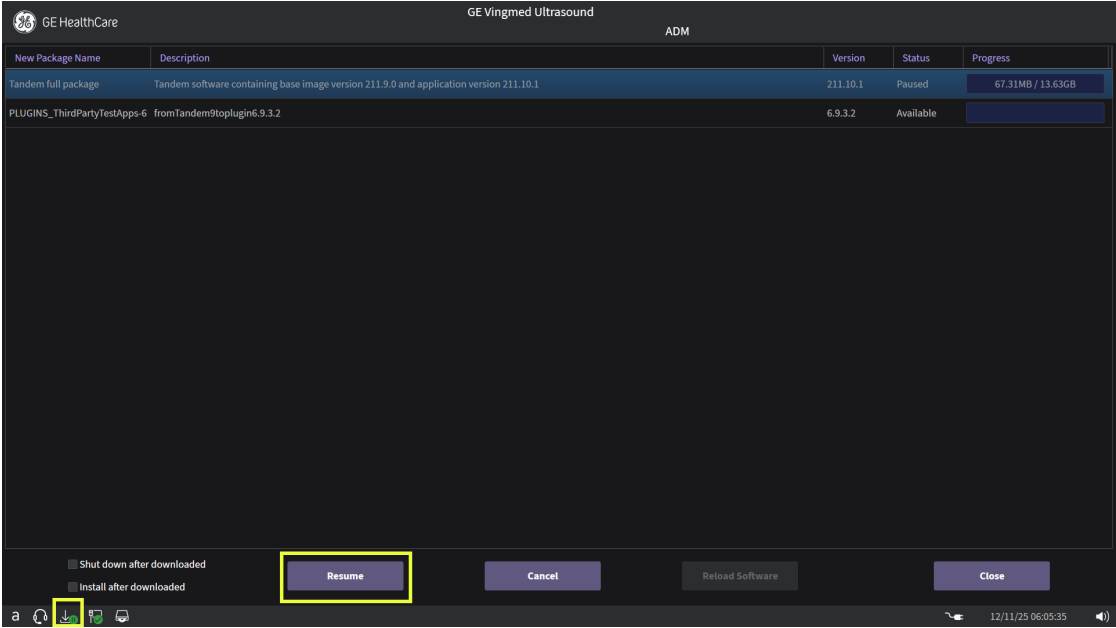


Figure 7-10 Select Resume to continue the SW download.

9. When the download is complete, the software is ready to be installed on the ultrasound system.

NOTE
Please make sure system power plug is connected before you press the Install button to avoid installation failures due to power loss.

NOTE
Only users with administration right (ADM) can perform the software installation.

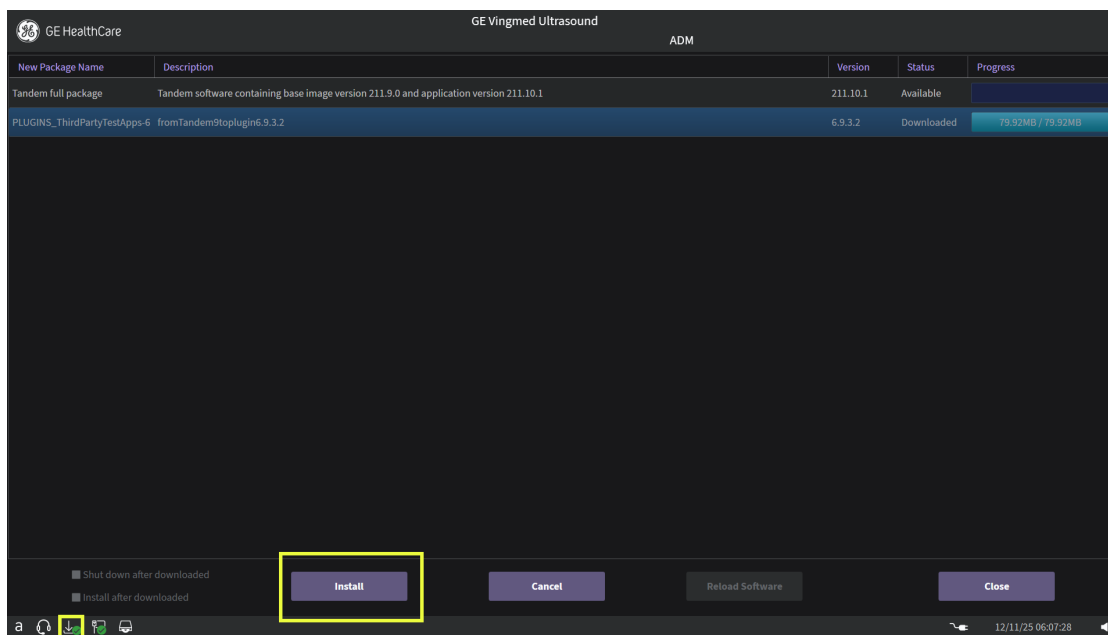


Figure 7-11 Press the **Install** button to install the new software

10. If not already done, log in as an administrator (ADM).
 11. Press the **Install** button to install the new software.
- A message is displayed on the screen.
Read the text in the popup warning and follow the instructions.

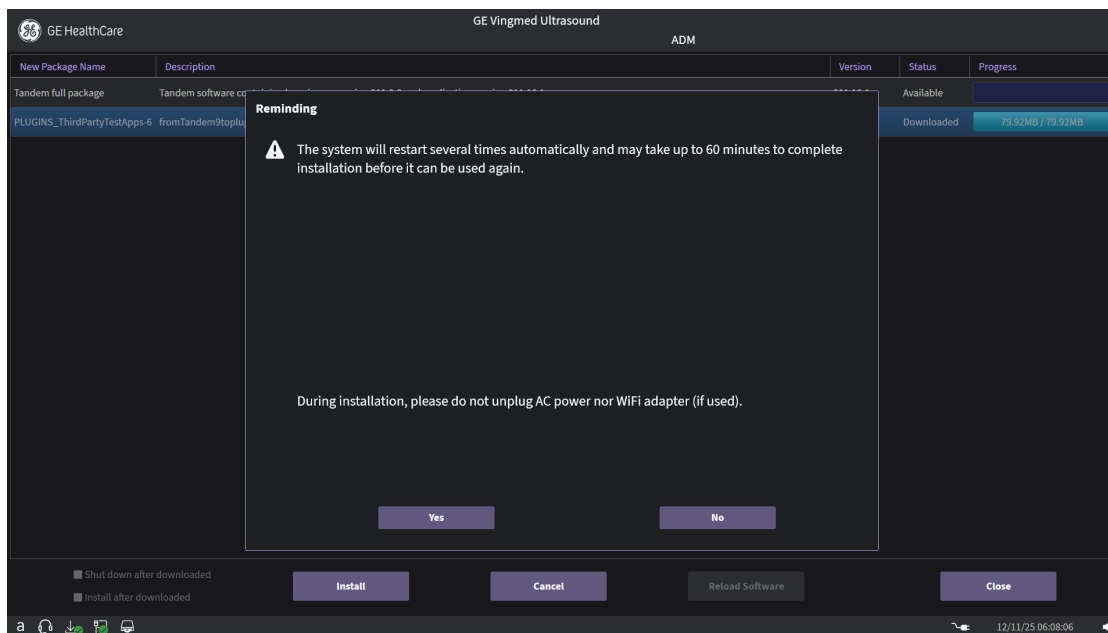


Figure 7-12 Popup warning

- If needed, click **No** to postpone the installation.
- Click **Yes** to start the software installation. The system starts the software installation.

After approximately ten minutes, you may need to start the system manually by pressing the **On/Off** button on the Operating Panel.

During the installation the ultrasound system will reboot several times before the installation is complete.

NOTE

Do not power off the system during the software installation.

NOTE

A typical full installation may take up to 60 minutes.

NOTE

A typical application software installation may take up to 20 to 30 minutes.

7.2.4 Software update via End-User Portal

Customers entitled to eDelivery updates get a customer account to download software within the End-User Portal.

Users are created for the account based on e-mail addresses provided by the customer at the point of sale. These e-mail addresses are the log-in credential for the End-User Portal along with a temporary password provided to the user through e-mail. When logging in to the End-User Portal the first time, the user is prompted to change the password and enter a secret question and answer for password retrieval.

**CAUTION**

Please back up presets and database before installation of the software (see [Data Backup and restore](#) on page 197).

Remote software download should not change user presets or affect customer database; however, it is always best practice to ensure patient data and presets are backed up before proceeding with any software installation.

Preparations

Ensure that the following information is available (recorded) on paper:

- wireless configuration
- Windows OS password (if changed from factory default)
- Trusted Certificates (for Dicom server with TLS)
- Detailed printer configurations.

Download instructions**CAUTION**

Before loading software, CONFIRM WITH THE CUSTOMER OR SITE ADMINISTRATOR, if the patient archive has been encrypted and if the customer has the password or the Key to unlock the data. Failure to obtain those items, will cause permanent DATA LOSS!

Follow the below instructions to download software from the portal:

1. Log on to the portal website which is provided to end user via a welcome email: <https://gehealthcare.flexnetoperations.com/flexnet/operationsportal>.
2. Log in using the user name (email) and password.

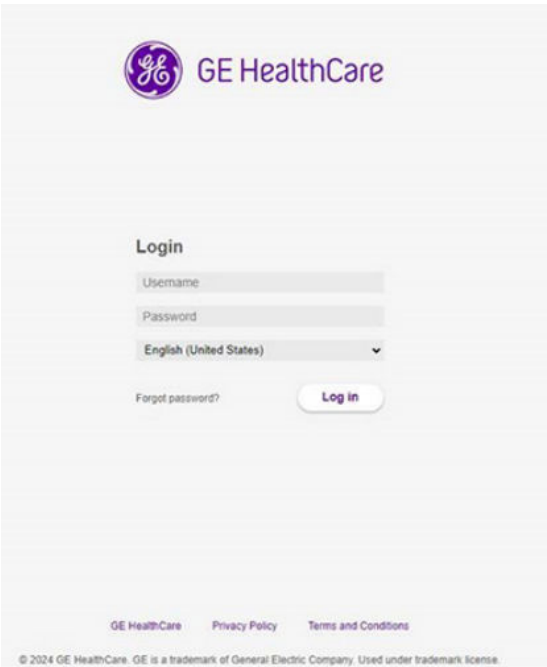


Figure 7-13 Login Window

3. The Software and License Delivery dashboard is displayed. Downloads can be found by browsing the products under “Your Downloads”.

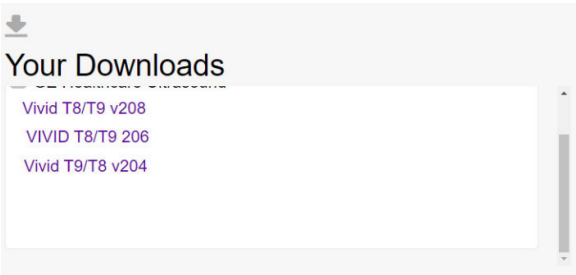


Figure 7-14 Software and License Delivery Page

4. Select the desired software to enter the **Downloads** page.



Figure 7-15 Downloads Package

5. To start the download, click the purple file name link and save the file to a local storage location (i.e. USB flash drive).

NOTE

At this point, you may be asked to download and install **Download manager**. Follow the on-screen instructions to download and install it. When done, return to this procedure and continue.



Figure 7-16 Downloads Page

NOTE

Ensure that the USB Flash Drive is formatted as NTFS format so that the eDelivery package can be successfully copied into it.

6. Check the box in front of the file you want to download, click the **Download Selected Files** button and save it in a local storage location (i.e. USB flash drive).
If you are unable to download using the **Download Selected Files** button, you can directly click the '.7z' file.
7. Load the software on the ultrasound system from the selected storage location. Refer to the Service Manual for detailed installation procedures.

7.2.5 Installing the Software from USB stick

To install software downloaded to a USB stick from the Customer Portal onto the Vivid Explorer/Vivid Advanced/Vivid Focus,

1. Power down the Vivid Explorer/Vivid Advanced/Vivid Focus and insert the USB Flash Drive with the software into one of the USB ports on the ultrasound system.

NOTE

Ensure that the system is USB Device Enabled (check Config->Service page).

2. Power on the Vivid Explorer/Vivid Advanced/Vivid Focus and wait for the dialog below.

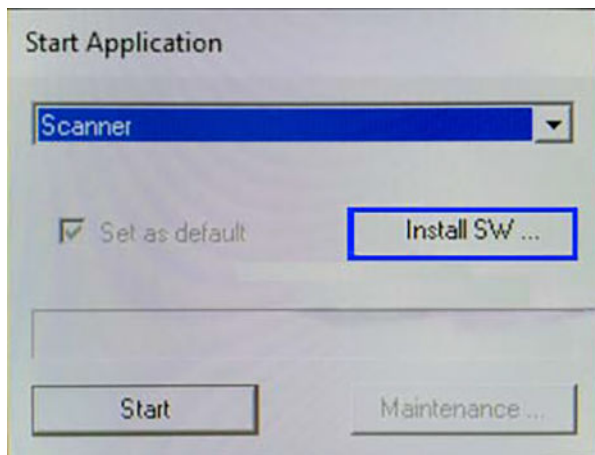


Figure 7-17 Select Install SW...

3. Select **Install SW ...**.
Select **OK** to proceed with installation or **Cancel** to cancel installation.

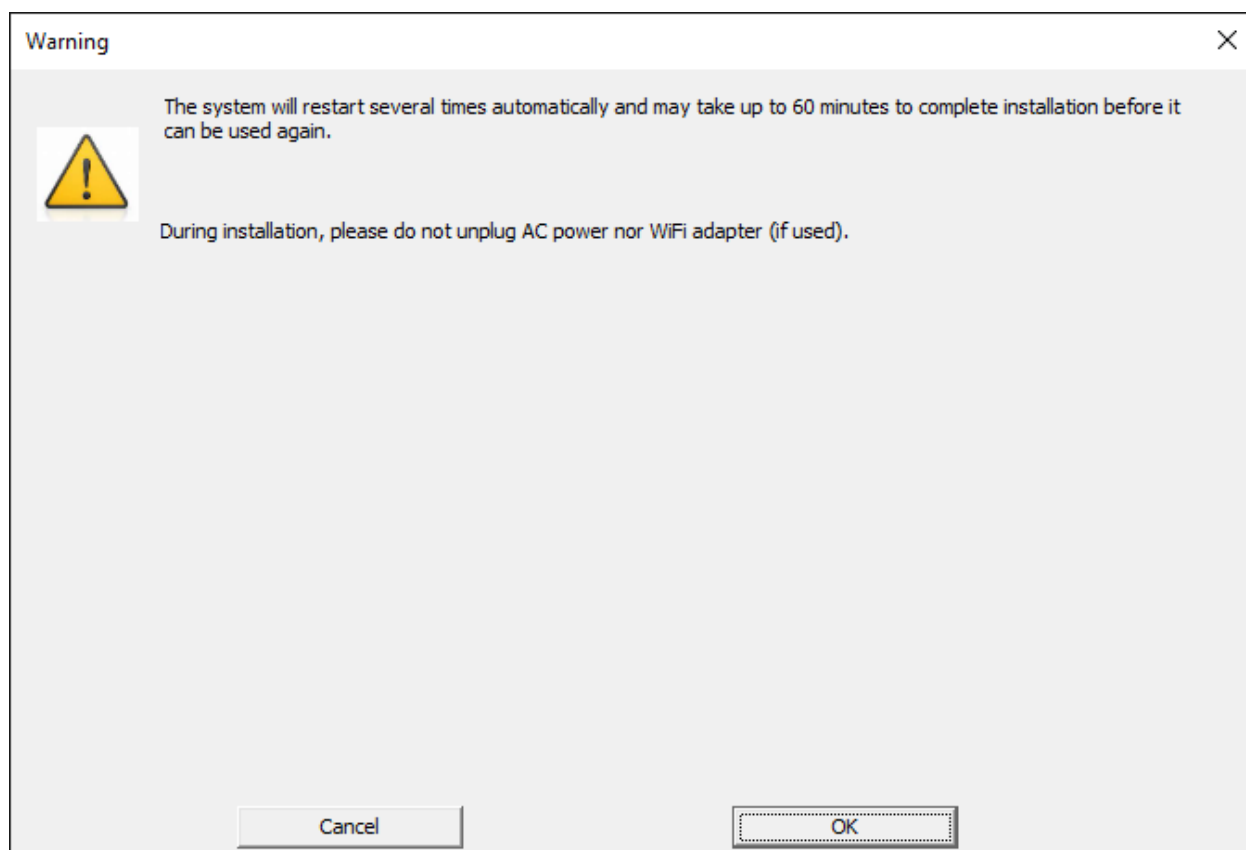


Figure 7-18 You are about to start software installation

4. Select the file in the list and click **Install**.

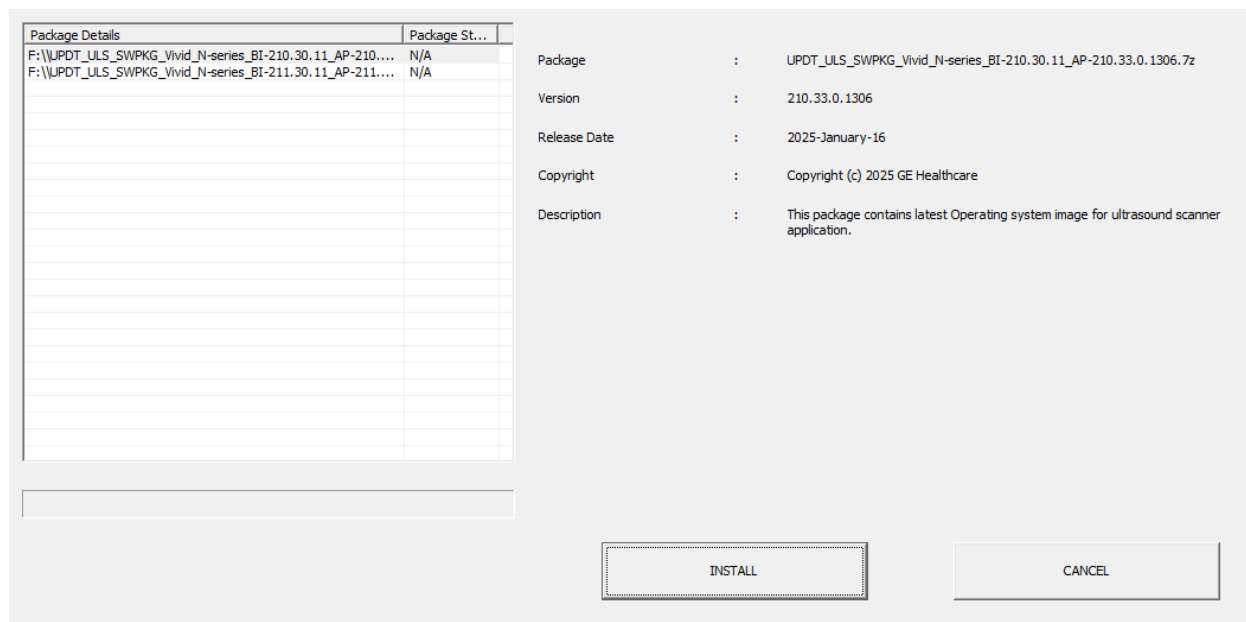


Figure 7-19 Select File - Click Install
Installation commences.

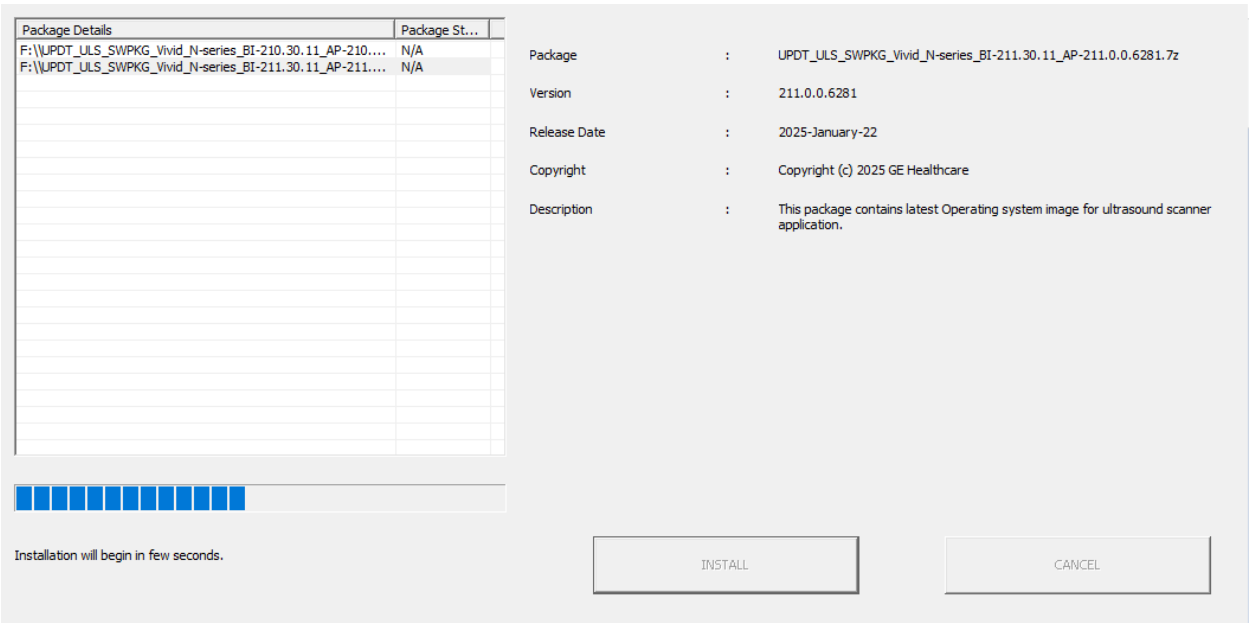
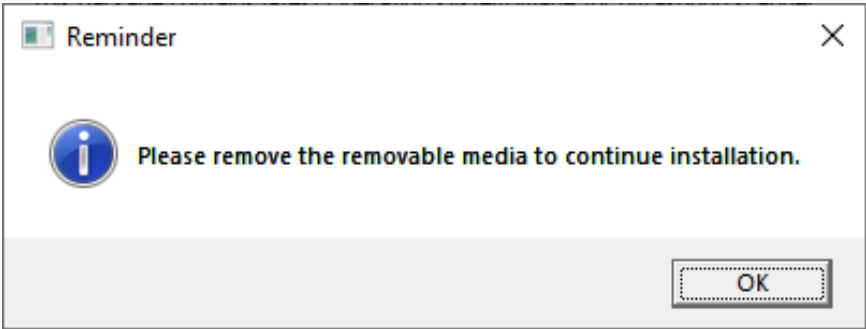


Figure 7-20 Installation starts
The progress bar will fill up multiple times and the system will automatically restart multiple times.



NOTE
Do not power off the system during the software installation.

NOTE
A typical full installation may take up to 60 minutes

5. Remove the USB stick and press **OK** to continue the installation.
The reminder below is displayed:

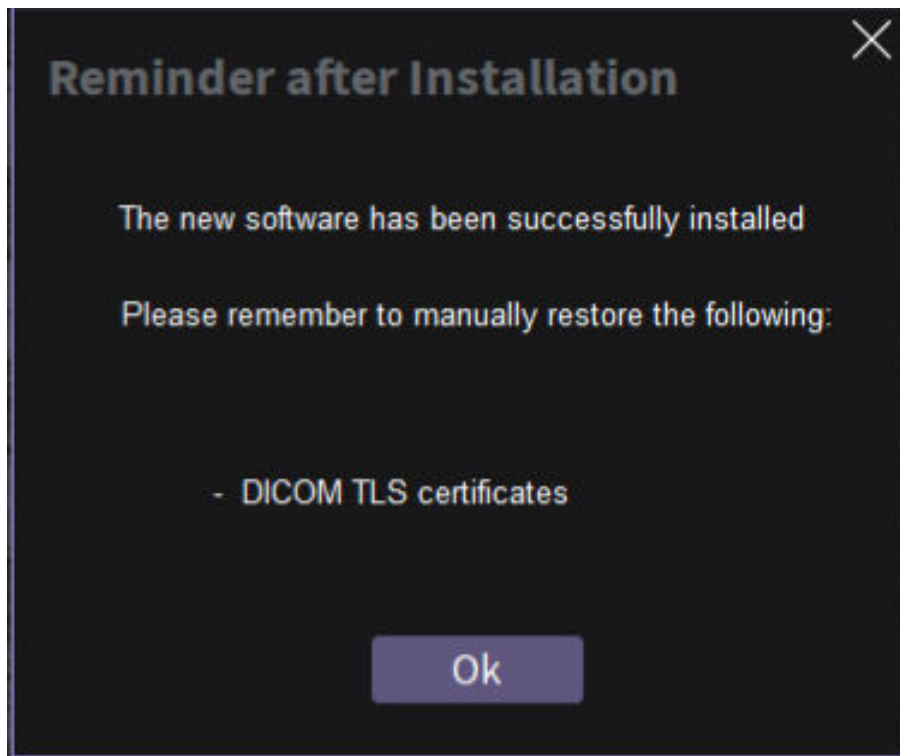


Figure 7-21 Reminder

6. Select **OK**. The software update is complete.
7. Manually restore:
 - WPA-Enterprise WiFi configuration
 - DICOM TLS certificates

7.2.6 Reload Software

Software Reload displays versions of the base image, application software currently installed on the Vivid Explorer/Vivid Advanced/Vivid Focus. Software Reload provides specific software versions for users to reload.

NOTE

The system keeps at most two versions of software for reload. When a newer version is installed, the earliest version will be removed permanently.

A patch installation may also be available. The OS Patch package includes MS Patch only, while the regular eDelivery package includes the whole OS, Application, Patch, and other configurations.

- If the drives are encrypted, Software Reload will have a catastrophic effect if the customer does not have the recovery key. All patient data will be lost.
- To disable disk encryption navigate to **Config > Security > Disk Encryption** and click **Suspend**

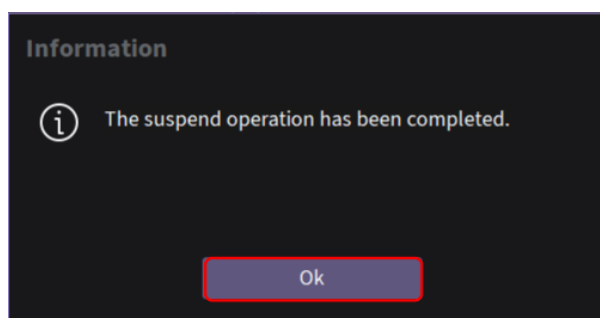
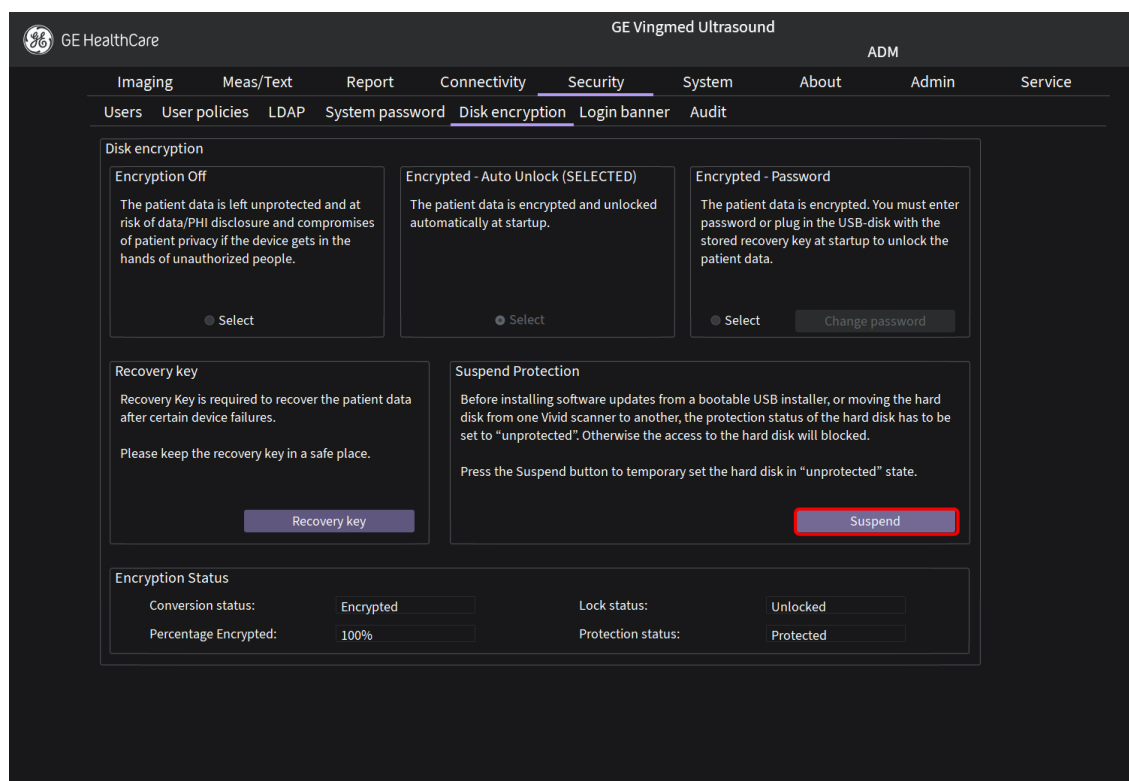


Figure 7-22 Suspend Success

- If the drives are not encrypted, software can be reloaded without a recovery key.

Before starting the Software Reload or Rollback, record the following information and configuration which must be manually restored after the selected software has been loaded.

- DICOM TLS certificates
- WPA-Enterprise configuration

To reload the software onto the Vivid Explorer/Vivid Advanced/Vivid Focus:



CAUTION

Reload software will remove all installed 3rd part applications and settings. They can be reinstalled after reload is finished.

NOTE

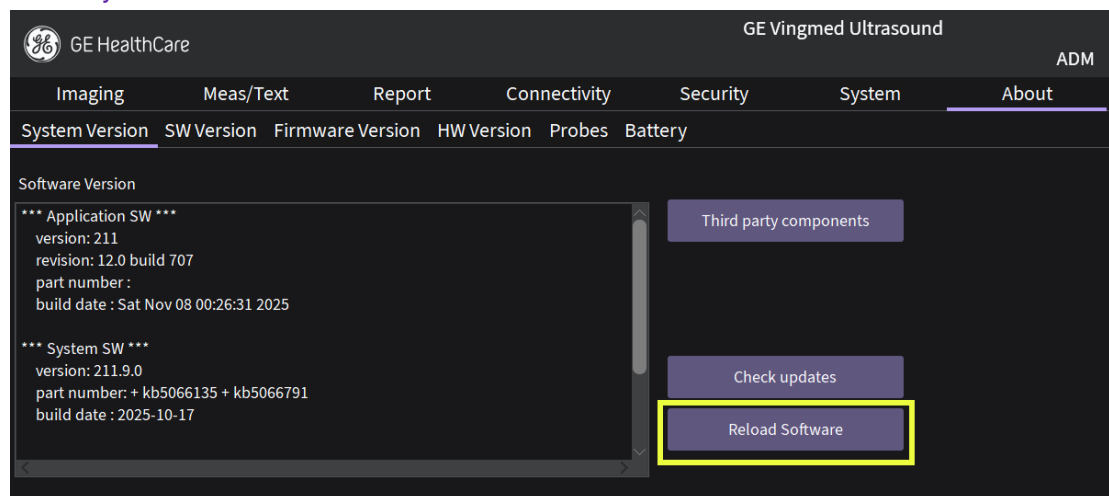
Ensure that AC power remains connected during the software loading.

NOTE

If used, keep the WiFi adapter plugged in.

1. Log on as ADM.
2. Select **About > System Version**.

Figure 7-23 System Version screen



3. Select **Reload Software**.
After selecting **Reload Software**, the following pop-up is displayed:

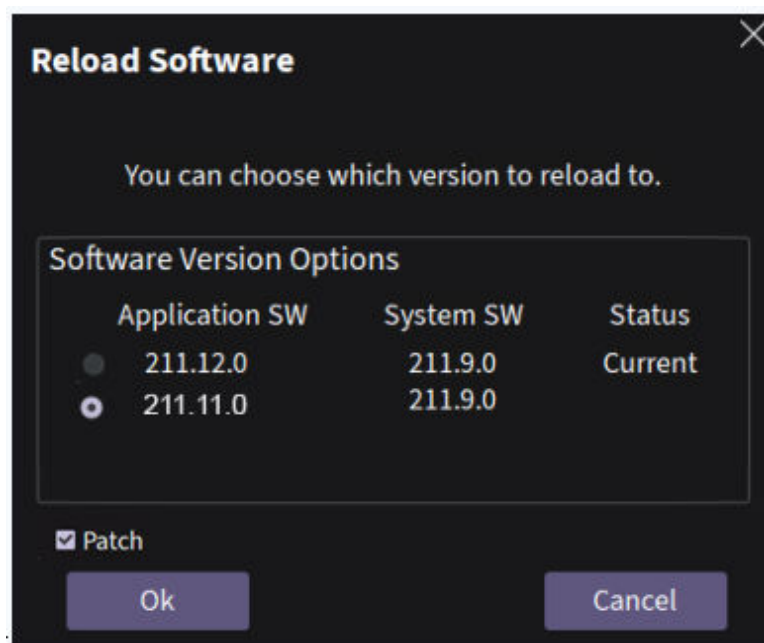


Figure 7-24 Software Version Options

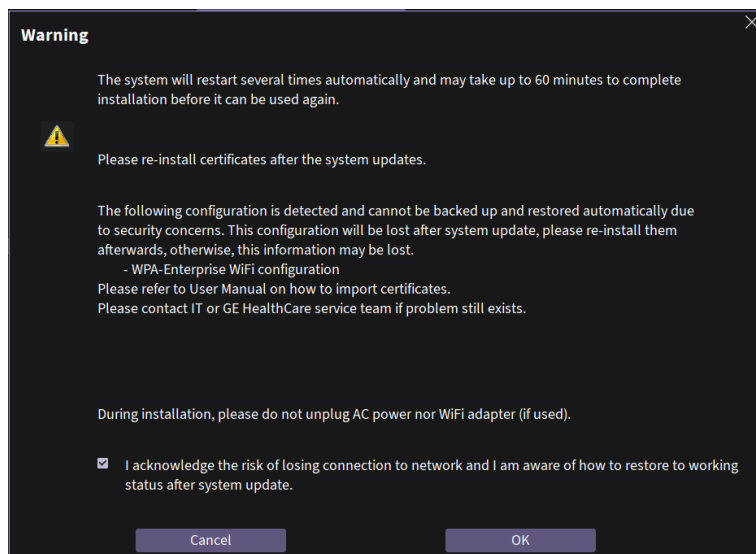
In the pop-up, you may select to reinstall the current installed software (Reload) or select the previous installed software (Rollback).

After installation of an OS Patch, patch software becomes available during reload of software. The **Patch** option is checked by default, which means the reload will include the separate patch only. To reload the full software, uncheck **Patch**.

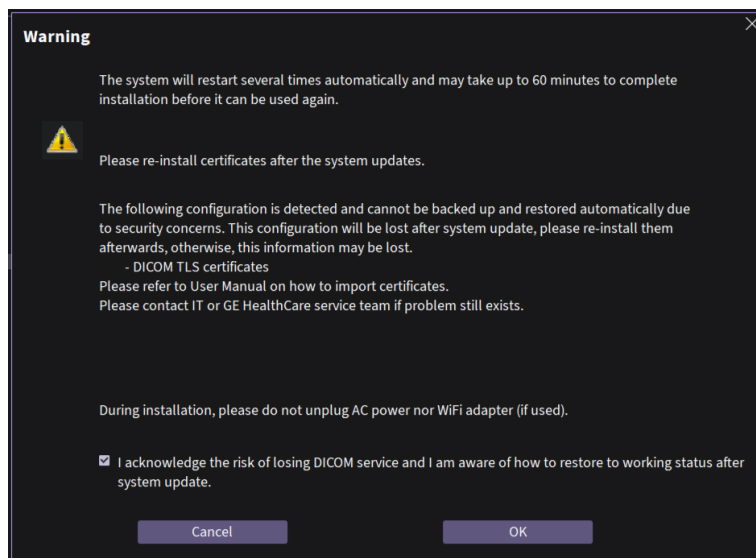
In the example [Figure 7-24 Software Version Options](#) on page 236 in , OS Patch based on 211.11.0 was installed and later updated to 211.12.0. To reload only the OS Patch, check **Patch** and press **Ok**. To reload the full software, uncheck **Patch** and press **Ok**. If a separate Patch based on 210.11.0 was not installed, the **Patch** option will not be available.

4. A new pop-up with reminder is displayed:

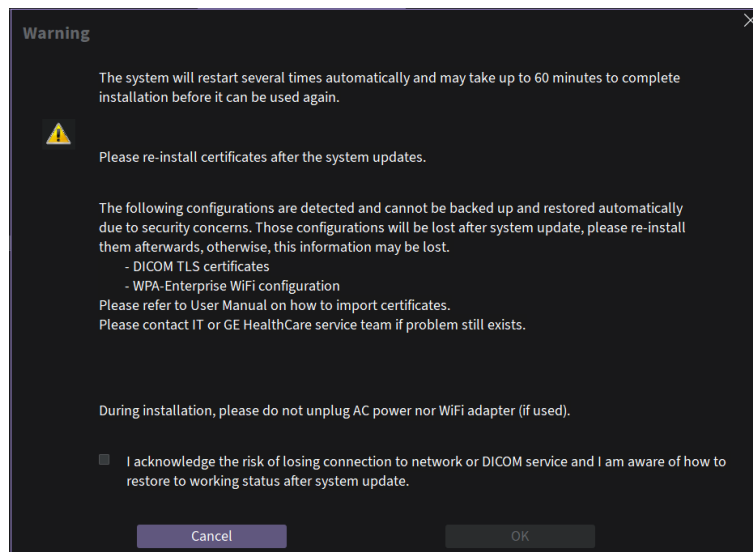
- If WPA is enabled:



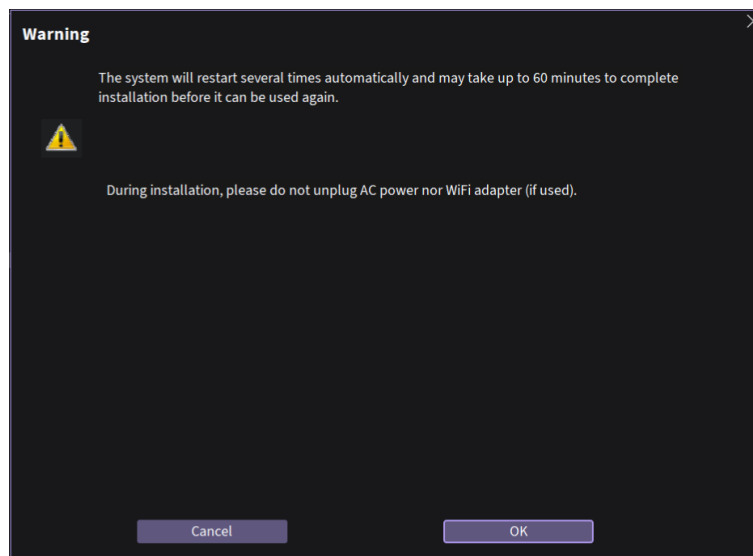
- If DICOM TLS is enabled:



- If DICOM TLS and WPA are enabled:



- If DICOM TLS and WPA are not enabled:



5. If you have already recorded the DICOM TLS certificates and the WPA-Enterprise configuration and you are sure the system can stay connected to mains power for the duration of the software loading (50 to 60 minutes), click **OK**.

If not, click **Cancel**.

After clicking **OK**, the system starts the software loading. The system will automatically power down. Wait ten seconds, then press the **ON** switch on the Operator Panel to restart the system. It will start to prepare the C: drive and load the selected software.

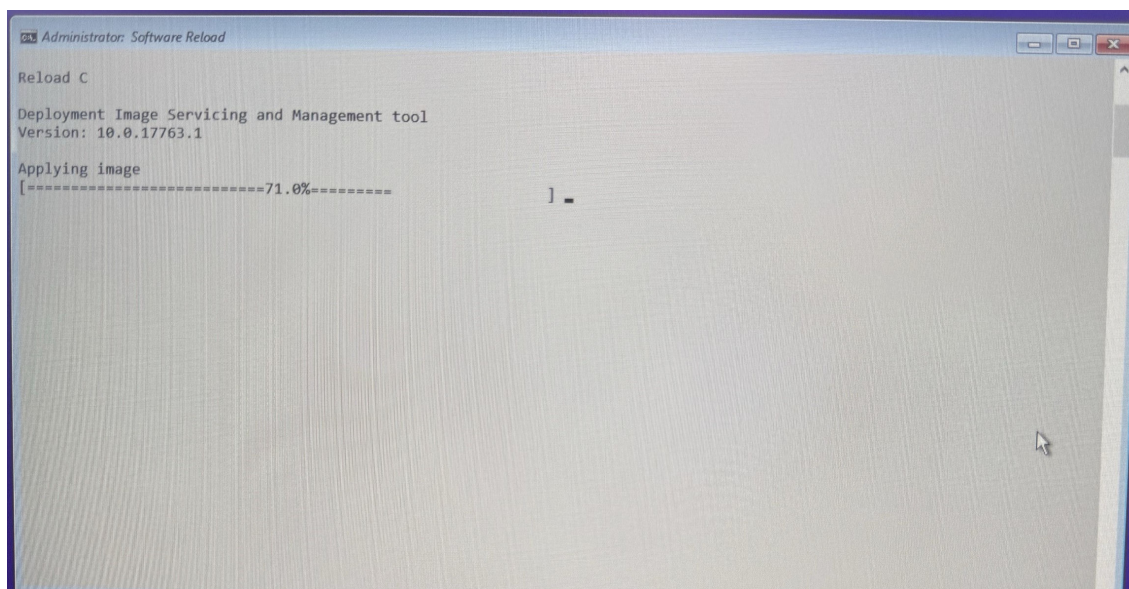


Figure 7-25 Starting the software loading

The system will restart several times during the installation process without any user actions, and the software loading will complete in 50 - 60 minutes.

When the software loading has completed, the ultrasound system remains ON.

6. Manually restore the DICOM TLS certificates and the WPA-Enterprise configuration that you recorded earlier.
7. Power down the ultrasound system.

7.3 Setup after software loading

7.3.1 Procedure for Setup After Software Loading

Use this procedure if all partitions on the hard drive have been erased during the software load.

Restore the Patient Archive and System Configurations from the backup you made before the software loading.

For instructions, see: "Data Backup and Restore" in the User Manual.

7.4 Backup and Restore

Please refer to:

- [Space Management - Moving All Images](#) on page 220
- [Backing up the Patient Archive and System Configurations](#) on page 221
- [Procedure for Setup After Software Loading](#) on page 240

Chapter 8

Theory

In this chapter

General Information.	244
Advanced Security Features.	248
Peripherals and/or optional Equipment.	250

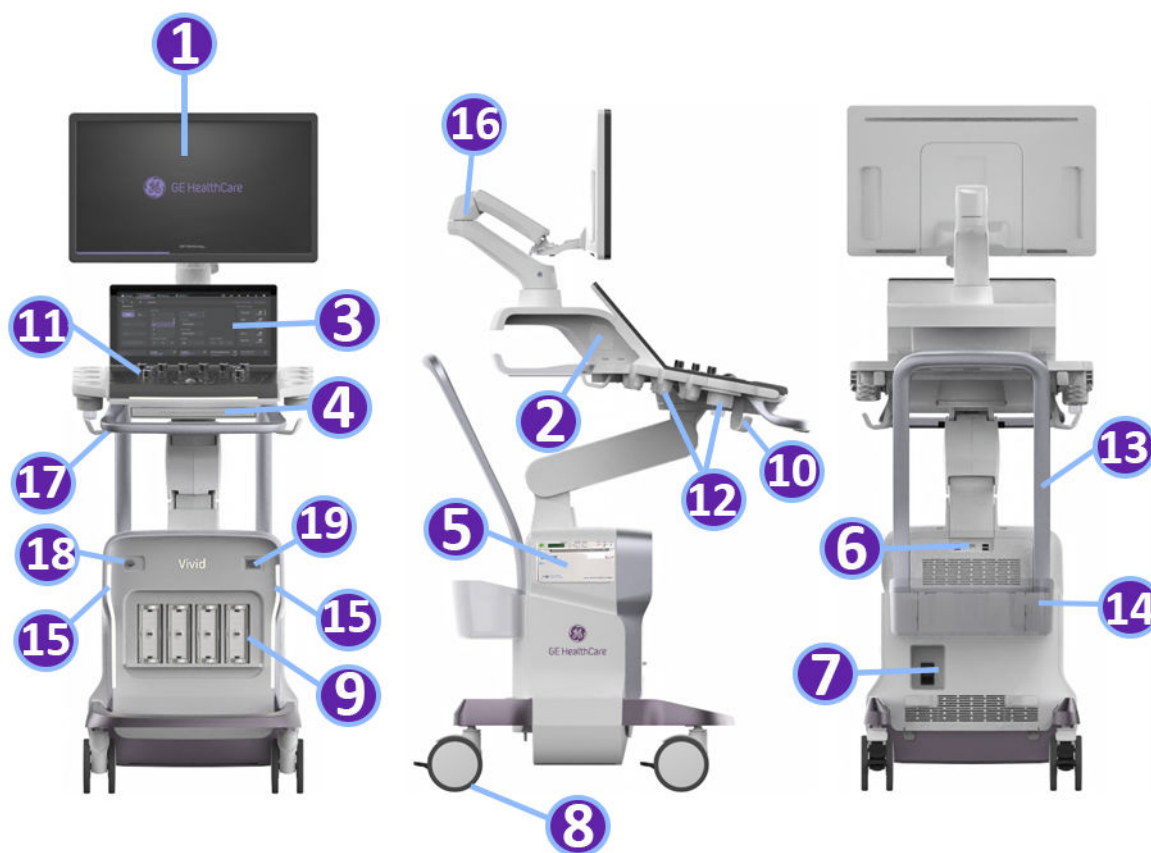
8.1 General Information

The Vivid Explorer/Vivid Advanced/Vivid Focus is a high performance digital ultrasound imaging system with total data management.

Modes of operation include 3D/ 4D Imaging mode (only for Vivid Explorer), B, M, PW Doppler, CW Doppler, Color Doppler, Color M Doppler, Power Doppler, Harmonic Imaging, Coded Pulse and Combined modes: B/M, B/Color M, B/PWD or CWD, B/Color/PWD or CWD, B/Power/PWD.

The fully digital architecture of the Vivid Explorer/Vivid Advanced/Vivid Focus system allows optimal usage of all scanning modes and probe types, throughout the full spectrum of operating frequencies.

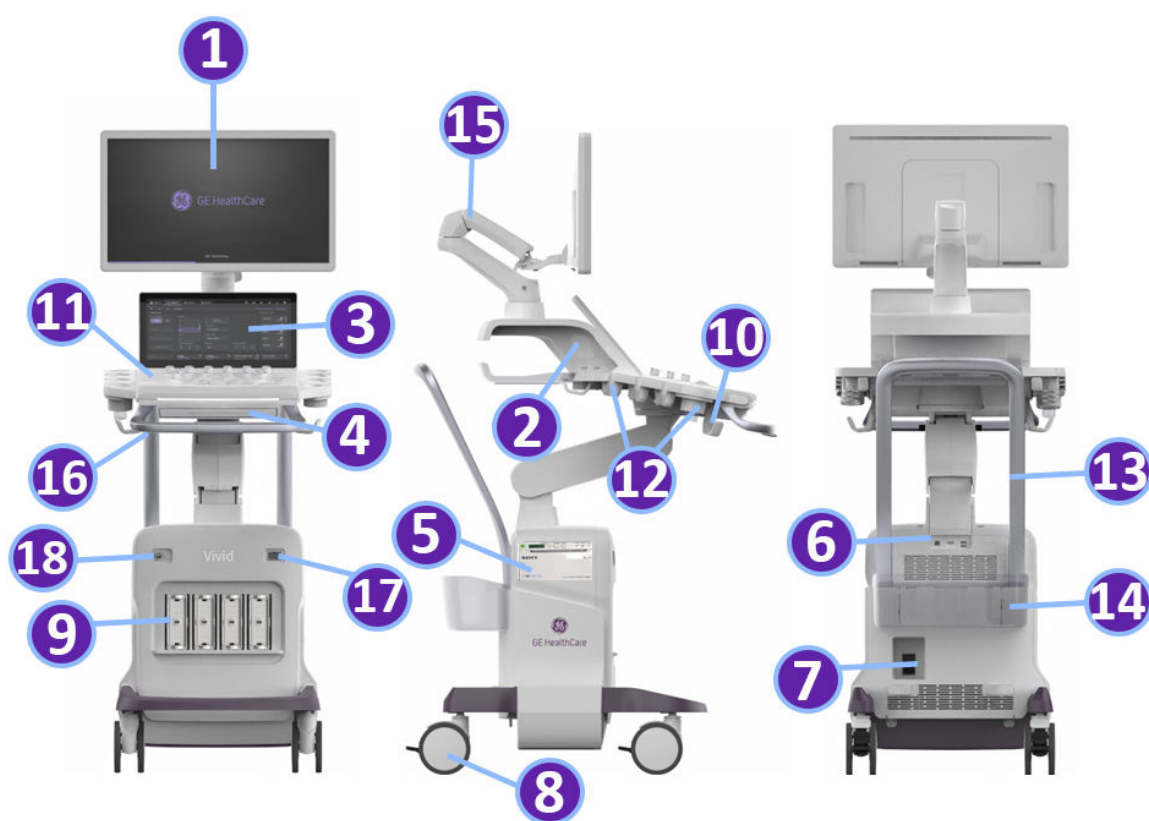
8.1.1 Getting to know the Vivid Explorer System



- | | |
|--|--|
| 1. Main display monitor | 10. Cable hooks |
| 2. Speakers | 11. Control Panel |
| 3. Touch Panel | 12. Probe holder and gel holder. |
| 4. Alphanumeric Keyboard (optional) | 13. Rear handle |
| 5. Printer (optional) | 14. Rear Storage Bin (optional) |
| 6. I/O panel (Network, HDMI and USB ports) | 15. Customizable light bar and stripes |
| 7. Power cable socket, circuit breaker | 16. Flexible arm |
| 8. Wheels | 17. Front handle |
| 9. DLP Probe ports | 18. Doppler probe port |
| | 19. ECG port |

Figure 8-1 Vivid Explorer

8.1.2 Getting to know the Vivid Advanced system



- | | |
|--|----------------------------------|
| 1. Main display monitor | 10. Cable hooks |
| 2. Speakers | 11. Control Panel |
| 3. Touch Panel | 12. Probe holder and gel holder. |
| 4. Alphanumeric Keyboard (optional) | 13. Rear handle |
| 5. Printer (optional) | 14. Rear Storage Bin (optional) |
| 6. I/O panel (Network, HDMI and USB ports) | 15. Flexible arm |
| 7. Power cable socket, circuit breaker | 16. Front handle |
| 8. Wheels | 17. ECG port |
| 9. DLP Probe ports | 18. Doppler probe port |

Figure 8-2 Vivid Advanced

8.1.3 Getting to know the Vivid Focus system



- | | |
|--|----------------------------------|
| 1. Main display monitor | 9. RS probe ports |
| 2. Speakers | 10. Cable hooks |
| 3. Touch Panel | 11. Control Panel |
| 4. Alphanumeric Keyboard (optional) | 12. Probe holder and gel holder. |
| 5. Printer (optional) | 13. Fixed arm |
| 6. I/O panel (Network, HDMI and USB ports) | 14. Rear hook |
| 7. Power cable socket, circuit breaker | 15. Front handle |
| 8. Wheels | 16. ECG port |
| | 17. Rear hook dummy left/right |

NOTE

Rear handle and flexible arm are optional for Vivid Focus.

NOTE

If the Rear handle is ordered, the Rear hook dummy left/right is not available.

Figure 8-3 Vivid Focus

8.1.4 Significant Features

Examples of significant features of the Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound system are the following:

- Integrated FrontEnd (uses advanced ASIC and FPGA technologies)
- Open connectivity using USB ports
- Bluetooth - wireless connectivity
- 23.8" HDU type monitor (Full HD, 1920x1080) for Vivid Explorer, 23.8" LCD display (1920x1080) for Vivid Advanced, 21.5" LCD display (1920x1080) for Vivid Focus

- Low profile, backlit Keyboard

Vivid Explorer/Vivid Advanced/Vivid Focus has a digital beam forming system (incorporated in the FrontEnd).

The internal electronics are divided into three:

- E-Cage (including Front-End and Back-End)
- Power Supply Unit

Interconnecting signals from FrontEnd, BackEnd, keyboard, monitor, and power distribution sub-systems are routed internally.

8.1.5 Major System Components

- FrontEnd processor
- BackEnd processor
- Control Console (User interface); System I/O with hard keys, and Touch Panel
- Monitor
- External I/O
- Peripherals and/or optional equipment
- Peripherals
- System mechanical chassis: trolley to keep all major components

The Vivid Explorer/Vivid Advanced/Vivid Focus supports a variety of linear-, curved-, phased array and pencil CW probes for various clinical applications.

For more detailed explanations of functions and controls refer to the Vivid Explorer/Vivid Advanced/Vivid Focus User Manual.

8.2 Advanced Security Features

The "Vivid system has multiple patient privacy and cybersecurity features as well as cybersecurity hardening of the system. This section will describe the features below in more detail:

- [Disk Encryption](#) on page 248
- [Whitelisting](#) on page 249
- [Secure Wipe](#) on page 249

8.2.1 Disk Encryption

Disk Encryption protects:

- the operating system
- the application software
- the user configuration
- patient data and patient images with all their corresponding measurements.

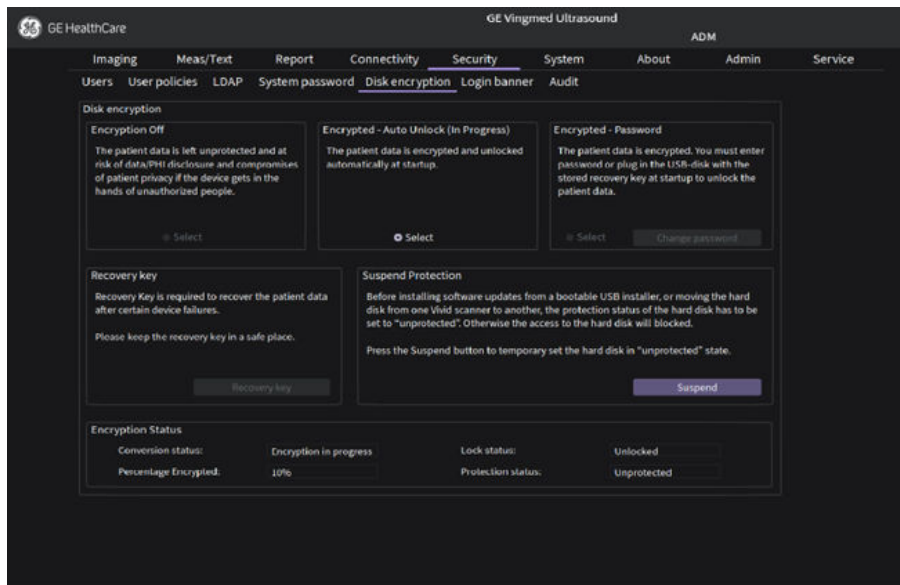


Figure 8-4 Disk Encryption Screen

The Disk encryption is set to one of three selectable modes where the selected mode applies to the patient data partition (the E:-drive) and not the other partitions:

- **Encryption Off:** The patient data is not encrypted. It is **not recommended** to use this mode, as patient data might be at risk of disclosure if the ultrasound system or the SSD is compromised.
- **Encryption - Auto Unlock:** The patient data is encrypted but unlocked automatically when the system is started. No user interaction is needed to unlock the patient data.
- **Encrypted - Password:** The patient data is encrypted. The user must define an encryption password for unlocking the patient data partition. In this mode the user must enter the encryption password to unlock the patient data and access the Local Archive on the system. Alternatively, the patient data may be unlocked with the disk encryption recovery key store on an attached USB storage device.

On Vivid Explorer/Vivid Advanced/Vivid Focus, all disk partitions, with the exception of the Z:-drive, are encrypted. In case **Encryption Off** is selected, the E:-drive will not be encrypted. The C:-drive encryption key is stored in the system's Trusted Platform Module (TPM) on the BackEnd processor, while the Windows operating system manages the encryption keys for the other drives.

As a result of this, if the SSD is separated from the TPM (for example if the BackEnd processor is to be replaced) it will not be possible to unlock and access any of the drives unless:

- The disc encryption is set to **Suspend** mode before separating the SSD from the BackEnd processor. This will allow access to all partitions the first time the system is booted after the **Suspend** mode is enabled. On consecutive boots, **Suspend** mode is disabled and the encryption will be enabled and activated by the Windows operating system.
- The user has a recovery key for the patient data partition. With the recovery key you can access the E:-drive where patient data and images are stored. This will make it possible to access the patient data from the SSD if connected to another system or if the software is reinstalled on the system without deleting existing data.
- The encryption of patient data (E:-drive) is changed from the default to either **Encryption Off** or **Encryption – Password** and the user has the password. This will make it possible to access the patient data from the SSD if connected to another system or if the software is reinstalled on the system without deleting existing data

**CAUTION**

Before replacing the BackEnd processor or moving the SSD from one system to another, ensure that the disk encryption is set to **Suspend** mode. Please note that during this time the disk encryption is temporarily disabled, leaving the data on the SSD unprotected and vulnerable to potential threats.

NOTE

Any executables and data stored on the internal SSD is encrypted by BitLocker configured with AES 256-bits key length.

NOTE

Disk encryption is enabled before the ultrasound system is shipped from the factory. However, the process of encrypting the disk might in some case not be fully complete before the system is shipped. This process will continue and execute seamlessly in the background when the system is started. The status of the encryption will be displayed in the Disk Encryption page in Config

**CAUTION**

Without the Encryption Password or the Recovery Key, accessing data stored on the system, including patient information, images, and the local archive, may become impossible. GE HealthCare cannot access this information or undo the encryption if the Encryption Password and Recovery Key are lost. Ensuring the protection and availability of the Encryption Password and Recovery Key is solely the user's responsibility.

8.2.2 Whitelisting

Whitelisting prevents the Vivid Explorer/Vivid Advanced/Vivid Focus system from installing unwanted software. It monitors all applications/services if they are known and accepted (white-listed). Any application/service that is not known on the system creates an error message on the screen.

NOTE

"Whitelisting" is part of the [Advanced Security Features](#) on page 248 option.

8.2.3 Secure Wipe

For more information about the Secure Wipe function see Chapter 5 [Secure Wipe](#) on page 206.

8.3 Peripherals and/or optional Equipment

See the Vivid Explorer/Vivid Advanced/Vivid Focus User Manual for more detailed information..

NOTE

Some peripheral devices (hardware options) may not be available and/or approved in all countries. Please contact Sales & Marketing in your region for information on availability.

8.3.1 Printer

Black & White Digital Printer

The B&W Digital Printer receives image data via the USB port. The print command is controlled by the keys on the Vivid Explorer/Vivid Advanced/Vivid Focus control panel.

8.3.2 DVD Drive



CAUTION

Laser radiation: Avoid exposure to the beam Class 3B laser product.

Class 3B laser radiation: When open avoid exposure to the beam.

DVD Drive

The DVD Drive (Writer) is used to backup images and reports. It is also used as the main source of software upgrades and other service utility operations. It is controlled by the BEP via USB port.

Table 8-1 Supported Media for Data Burning

Media supported for data burning
• DVD+R • DVD-R • DVD-RW • DVD+RW • CD-R • CD-RW

8.3.3 Foot Switch

The Foot Switch is used for comfortable system control when no hand is free.

8.3.4 ECG Module

The ECG module is used for acquiring an ECG signal to be displayed with the ultrasound image. This optional peripheral serve for gaining an ECG signal to mark the systolic and end diastolic moments in M-Mode and Doppler evaluations.

8.3.5 Wireless Network (WLAN) Adapter

The Vivid Explorer/Vivid Advanced/Vivid Focus supports a Wireless Network USB Adapter based on industry standards to provide easy-to-use and compatible high-speed wireless connectivity.

The Wireless Network (WLAN) USB Adapter provides a mobile network connection to the local area network.

Chapter 9

System Diagrams

In this chapter

System Diagram.	252
-------------------------	-----

9.1 System Diagram

There is currently no information available for this chapter.

Chapter 10

Troubleshooting

In this chapter

Collect Vital System Information.	254
Service desktop.	259
Request for Service (RFS).	307
Check Point Voltages.	309
Battery Check.	310
Monitor Troubleshooting.	315
Printer Malfunction.	316

10.1 Collect Vital System Information

The following information is necessary in order to properly analyze data or images being reported as a malfunction or being returned to the manufacturer:

NOTE

This information is normally collected with Alt+D or **Gather Logs**

- Product Name = [Product Name]
- Navigate to **Utility > Config > About**
- Under **Software Version**, record:
 1. Software Version
 2. Software Revision
 3. Software Part number
 4. Build Name
 5. Build View
 6. Build Date

10.1.1 Screen Captures and Logs

If the Ultrasound system malfunctions, simultaneously press the Alt+D keys. Alt+D is always available and collects a screen capture of the image monitor, user-defined presets, and these logs:

- Keyboard Shadow Log (protected information)
- Error Logs
- Crash Log (protected information)
- Vital Product Data
- DICOM Logs
- Windows Event Logs
- Windows Modem Log
- Diagnostic Logs
- Service Logs

For a detailed list of Service Logs captured, see: [Capturing Network Logs](#) on page 257.

When Alt+D is pressed, a dialog box opens. Enter the following information:

- System ID serial number
- Software version
- System date and time of occurrence
- Sequence of events leading to issue
- Whether the issue is repeatable
- Imaging mode, probe, preset/application
- Media brand, speed, capacity, and type
- To authorize the collection of protected information, check Include Protected Information, requires Admin privileges. If this box is checked, the system creates two log files with the following format name:
 - log_<SN>_<DATE>_<TIME>_DB.zip
 - log_<SN>_<DATE>_<TIME>_ProtectedInfo_DB.zip

Where <SN> is the serial number, <DATE> is the date in format YYMMDD and <TIME> is the time in format HHMMSS.

- To include a duration, check Time Duration and select a duration from the dropdown.
- Under Destination, select a storage media or Service directory for remote viewing.
- Click the Store button.

Figure 10-1 ALT+D Dialog Box

NOTE

To save to a CD/DVD or USB Flash Drive, you MUST select CD/DVD Recordable or USB drive as the destination device, otherwise the data is written to the default Export/Service directory on the hard drive. The Export/Service directory is only used for remote service and is not intended for images or report storage use.

For a CD/DVD, the Ultrasound system automatically formats an un-formatted disk, gathers logs and writes it out to the disk.

The subsequent file is compressed, and time stamped. The screen capture is a bitmap which eliminates the possibility of artifacts from compression.

Double check the media that you made to ensure it contains the file. An example is shown in the figure below:

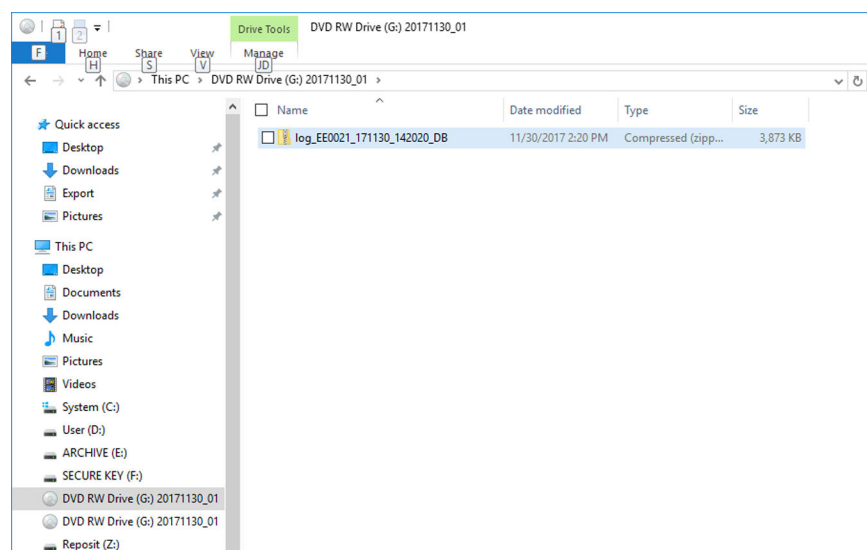


Figure 10-2 Example of Zipped Trouble Image and Logs File

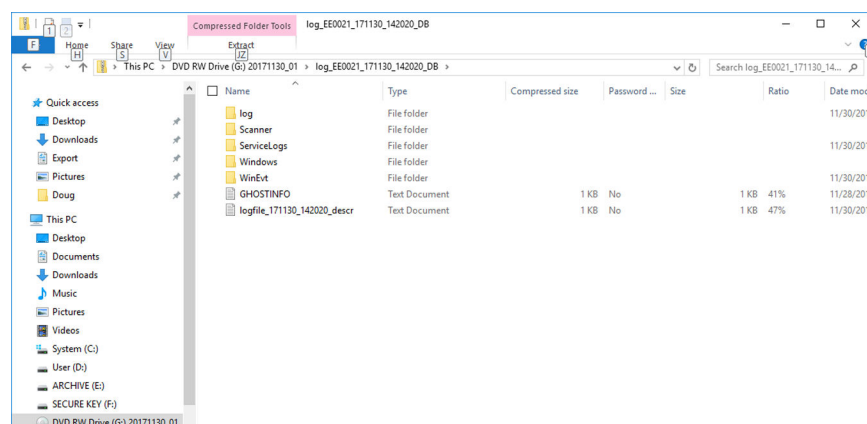


Figure 10-3 Contents of the Zipped Trouble Image and Logs File

10.1.1.1 Bookmarking a system malfunction

1. If a system malfunction is observed, press Alt - B.

A bookmark will be created when creating a log file.

10.1.1.2 Ctrl+PrintScreen shortcut

A Ctrl+PrintScreen shortcut is available for quickly capturing the image displayed on the Ultrasound system. Images captured using this shortcut are saved in the D:\export directory using both the JPEG (.jpg) and raw DICOM (.dcm) formats.

The InSite connection will have access to the export folder on the “D:” drive to retrieve these images. This feature will allow the customer to quickly and easily acquire images that can then be viewed by the Online Centre (OLC).

10.1.2 Needed Data in Case of System Crash or Errors

NOTE

In case of a system crash or errors that are probably software bugs, provide as much detail information as possible before contacting the OLC (**O**n-**L**ine **C**enter).

1. Contact person (+ telephone number and/or email address)
2. Vital system data as a basic need on every failure investigation
 - System Type
 - System Serial number (also visible on label on the back of the system)
 - Application Software version
 - additional information such as:
 - Hardware ID
 - Hardware Configuration (board versions)
 - Console version
3. Detailed problem description or error message (dump-file).
 - Address
 - Application Software version
 - Exact date and time of the occurrence
 - Problem frequency
 - screenshot (+)
 - Reproducible? (button by button workflow)
 - DICOM issues (provide sniff)
 - Image quality issues
4. If the bug or error is difficult to reproduce, record an Auto Tester file (ending at fault occurrence).

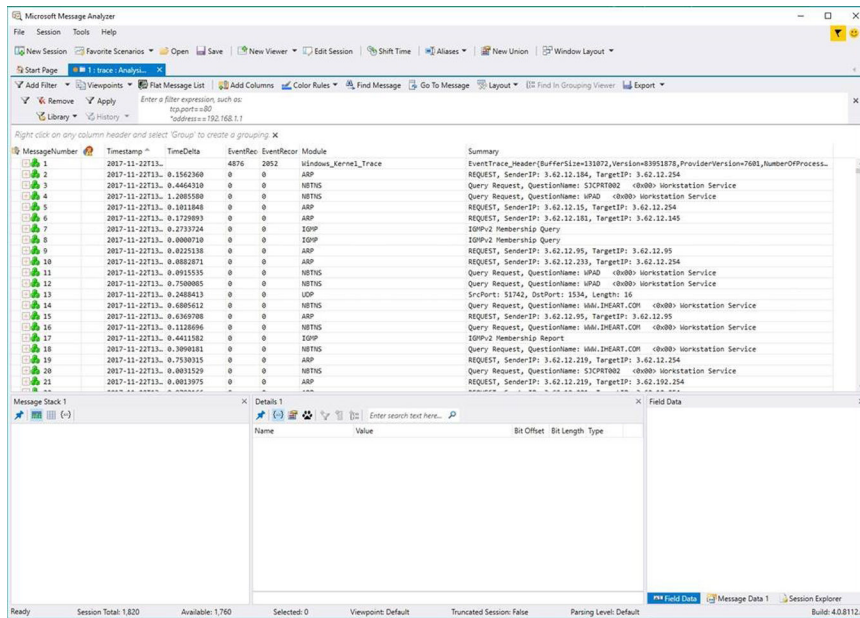
10.1.3 Capturing Network Logs

Use the Network Capture utility to capture network logs. Once the logs are captured and extracted from the system, use Microsoft Message Analyzer to convert the output, and then Wireshark (or some other network sniffer) to analyze it.

To troubleshoot network capture logs:

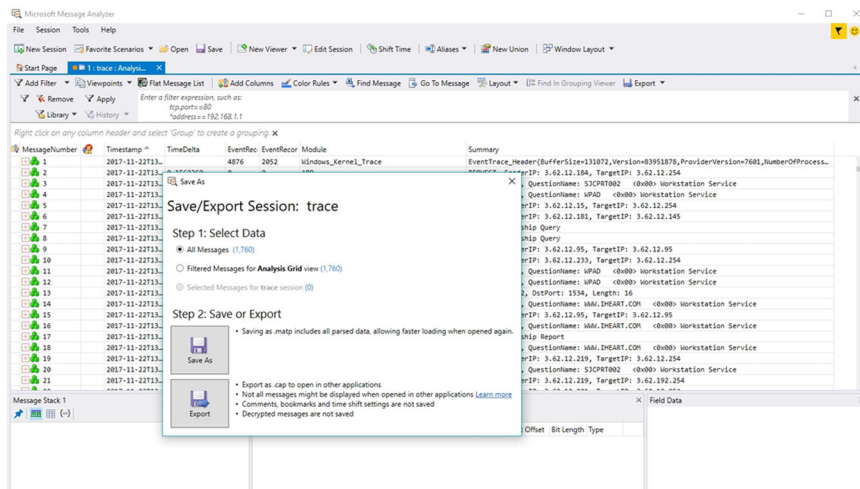
1. On the Ultrasound system, run the Network Capture utility to generate .etl and .cab files for analysis. For more information, see [Network Capture](#) on page 297.
2. Locate the .etl file at D:\Service.
3. Transfer these files to a laptop.
4. On the laptop, download Microsoft Message Analyzer (if not already done).
5. In Microsoft Message Analyzer, open the .etl file.

If the DICOM connection is encrypted, you will not be able to see the contents of the packets. Microsoft Message Analyzer does not have a plug-in to decode DICOM packets



1. To export the .etl file to .cab, select File > Save As and then select Save As or Export.

Figure 10-4 Opened .etl file



If the network capture was of a wireless connection, Microsoft Message Analyzer creates a .pcap file that does not have any of the packets and is probably useless.

1. Use Wireshark or some other third-party network sniffer to examine the .cab file.

Figure 10-5 File > Save As or Export

10.2 Service desktop

There are the different levels of access to the Service desktop:

- Service Basic access (Class A) – a user locally logged into the machine with Local Service Access privilege. This level provides limited access to Service desktop widgets and utilities.
- Service Expert, Pro, and Advanced access (Class C) Local – Depending on the purchase level, includes an option string to control access.
- GE Service access (Class M) and an SSA key. For users with local Service Access privileges, this level provides unrestricted access to all Service desktop widgets and utilities.
- Remote access – a user remotely accessing the Ultrasound system. This level provides unrestricted access to all Service desktop widgets and utilities. Disruptive mode is limited to the user access privileges to Remote Service Access.

10.2.1 Disruptive mode

Disruptive mode is a way to control interruptions to operation of the Ultrasound system. Disruptive mode is required whenever service performs a function that may disrupt a normal scan. Activating Disruptive mode results in a red message displayed on the task bar. This message indicates that the Ultrasound system needs to be restarted once the service activity is complete. The message remains until the Ultrasound system is restarted. This prevents patient scanning while the Ultrasound system is not operating at an optimal status. For example, running a diagnostic may leave the Ultrasound system in a state that is not good for imaging.

Specifically, Disruptive mode is required to run diagnostics, clean presets, and reset the patient database, and turn on Virtual Console Observation (VCO).

- When Disruptive mode is On, all service functionality on the Service desktop is allowed but user operation of the Ultrasound system may be limited.
- When Disruptive mode is Off, some service functionality on the Service desktop is not available and user operation of the Ultrasound system is normal.

Additionally, the ability to enable Disruptive mode depends on the logged in user.

- Local user – a user locally logged into the machine will be able to set the Ultrasound system to Disruptive mode or allow a Disruptive mode request from a remote user through the Service desktop. The local user must have Authorize Remote Service Access to allow Disruptive mode. If the local user does not have this right, the remote user's request will be automatically denied.
- Remote user – a user remotely accessing the Ultrasound system will not be able to automatically switch Disruptive mode to On. The logged in user (local user logged on to the Ultrasound system) must have the ability to grant remote access. The logged in user will be notified through a dialog box and asked to allow Disruptive mode.

NOTE

Change Password and Disk Defragment are not available for the remote user whether Disruptive mode is On or Off.

For more information, see:

[Disruptive Mode](#) on page 295

10.2.2 Color statuses

Throughout the Service desktop, colors indicate the following:

- Green – Status is normal
- Orange – Status is a warning
- Red – Status is an error

10.2.3 Licenses

With Service Basic Access (Class A), these are the available options:

- HOME
- Utilities
 - Change Password
 - Data Transfer
 - Delete Files
 - Gather Logs
 - Network Capture
 - SSA License
 - Third Party Licenses
- Options
- Agent Configuration

10.2.4 Home

10.2.4.1 Home Access

Home configurations vary depending upon the purchased service level.

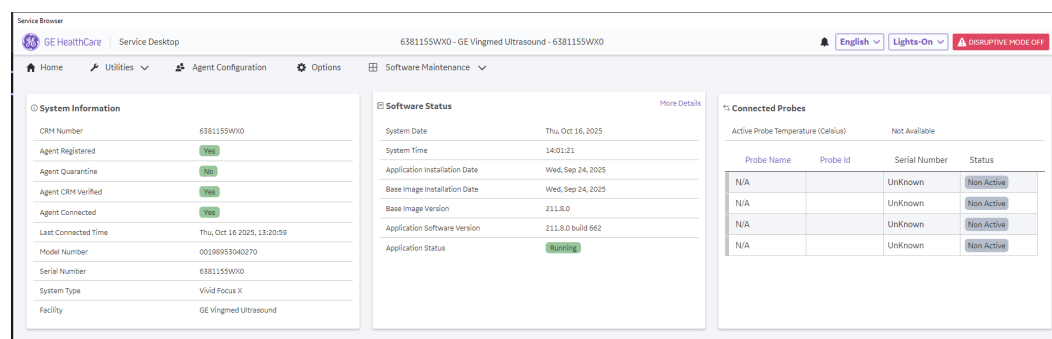


Figure 10-6 Home with Class A Access

For more information, see:

- [System Information](#) on page 260
- [Software Status](#) on page 262
- [System Status](#) on page 263
- [Disk Drive Space](#) on page 271
- [Network Configuration](#) on page 273
- [Connected Probes](#) on page 276

10.2.4.2 System Information

System Information displays general information about the Ultrasound system. When the ultrasound system has been successfully configured with the back office, these elements will have the corresponding values:

- Agent Registered will be Yes
- Agent Quarantine will be No
- Agent CRM Verified will be Yes

The information on System Information is available to all service class licenses.

To access System Information, navigate to Home.

① System Information	
CRM Number	6381155WX0
Agent Registered	Yes
Agent Quarantine	No
Agent CRM Verified	Yes
Agent Connected	Yes
Last Connected Time	Mon, Oct 13 2025, 05:10:16
Model Number	00198953040270
Serial Number	6381155WX0
System Type	Vivid Focus X
Facility	GE Vingmed Ultrasound

Figure 10-7 System Information

This table shows all the elements available on System Information with descriptions.

Table 10-1 System Information

Element	Description
CRM Number	Customer Relationship Management (CRM) number. System identifier assigned to the customer unit by the service region.
Agent Registered	Registered status of the agent. Valid values are: - Yes – The agent is registered in the back office. - No – The agent is not registered in the back office. - Not Available – The agent is not running or has not been configured.
Agent Quarantine	Quarantine status of the agent. Valid values are: - Yes – The agent has more than one device registered with the same CRM Number in the back office. - No – The agent has one device registered with the listed CRM Number in the back office. - Not Available – The agent is not running or has not been configured.
Agent CRM Verified	CRM verified status of the agent. Valid values are: - Yes – The agent is verified in the back office. - No – The agent is not verified in the back office. - Not Available – The agent is not running or has not been configured.
Agent Connected	Connected status of the agent. Valid values are: - Yes - The agent is connected in the back office. - No - The agent is not connected in the back office. - Not Available - The agent is not running or has not been configured.
Last connected Time	Indicate the last conencted time of agent.
Model Number	GE HealthCare part number for the ultrasound system. The same number as listed on the rating plate.

Element	Description
Serial Number	Serial number of the ultrasound system. The same number as listed on the rating plate.
System Type	Product name of the ultrasound system.
Facility	Name of the hospital or facility where the ultrasound system is installed.

For more information, see:

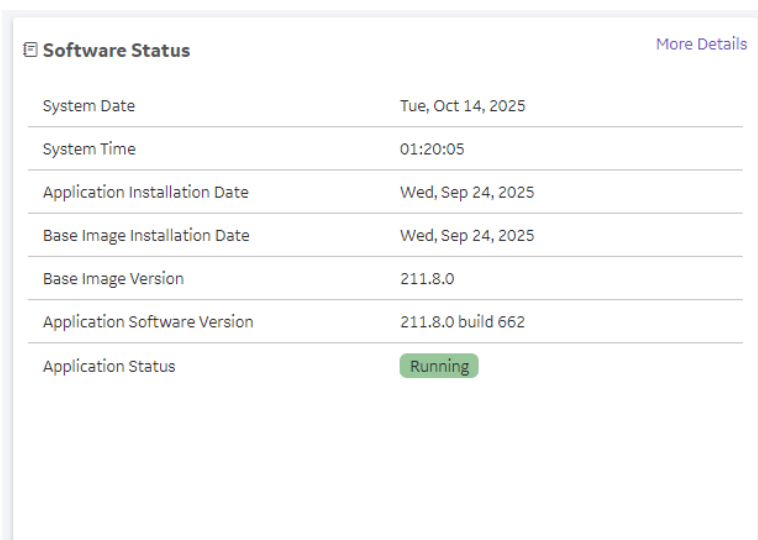
[Home](#) on page 260

10.2.4.3 Software Status

Use Software Status to view general information about the software installed on the ultrasound system.

The information on Software Status is available to all service class licenses.

To access Software Status, navigate to: Home > Software Status.



The screenshot shows a window titled "Software Status" with a "More Details" link in the top right corner. The window contains a table with the following data:

System Date	Tue, Oct 14, 2025
System Time	01:20:05
Application Installation Date	Wed, Sep 24, 2025
Base Image Installation Date	Wed, Sep 24, 2025
Base Image Version	211.8.0
Application Software Version	211.8.0 build 662
Application Status	Running

This table shows all the elements available on Software Status with descriptions.

Figure 10-8 Software Status

Table 10-2 Software Status

Element	Description
System Date	Current date in the format <day>, <month> <date> <year>.
System Time	Local time based on the last time the system desktop was refreshed in the format <hh:mm:ss>.
Application Installation Date	Date the application software was installed. The application software includes the Ultrasound system's product-specific software.
System software (base image) Installation Date	Date the system software (base image) was installed. The system software (base image) includes the Windows operating system and other supporting software.
System software (base image) Version	Version number of the system software (base image).
Application Software Version	Version number of the application software.

Element	Description
Application Status	Status of the application. Valid values are: - Running - Stopped

For more information, see:

- [Home](#) on page 260

10.2.4.4 System Status

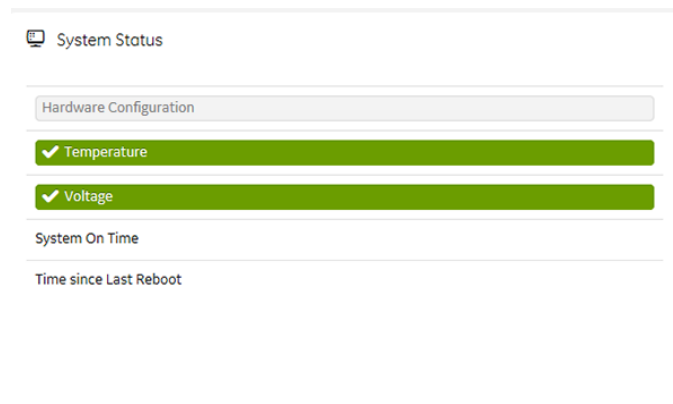
10.2.4.4.1 System Status overview

Use System Status to view status information on the Ultrasound system. Specifically, do the following:

- View the hardware configuration file
- View temperatures for FRUs, fans, and a graphical representation of the monitored temperatures
- View statuses of the batteries
- View voltages for FRUs and AC
- View issues with the boot process
- View alert logs
- View the amount of time the Ultrasound system has been running
- View the amount of time since the Ultrasound system has last been rebooted

The information on System Status is available to Class C and Class M licenses.

To access System Status, navigate to: Home > System Status.



This table shows all the elements available on System Status with descriptions.

Figure 10-9 System Status

Table 10-3 System Status

Element	Description
Hardware Configuration	Displays the contents of the hardware configuration file.
Temperature	Displays the temperature monitoring of the Ultrasound system as normal, warning, or error using the standard colors and icons. Includes temperature out-of-spec events including an overall count of event warnings and errors for the last 30 days or the last status checkpoint whichever is sooner. Temperature displays temperature a graph for the configured monitoring points.

Element	Description
Battery Status	Displays the battery monitoring of the Ultrasound system.
Voltage	Displays the voltage monitoring of the Ultrasound system as normal, warning or error using the standard colors and icons. Includes voltage out-of-spec events including an overall count of the event warnings and errors for the last 30 days or last status checkpoint whichever is sooner.
Alert Logs	Based on the latest occurrence, displays all alerts generated by the Ultrasound system for either the past 30 days or the latest check point.
GPU	
System On Time	Displays the time the Ultrasound system was turned on.
Time Since Last Reboot	Displays the time the Ultrasound system was last rebooted.

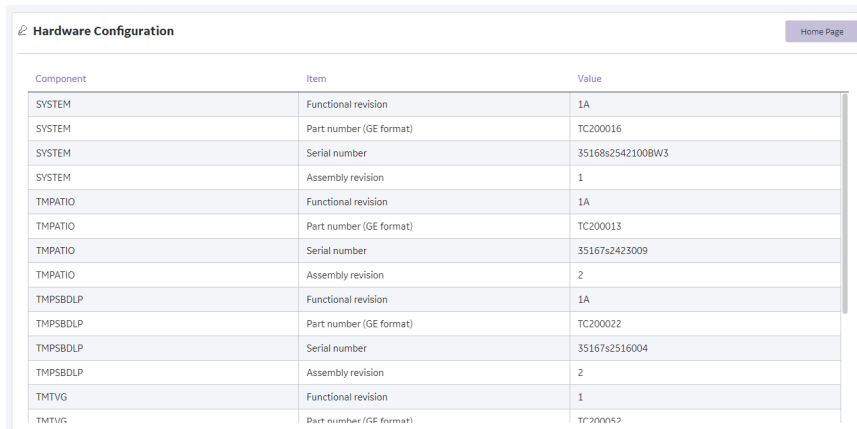
For more information, see:

- [Hardware Configuration](#) on page 264
- [Temperature](#) on page 265
- [Battery Status](#) on page 267
- [Voltage](#) on page 268
- [Alert Logs](#) on page 270
- [Home](#) on page 260

10.2.4.4.2 Hardware Configuration

Hardware Configuration displays the contents of the hardware configuration file which includes all the vital product data (VPD).

To access Hardware Configuration, navigate to: Home, and then under System Status, select Hardware Configuration.



Component	Item	Value
SYSTEM	Functional revision	1A
SYSTEM	Part number (GE format)	TC200016
SYSTEM	Serial number	35168s2542100BW3
SYSTEM	Assembly revision	1
TMPATIO	Functional revision	1A
TMPATIO	Part number (GE format)	TC200013
TMPATIO	Serial number	35167s2423009
TMPATIO	Assembly revision	2
TMPSBDLP	Functional revision	1A
TMPSBDLP	Part number (GE format)	TC200022
TMPSBDLP	Serial number	35167s2516004
TMPSBDLP	Assembly revision	2
TMTVG	Functional revision	1
TMTVG	Part number (GE format)	TC200012

This table shows all the elements available on Hardware Configuration with descriptions.

Figure 10-10 Hardware Configuration

Table 10-4 Hardware Configuration

Element	Description
Component	Hardware component.
Item	Parameter with the vital product data (VPD).
Value	Actual value of the Item.

For more information, see:

- [System Status](#) on page 263
- [Home](#) on page 260

10.2.4.4.3 Temperature

Use Temperature to troubleshoot problems with monitored FRUs and the fans. Specifically, use these pages:

- FRU Status
- Graphs
- Fan Status

To access Temperature, navigate to: Home, and then under System Status, select Temperature.

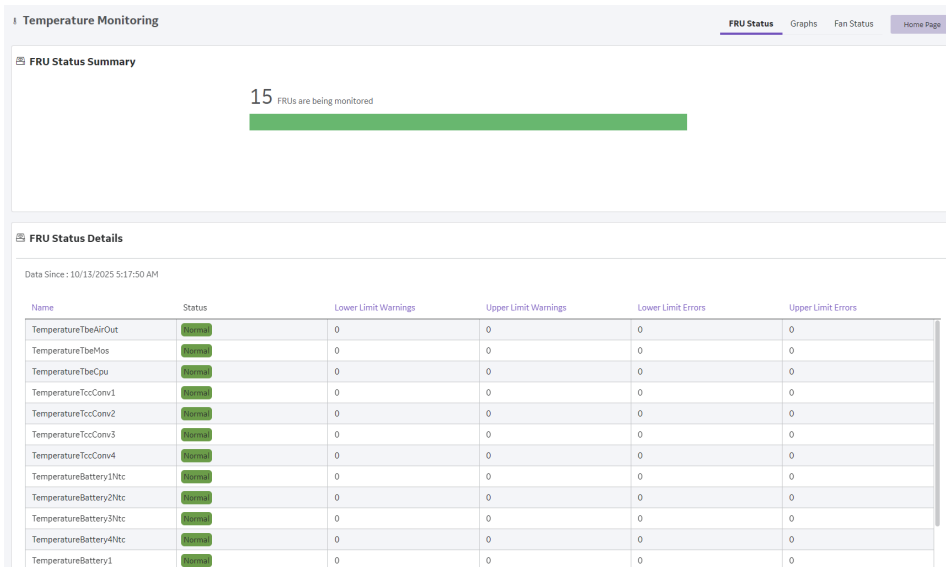
For more information, see:

- [FRU Status](#) on page 265
- [Graphs](#) on page 266
- [Fan Status](#) on page 266
- [System Status](#) on page 263
- [Home](#) on page 260

10.2.4.4.4 FRU Status

FRU Status displays a summary of the FRUs being monitored and the details for each FRU.

To access this page, under System Status, select Temperature and then select FRU Status.



This table shows all the elements available on FRU Status with descriptions.

Figure 10-11 FRU Status

Table 10-5 FRU Status

Element	Description
FRU Status Summary	Number of the FRUs being monitored.
FRU Status Details	

Element	Description
Name	Name of the FRU. - ETX3 – Represents the ETX 64 transmitter board. - ETX4 – Represents the ETX 128 transmitter board.
Status	Status of the FRU. Valid values are: - Normal – Indicates that the FRU is operating within the allowable range. - Warning – Indicates that the FRU is operating close to the limit of the allowable range. - Error – Indicates that the FRU is operating outside the allowable range.
Low Limit Warnings	Number of low temperature warnings.
Upper Limit Warning	Number of high temperature warnings.
Low Limit Errors	Number of low limit temperature errors
Upper Limit Errors	Number of high limit temperature errors.

For more information, see:

- [Temperature](#) on page 265
- [Home](#) on page 260

10.2.4.4.5 Graphs

Graphs displays trend graphs for selected elements which have been selected as key indicators of overall temperature status inside the card rack.

To access this page, under System Status, select Temperature and then select Graphs.

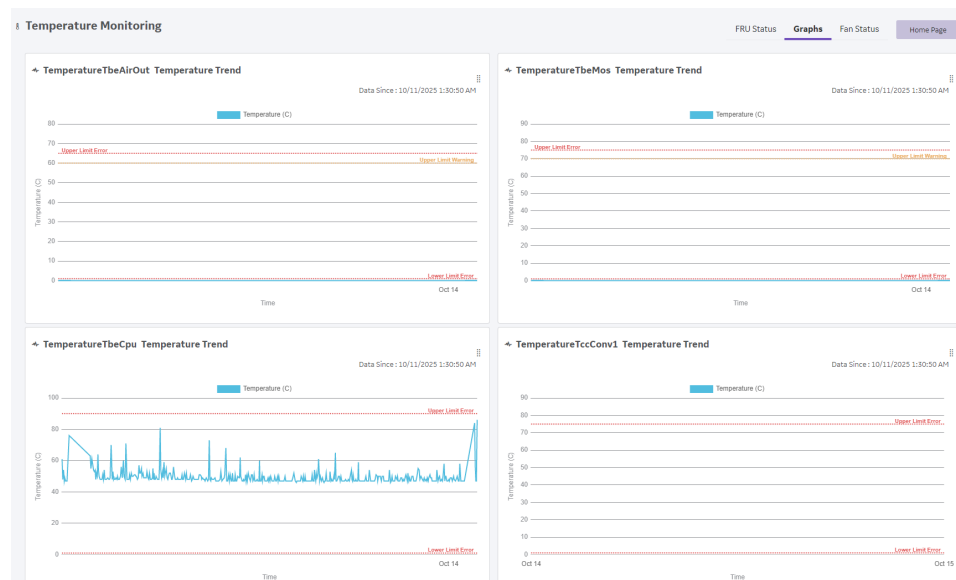


Figure 10-12 Graphs

For more information, see:

- [Temperature](#) on page 265
- [Home](#) on page 260

10.2.4.4.6 Fan Status

Fan Status displays the details for each fan. The peak level of operation for the fans is 2500 RPM.

To access this page, under System Status, select Temperature and then select Fan Status.

Monitoring Point	Device	Status	Lower Limit Warnings	Upper Limit Warnings	Lower Limit Errors	Upper Limit Errors	Latest Value	Latest Check Point
N/A	BepFan0MeasuredSpe...	Normal	0	0	0	0	98.3592	2025-10-14T01:29:55
N/A	BepFan1MeasuredSpe...	Normal	0	0	0	0	7.00649e-43	2025-10-14T01:29:55

This table shows all the elements available on Fan Status with descriptions.

Figure 10-13 Fan Status

Table 10-6 Fan Status

Element	Description
Monitoring Point	Name of the fan. - Fan_control – Indicates the overall control of fan speed. - Rack Fan # – Indicates individual fan on the fan tray.
Device	Device monitoring the parameters.
Status	Status of the fan. Valid values are: - Normal – Indicates that the fan is operating within the allowable range. - Warning – Indicates that the fan is operating close to the limit of the allowable range. - Error – Indicates that the fan is operating outside the allowable range.
Lower Limit Warnings	Number of low limit temperature warnings.
Upper Limit Warnings	Number of high limit temperature warnings.
Lower Limit Errors	Number of low limit temperature errors.
Upper Limit Errors	Number of high limit temperature errors.
Latest Value	Latest fan speed recorded in log.
Latest Check Point	Timestamp to collect the latest value of fan speed.

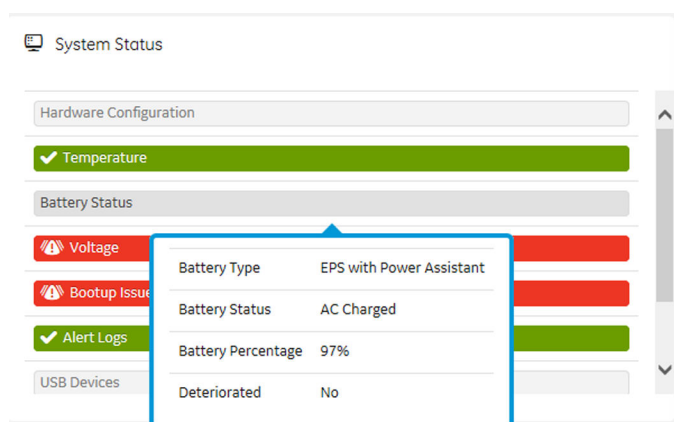
For more information, see:

- [Temperature](#) on page 265
- [Home](#) on page 260

10.2.4.4.7 Battery Status

Battery Status displays details for each battery.

To access Battery Status, navigate to Home, and then under System Status, select Battery Status.



This table shows all the elements available on Battery Status with descriptions.

Figure 10-14 Battery Status

Table 10-7 Battery Status

Element	Description
Battery Type	Type of the battery. Valid values are: - No battery – Indicates that no battery is present. - EPS only – Indicates that the extended power shutdown batteries are used. - EPS with Power Assistant – Indicates that the power assistant batteries are used.
Battery Status	Status of the battery. Valid values are: - AC Charged – Indicates that batteries are charging or fully charged. - Other – Indicates that the batteries are discharging.
Battery Percentage	Percentage of the battery charge.
Deteriorated	Deterioration state of the battery. Valid values are: - Yes – Indicates that there is a problem with the battery. - No – Indicates that the battery is operating normally.
For more information, see:	System Status on page 263 Home on page 260

10.2.4.4.8 Voltage

Use Voltage to troubleshoot problems with monitored FRUs and the AC voltage of the Ultrasound System. Specifically, use these pages:

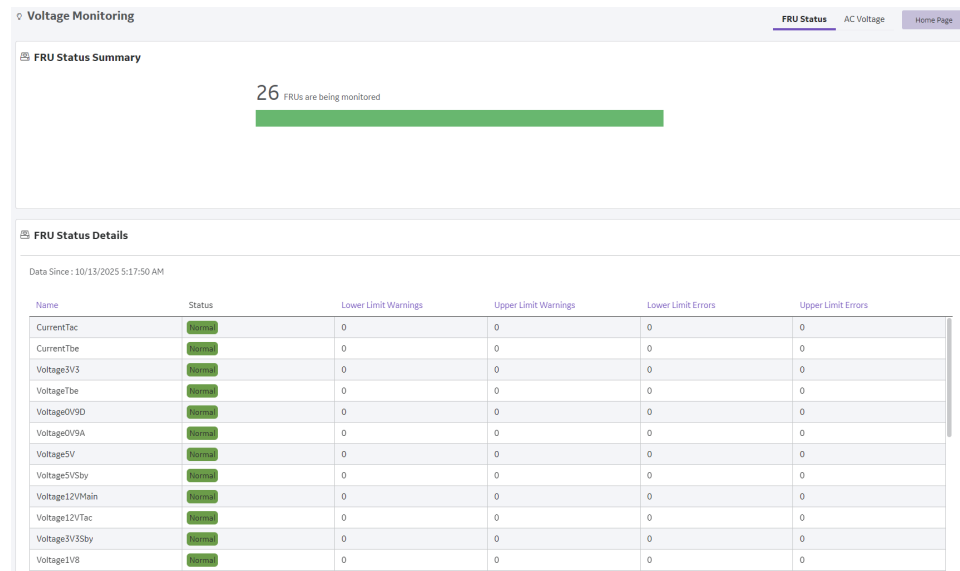
- FRU Status
- AC Voltage

For more information, see:	FRU Status on page 265 AC Voltage on page 269 System Status on page 263 Home on page 260
----------------------------	---

10.2.4.4.9 FRU Status

FRU Status displays a summary of the voltages for the FRUs being monitored and the details for each FRU.

To access this page, under System Status, select Voltage and then select FRU Status.



This table shows all the elements available on FRU Status with descriptions.

Figure 10-15 FRU Status

Table 10-8 FRU Status

Element	Description
Name	Name of the FRU.
Status	Status of the voltage for the FRU. Valid values are: - Normal – Indicates that the voltage is within the allowable range. - Warning – Indicates that the voltage is close to the limit of the allowable range. - Error – Indicates that the voltage is outside the allowable range.
Lower Limit Warnings	Number of low limit voltage warnings.
Upper Limit Warnings	Number of high limit voltage warnings.
Lower Limit Errors	Number of low limit voltage errors.
Upper Limit Errors	Number of high limit voltage errors.

For more information, see:

- [Voltage](#) on page 268
- [Home](#) on page 260

10.2.4.4.10 AC Voltage

AC Voltage graphically displays the AC voltage input over a period to help identify instabilities, or fault conditions, over time.

To access this page, under System Status, select Voltage and then select AC Voltage.

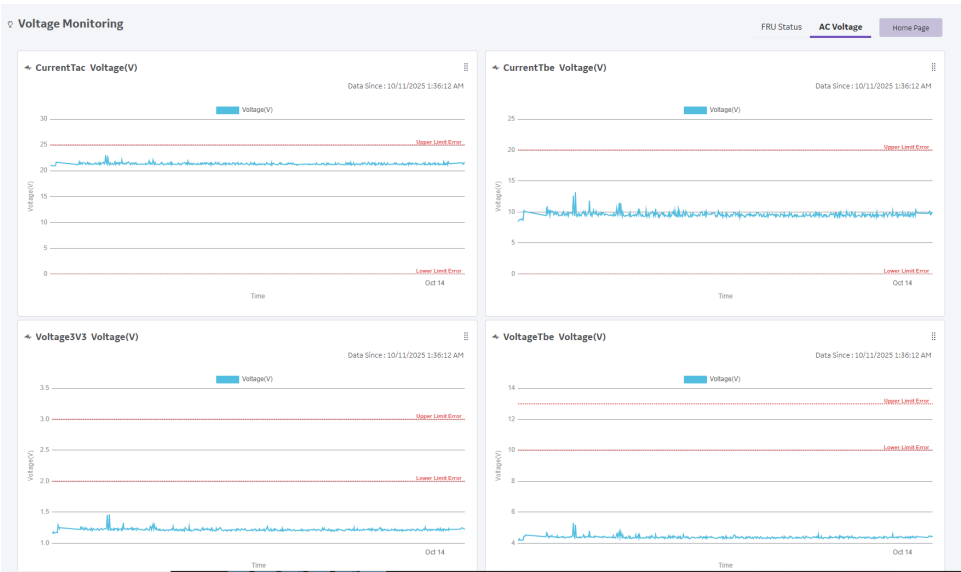


Figure 10-16 AC Voltage

For more information, see:

- [Voltage](#) on page 268
- [Home](#) on page 260

10.2.4.4.11 Alert Logs

Alert Logs displays the details for each alert. Alerts are status logs typically reported because of an error. Alerts are sent to the back office to be managed and reported to service.

To access this page, under System Status, select Alert Logs.



This table shows all the elements available on Alert Logs with descriptions.

Figure 10-17 Alert Logs

Table 10-9 Alert Logs

Element	Description
Alert ID	Identifier for the alert log that has been sent to the back office.
Sub System	Hardware component.
Rule	Code for the alert. Alerts are filtered in the back office using rule codes.
Description	Description of the alert log.
Value	Current value read for the associated hardware component.
Lower Limit	Value for the lower limit for the associated hardware component.
Upper Limit	Value for the upper limit for the associated hardware component.

Element	Description
Severity	Severity of the alert. These are the valid values: 1 - Error. Out of specification 2 - Warning 3 - Normal
First Alert Time	Time the alert first occurred.
Last Alert Time	Time the alert last occurred.
Last Alert Local Time	Time (local) the alert last occurred.
Last Alert Sent Time	Time the alert was last sent to the back office.
Count	Number of times that the alert has occurred.

For more information, see:	System Status on page 263 Home on page 260
----------------------------	---

10.2.4.5 Disk Drive Space

Use Disk Drive Space to view the hard drive partitions including total size and available free space in GB. Includes the overall health of the drive in one of these colors:

- Red – Available free space is less than 10% of the total size.
- Orange – Available free space is more than 10% and less than 20% of the total size.
- Green – Available free space is more than 20% of the total size.

Each partition displays these encryption statuses:

- Conversion status
- Percentage encrypted
- Encryption method
- Protection status
- Lock status

The information on Disk Drive Space is available to Class C and Class M licenses.

To access Disk Drive Space, navigate to Home.

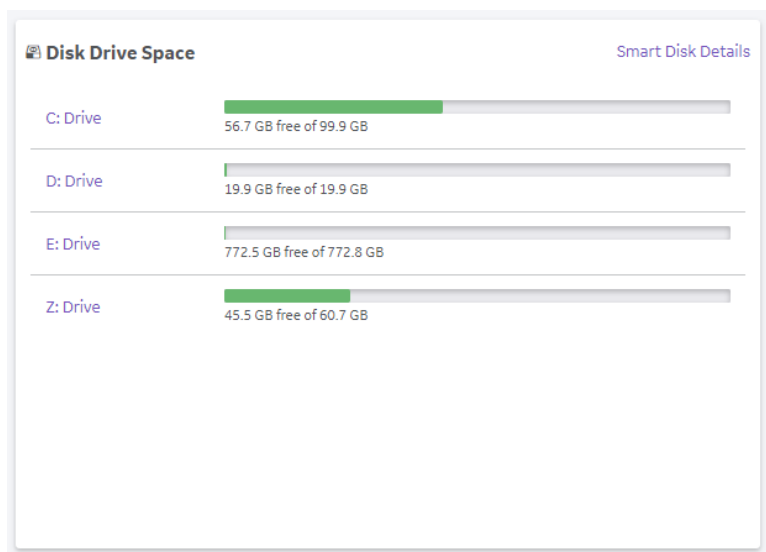
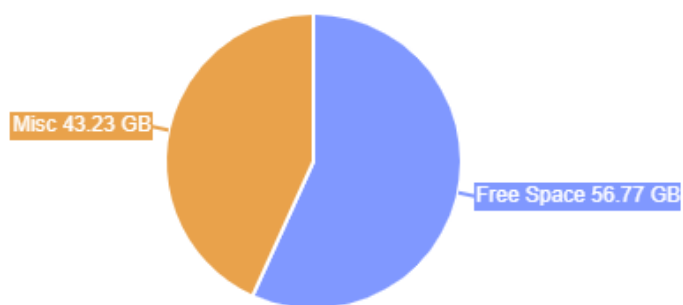


Figure 10-18 Disk Drive Space

C: Drive



Encryption Status

Conversion status	Used Space Only Encrypted
Percentage encrypted	100.0%
Encryption method	AES 256
Protection status	Protection On
Lock status	Unlocked

This table shows all the elements available on Disk Drive Space with descriptions.

Figure 10-19 Disk Drive Space

Table 10-10 Disk Drive Space

Element	Description
C: Drive	C: partition shows the graphical representation of the following: • Used space in GB • Available free space in GB
D: Drive	D: partition shows the graphical representation of the following: • Service • Logs • Temp • Export • Service configuration • Misc • Available free space in GB
E: Drive	E: partition shows the graphical representation of the following: • Print spooler • Clipboard • Dicom spooler • Misc • Available free space in GB
Z: Drive	Z: partition shows the graphical representation of the following: • Package repository • Misc • Available free space in GB
For more information, see:	• Home on page 260

10.2.4.6 Network Configuration

Use Network Configuration to view network (wired and wireless) information (the full ipconfig details) for the network configured with the Ultrasound system.

The information on Network Configuration is available to Class C and Class M licenses.

To access Network Configuration, navigate to Home.

Network Configuration

More Details

Host Name

VF-6381155WX0

NETGEAR A8000 WiFi 6 & 6E Adapter

IPv4 Address

172.24.0.30

Subnet Mask

255.255.224.0/64

Default Gateway

172.24.0.1

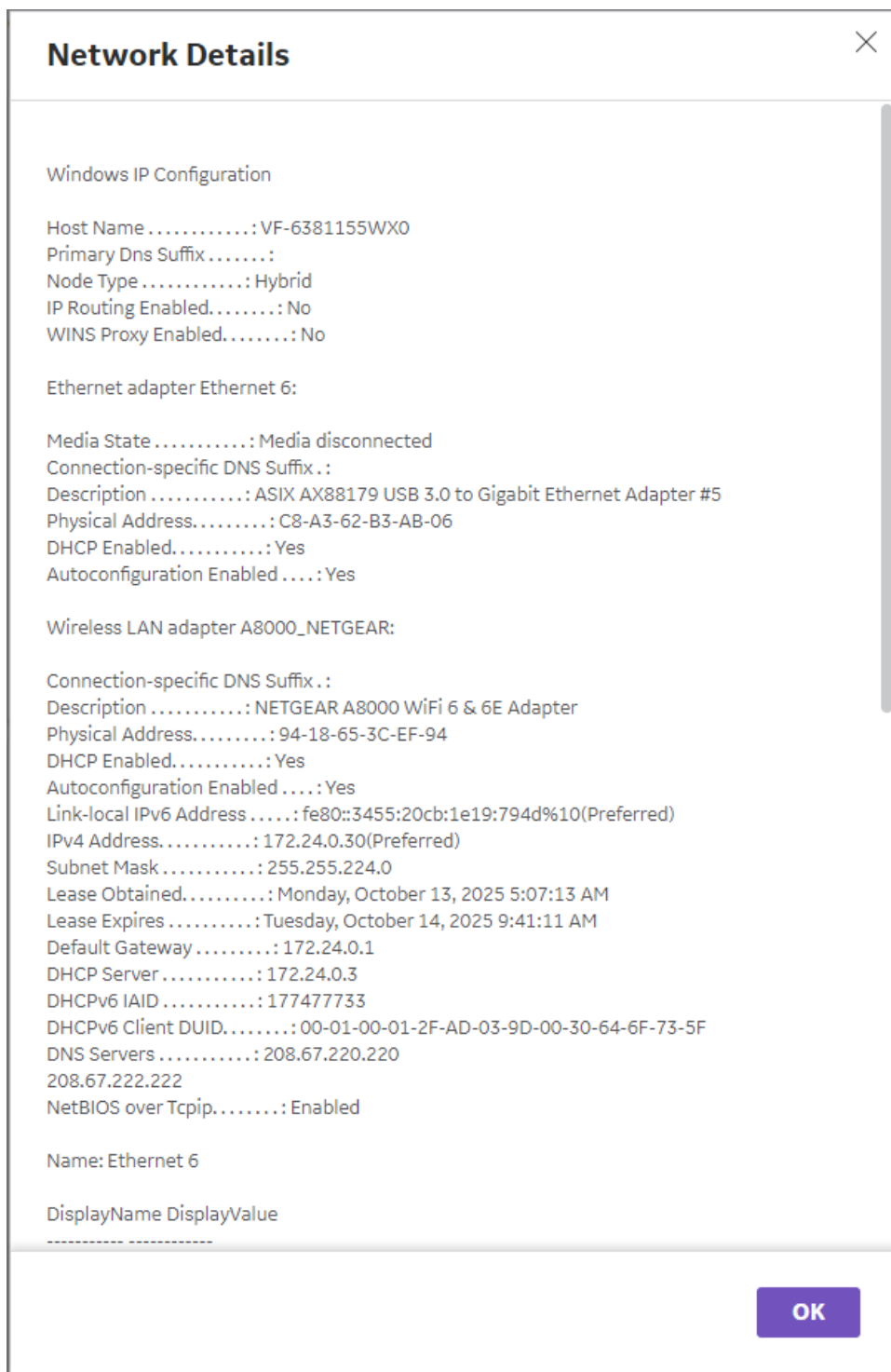
MAC Address

94:18:65:3C:EF:94

Speed (in Mbps)

173.3

Figure 10-20 Network Configuration



This table shows all the elements available on Network Configuration with descriptions.

Figure 10-21 Network Configuration with More Details

Table 10-11 Network Configuration

Element	Description
Host Name	Name of the local host for the Ultrasound system.
More Details	Displays the ipconfig details.
Wired Connection	
IPv4 Address	Local IP address for the wired network connection.
IPv6 Address	Local IP address for the wired network connection.
Subnet Mask	Local subnet mask for the wired network.
Default Gateway	Default gateway for the wired network.
MAC Address	Address for the MAC for the wired network.
Speed (Mbps)	Speed of the wireless connection.
Wireless Connection	
IP Address	IP address for the wireless network connection.
Subnet Mask	Subnet mask for the wireless connection.
Default Gateway	Default gateway for the wireless connection.
MAC Address	Address for the MAC for the wireless connection.
Speed (Mbps)	Speed of the wireless connection.
Status	Status of the wireless connection.
For more information, see:	• Home on page 260

10.2.4.7 Connected Probes

Connected Probes shows probes connected to the Ultrasound system. The order on the user interface is top-down matching the left-to-right order on the Ultrasound system.

The information on Connected Probes is available to all service class licenses.

To access Connected Probes, navigate to Home.

Connected Probes			
Active Probe Temperature (Celsius)		Not Available	
Probe Name	Probe Id	Serial Number	Status
12S	fpa_96_12.0_12S	UnKnown	Active
N/A		UnKnown	Non Active
N/A		UnKnown	Non Active
N/A		UnKnown	Non Active

This table shows all the elements available on Connected Probes with descriptions.

Figure 10-22 Connected Probes

Table 10-12 Connected Probes

Element	Description
Active Probe Temperature (Celsius)	When available, temperature of the active probe. Not all probes report temperature. The most common probe to report temperature is the TEE probe.
Probe Name	Name of the probe connected to the Ultrasound system.
Probe ID	Identifier of the probe connected to the Ultrasound system.
Serial Number	Serial number of the probe connected to the Ultrasound system. If the serial number of the probe is not available, then N/A displays.
Status	Statuses of the probe connected to the Ultrasound system. Valid values are: - Active - Non-Active
For more information, see:	Home on page 260

10.2.5 Diags

Use Diags to troubleshoot functionality with hardware components. With an active Service PRO license, probe diagnostics will be available. Specifically, use these pages:

- Run Diags – Run diagnostics on hardware components.
- Diag History – View a history of the diagnostics you have run and the results.

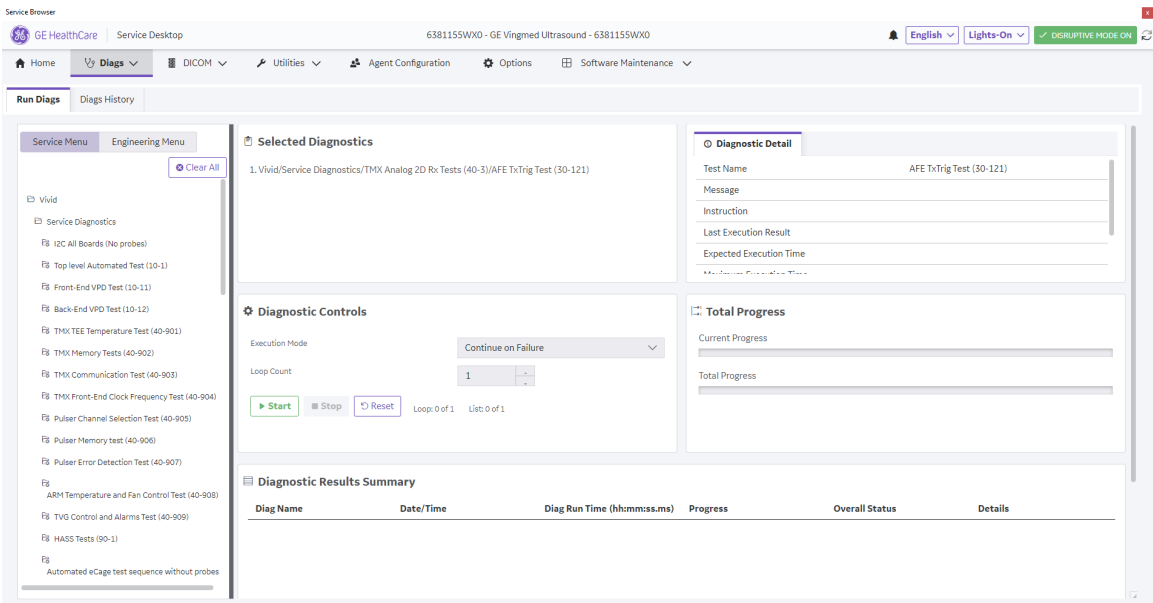
NOTE

Reboot the system after performing any diagnostics before returning the Ultrasound system to customer use.

10.2.5.1 Run Diags

The information on Run Diags is available to Class C and Class M licenses.

- 1. Navigate to **Utilities > Disruptive Mode Utility** and set Disruptive mode to **On**.
- 2. Navigate to **Diags > Diags Home > Run Diags**



The table below shows all the elements available on Run Diags with descriptions.

Figure 10-23 Run Diags - need new image

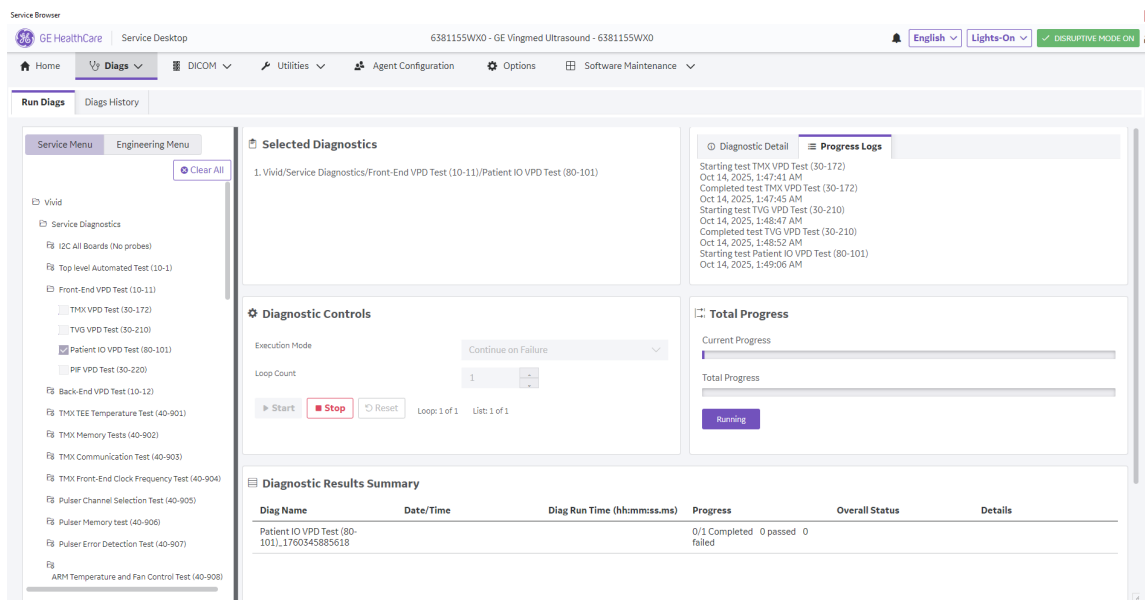
Table 10-13 Run Diags

Element	Description
Service Menu	Available service diagnostics in a diagnostics tree.
Engineering Menu	Available engineering diagnostics in a diagnostics tree.
Clear All	Resets the selections within the diagnostics tree.
Selected Diagnostics	Diagnostics that have been selected.
Diagnostic Detail	
Test Name	Name of the diagnostic.
Message	
Last Execution Result	Result of the diagnostic run.
Expected Execution Time	Time expected to run the diagnostic.
Maximum Execution Time	Maximum amount of time allowed to run the diagnostic.
Instruction	Special instructions needed to perform the diagnostic.
Diagnostic Controls	

Element	Description
Execution Mode	Way the diagnostic behaves when a failure occurs. Valid values are: Continue on Failure, Finish Loop and Stop on Failure.
Loop Count	Number of times to run the selected diagnostic.
Start	Starts the selected diagnostic.
Stop	Stops the selected diagnostic.
Reset	Clears the current diagnostic information.
Total Progress	
Current Progress	Progress the current diagnostic.
Total Progress	Progress for all the selected diagnostics.
Diagnostic Results Summary	
Diag Name	Name of the diagnostic.
Date/Time	Date and time of the diagnostic run.
Diag Run Time	Amount of time taken to run the diagnostic.
Progress	
Overall Status	Status of the diagnostic.
Details	
Pass/Fail	Whether the selected diagnostic passed or failed. Green - Diagnostic passed, Red - Diagnostic failed.

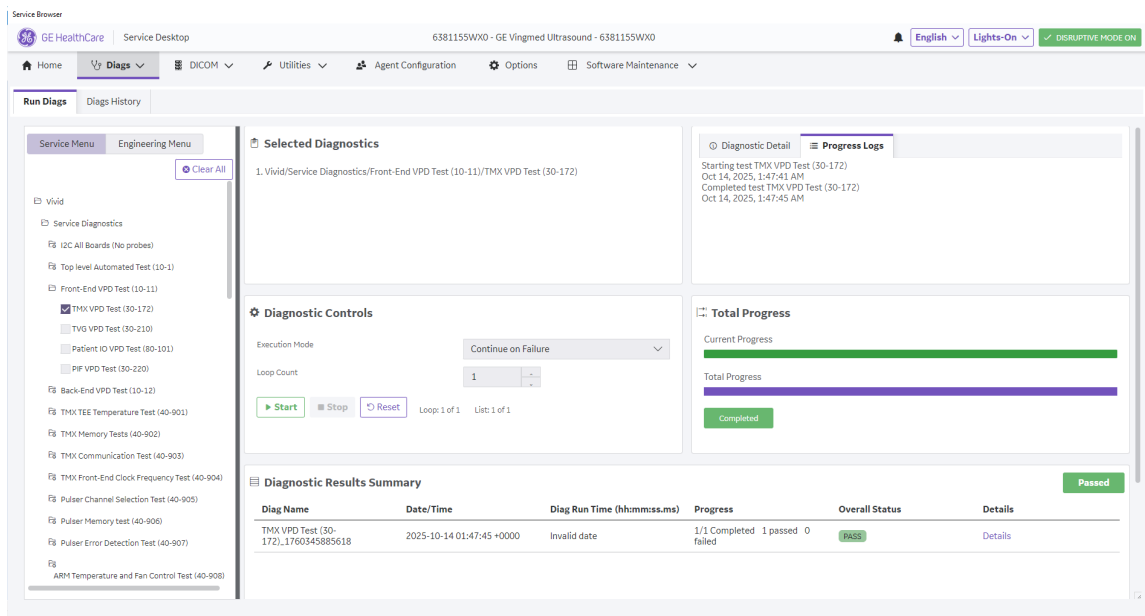
To run a diagnostic:

1. Navigate to **Utility (second page) > Utilities > Disruptive Mode Utility** and set Disruptive mode to **On**.
2. Navigate to **Diags > Diags home > Run Diags**.
3. From the **Service Menu** tab, select the wanted diagnostic test. The Selected Diagnostics and Diagnostic Detail areas populate with information specific to the selected diagnostic.
4. In **Execution Mode**, select the desired mode.
5. In **Loop Count**, select the number of times to run the diagnostic.
6. Click **Start**.



7. View progress in the **Total Progress** area.

8. View the results (Passed or Failed) in the **Diagnostic Results Summary** area.



9. To view details about the diagnostics run, see **Diag History**

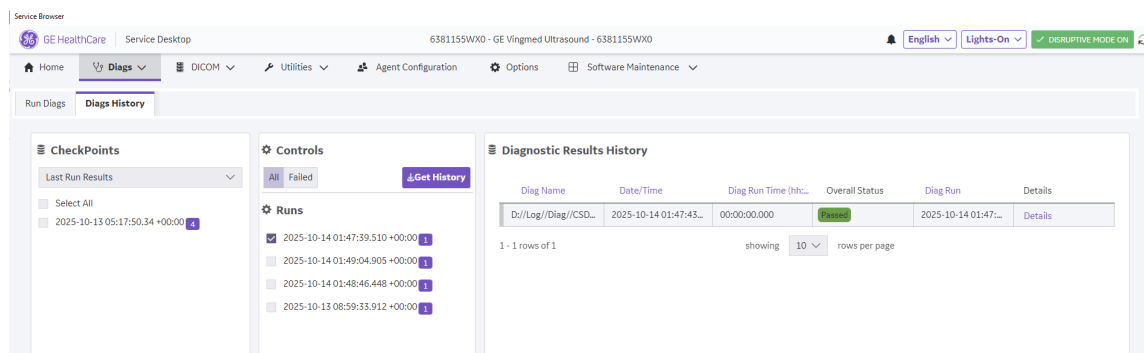
10.2.5.2 Diag History

Use Diag History to view a history of the diagnostics run on the Ultrasound system and the results.

The information on Diag History is available to Class C and Class M licenses.

To access Diag History:

- Navigate to **Diags > Diags Home > Diag History**.



The table [Table 10-14 Diag History Elements](#) on page 281 shows all the elements available on Diag History with descriptions.

Figure 10-24 Diag History

Table 10-14 Diag History Elements

Element	Description
Checkpoints	
Select All	Selects all the available checkpoints in the list.
Controls	
All	Filters results to show all the diagnostics.
Failed	Filters results to show failed diagnostics.
Get History	Select to display a history of the run diagnostics.
Runs	
	List of diagnostics run on the Ultrasound system
Diagnostic Results History	
	Details about the diagnostics run.

To view a history of diagnostics:

1. Navigate to **Utility (second page) > Service > Diag History**.
2. Select the options that you want to view.
3. Click **Get History**.
4. View the results of your query.
5. To view specific details, click **Details**.

Diagnostic Result

Diag Name : EPM FanSpeed Test

Last Start
Time : Tuesday, December 19, 2017 15:24:13

End Time : Tuesday, December 19, 2017 15:24:44

Diag Run Time : 31112ms

Overall Status : PASS

✓ EPM Fan Speed Test

Name	Duty Cycle(%)	Measured RPM	Min RPM	Max RPM	Result
FAN 0	0	<=82	0	100	Passed
FAN 1	0	<=82	0	100	Passed
FAN 2	0	<=82	0	100	Passed
FAN 3	0	<=82	0	100	Passed
FAN 0	22	2065	100	2500	Passed
FAN 1	22	2050	100	2500	Passed
FAN 2	22	2249	100	2500	Passed
FAN 3	22	2236	100	2500	Passed
FAN 0	100	2854	2000	4000	Passed
FAN 1	100	2846	2000	4000	Passed
FAN 2	100	3052	2000	4000	Passed
FAN 3	100	3056	2000	4000	Passed

OK

For more information, see:

- [Diags](#) on page 277
- [Run Diags](#) on page 278

10.2.6 DICOM

Use DICOM to troubleshoot DICOM components and connectivity.

For more information, see:

- [DICOM Spooler](#) on page 282

10.2.6.1 DICOM Spooler

DICOM Spooler provides a summary of the current and recent outgoing jobs.

The information on DICOM Spooler is available to Class C and Class M licenses.

10.2.7 Utilities

Utilities configurations vary depending upon the service class.

For more information, see:

- [Disk Defragment](#) on page 283
- [Gather Logs](#) on page 284
- [Delete Files](#) on page 285
- [Virtual Console Observation](#) on page 286
- [Change Password](#) on page 287
- [Reset Patient Database](#) on page 288
- [Clean Userdefs](#) on page 289
- [Third Party Software Licenses](#) on page 290
- [SSA License](#) on page 291
- [System Shutdown](#) on page 292
- [Checkpoints](#) on page 293
- [Disruptive Mode](#) on page 295
- [Data Transfer](#) on page 295
- [SSH](#) on page 296
- [Network Capture](#) on page 297

10.2.7.1 Disk Defragment

Disk Defragment provides a way to launch the Windows Disk Defragmenter tool to use to defragment the disks. Disk fragmentation can reduce the amount of disk space available, and slow computing speed. Use the disk defragmenter to restore optimum disk space and speed performance. If the Ultrasound system's performance would be significantly reduced when the disk is being defragmented, a warning message is displays. It is not possible to analyze the SSH drives.

The information on Disk Defragment is available to Class C and Class M licenses. Disk Defragment is not available through a remote connection.

To access Disk Defragment, select **Utility > Disk Defragment**.

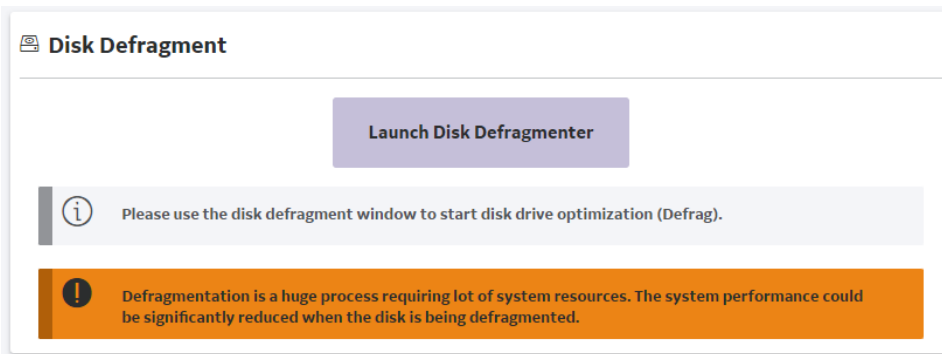
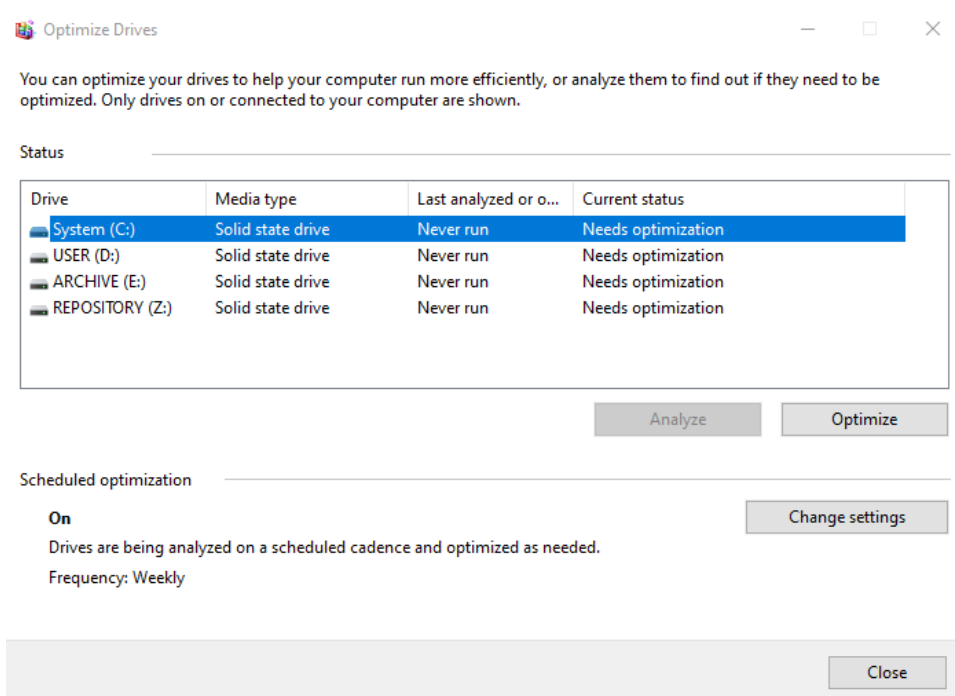


Figure 10-25 Disk Defragment

To analyze, optimize, or schedule optimization for a disk:

1. Navigate to **Utilities > Disk Defragment**.
2. Click **Launch Disk Defragmenter**.
3. In the **Optimize Drives** dialog box, select the drive.



4. To analyze the selected drive, click **Analyze**.
5. To optimize the selected drive, click **Optimize**.
6. To set up a schedule, click **Change settings** and select the schedule.

For more information, see:

- [Utilities](#) on page 282

10.2.7.2 Gather Logs

Gather Logs provides a way to collect system logs and place the log files in the D:\Service directory for retrieval by the online center. These log files do not include protected data such as crash dumps and keyboard shadow logs. The customer can collect logs (including protected data) using Alt+D when Protected Data is checked. Log files are compressed into a .zip file and the file path and name display. If the application software is not running, use the Gather Logs shortcut on the Windows desktop.

The information on Gather Logs is available to all service class licenses.

To access Gather Logs, select **Utilities** > **Gather Logs**.

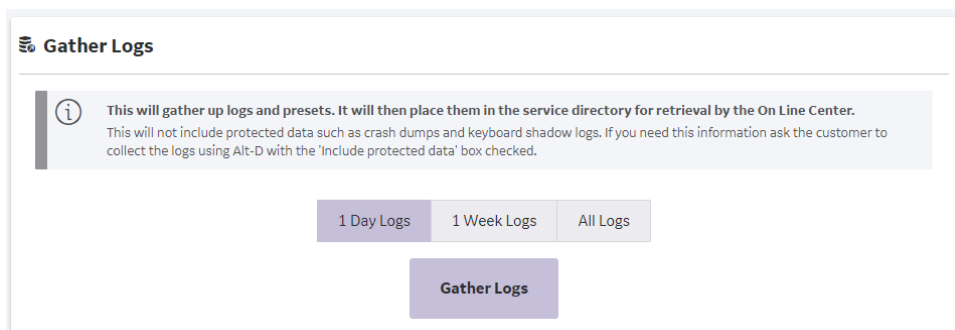


Table [Gather Logs](#) on page 284 shows all the elements available on Gather Logs with descriptions.

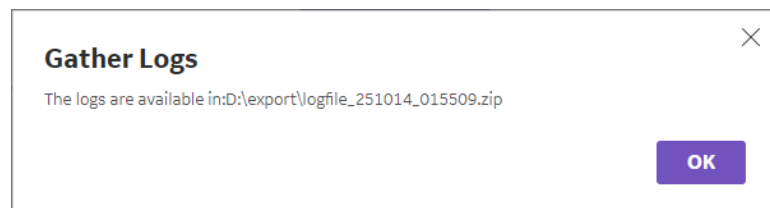
Figure 10-26 Gather Logs

Table 10-15 Gather Logs

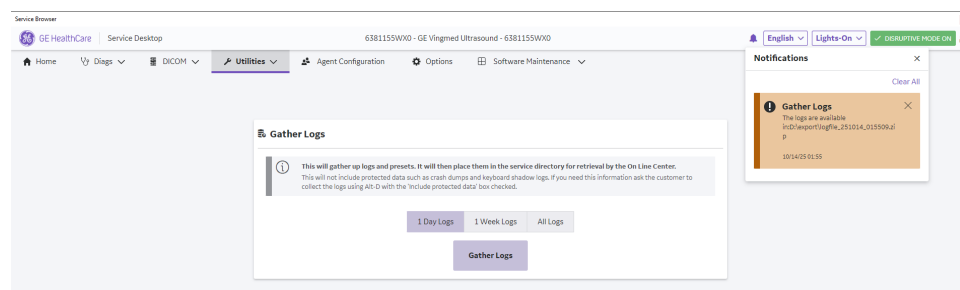
Element	Description
1 Day Logs	When selected, gathers log files for one day.
1 Week Logs	When selected, gathers log files for one week.
All Logs	When selected, gathers all available log files.
Gather Logs	Select to gather the log files for the selected period.

To gather log files:

1. Navigate to **Utilities > Gather Logs**.
2. Select one of the following:
 - 1 Day Logs
 - 1 Week Logs
 - All Logs
3. Click **Gather Logs**. In the resulting dialog box, record the location of the log files and click **OK**.



4. When the gather log operation is complete, click the notification icon in the banner to view the location of the log files.



For more information, see:

- [Utilities](#) on page 282

10.2.7.3 Delete Files

Delete Files displays all the files and folders present in the D:\Service folder and allows for their deletion. Deleting unneeded files improves performance and reduces the need to defragment the disk drive.

The information on Delete Files is available to all service class licenses.

To access Delete Files, select **Utilities > Delete Files**.

Delete Files

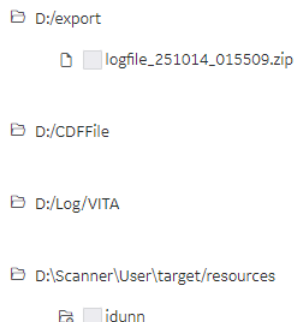


Table [Delete Files](#) on page 285 shows all the elements available on Delete Files with descriptions.

Figure 10-27 Delete Files

Table 10-16 Delete Files

Element	Description
Delete Files	Displays the files that are available for deletion.
Delete	Deletes the selected files.

To delete files:

1. Navigate to **Utilities > Delete Files**.
2. Under **Delete Files**, select the available folders and files that you want to delete.
3. Click **Delete**.
4. In the resulting dialog box, click **Delete** and then click **OK**.

For more information, see:	• Utilities on page 282
----------------------------	---

10.2.7.4 Virtual Console Observation

Virtual Console Observation displays the status of the VNC server. Virtual console observation (VCO) is normally used to remotely access and modify settings and programs.

The information on Virtual Console Observation is available to Class M licenses. Virtual Console Observation is not available through a remote connection.

To access Virtual Console Observation, select **Utilities > Virtual Console Observation**.

This table shows all the elements available on Virtual Console Observation with descriptions.

Table 10-17 Virtual Console Observation

Element	Description
Status	When Disruptive mode is on, displays the status of the VNC server. Valid values are: • VCO is stopped • VCO is running

Element	Description
Start	When Disruptive mode is ON, starts the VNC server.
Stop	When Disruptive mode is ON and the VNC server is running, stops the VNC server.

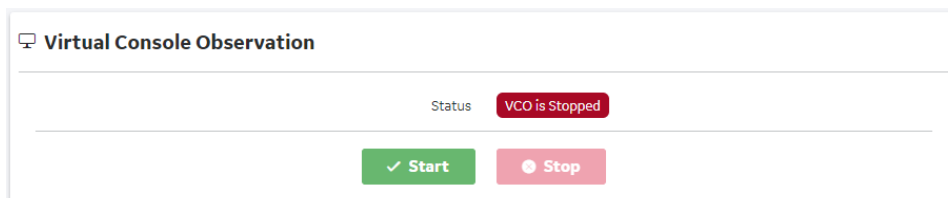


Figure 10-28 Virtual Console Observation

To start Virtual Console Observation:

1. Navigate to **Utilities > Disruptive Mode Utility** and set Disruptive mode to **On**.
2. Navigate to **Utilities > Virtual Console Observation**.
3. Select **Start**. When VCO starts, the Windows task bar and Windows desktop display.
4. Modify settings and programs as needed.
5. To stop Virtual Console Observation, click **Stop**. When VCO is stopped, the Windows desktop is hidden.
6. Restart the Ultrasound system.

For more information, see:	• Utilities on page 282
----------------------------	---

10.2.7.5 Change Password

Change Password allows you to change the password for a specified user type.

The information on Change Password is available to all service class licenses. Change Password is not available through a remote connection.

To access Change Password, select **Utilities > Change Password**.

This table shows all the elements available on Change Password with descriptions.

Figure 10-29 Change Password

Table 10-18 Change Password

Element	Description
User Type	Type of user for the password reset.
New Password	Password
Confirm Password	Password
Update Password	Select to update the password.
Reset	Select to reset the information.

To change the password:



CAUTION

Risk of losing data.

If the password is lost, GE HealthCare will not be able to recover or reset it.

Loss of a password may result in the loss of patient data.

Securely record the new password.

1. Navigate to **Utilities > Change Password**.
2. Under **User Type**, select the user.

NOTE

Make sure the Ultrasound system is connected to the network and the agent is configured. The GEService password is used to perform portions of remote service. If the password is changed and the system information is not updated, it may slow down remote service. Both file transfer and SSH depend on the GEService password.

3. In **New Password** and **Confirm Password**, enter the new password.
4. Click **Update Password**.
5. When a Service User's password has been changed, reboot the Ultrasound system to reflect the password change.

For more information, see:

- [Utilities](#) on page 282

10.2.7.6 Reset Patient Database

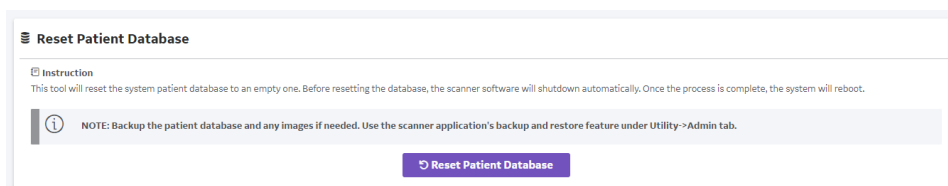
Reset Patient Database resets the Ultrasound system patient database. This operation automatically shuts down the application before cleaning the patient database. Once the operation is complete, the Ultrasound system reboots with an empty database. Disruptive mode must be On to reset the patient database. If Disruptive mode is Off, a warning message displays asking to turn on Disruptive mode.

Reset Patient Database deletes the contents of the following folders:

- E:\Scanner\GEMS_IMG
- E:\Scanner\GEMS_REP

The information on Reset Patient Database is available to Class C (Service Expert) and Class M licenses.

To access Reset Patient Database, select **Utilities > Reset Patient Database**.



This table shows all the elements available on Reset Patient Database with descriptions.

Figure 10-30 Reset Patient Database

Table 10-19 Reset Patient Database

Element	Description
Instruction	Information about the database reset.
Reset Patient Database	When Disruptive mode is ON, resets the Ultrasound system's patient database.

To reset the patient database:

1. Back up the patient database.
2. Navigate to **Utilities > Disruptive Mode Utility** and set Disruptive mode to **On**.
3. Navigate to **Utilities > Reset Patient Database**.
4. Click **Reset Patient Database**.
5. Confirm the patient database has been emptied.

For more information, see:	• Utilities on page 282
----------------------------	---

10.2.7.7 Clean Userdefs

Clean Userdefs provides a way to delete user-defined data when user presets may be inhibiting proper operation. User-defined data returns to the user-defined settings to the factory default values. Clean Userdefs provides information to back up the current settings if you want to use them later. Disruptive mode must be On to clean user-defined data. If Disruptive mode is Off, a warning message displays asking to turn on Disruptive mode.

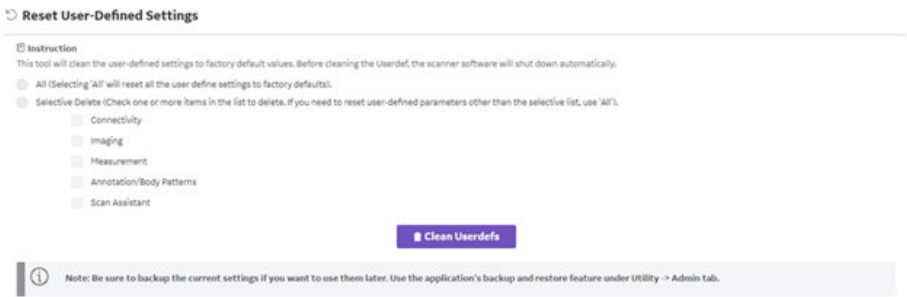
Before cleaning user-defined data, the application software automatically shuts down. Once the operation is complete, the system reboots.

Clean Userdefs provides a way to selectively delete the following user-defined data:

- Connectivity – Deletes this file: %TEST_ROOT%\resources\userdefs\connectivity.res
- Imaging – Deletes this folder: %TEST_ROOT%\resources\userdefs\ImagingPresets\
- Measurement – Delete this folder: %TEST_ROOT%\resources\userdefs\EchoMeasure\
- Annotation/Bodymark – Deletes the folder: %TEST_ROOT%\resources\userdefs\EchoAnnotation\
- Scan Assistant – Deletes the folder: %TEST_ROOT%\resources\userdefs\EchoExamProtocol

The information on Clean Userdefs is available to Class C (Service Expert) and Class M licenses.

To access Clean Userdefs, select **Utilities > Clean Userdefs**.



This table shows all the elements available on Clean Userdefs with descriptions.

Figure 10-31 Clean Userdefs

Table 10-20 Clean Userdefs

Element	Description
All	When Disruptive mode is ON, deletes all the user-defined data and returns the Ultrasound system to factory default settings.
Selective Delete	When Disruptive mode is ON, deletes the selected user-defined data.
Connectivity	When Selective Delete is selected, deletes this file: %TEST_ROOT%\resources\userdefs\connectivity.res
Imaging	When Selective Delete is selected, deletes this folder: %TEST_ROOT%\resources\userdefs\ImagingPresets\
Measurement	When Selective Delete is selected, delete this folder: %TEST_ROOT%\resources\userdefs\EchoMeasure\
Annotation/Body Patterns	When Selective Delete is selected, deletes this folder: %TEST_ROOT%\resources\userdefs\EchoAnnotation\
Scan Assistant	When Selective Delete is selected, deletes this folder: %TEST_ROOT%\resources\userdefs\EchoExamProtocol
Clean Userdefs	Restarts the application software with a fresh database.

To clean user-defined data:

1. If you want to use the current settings later, back them up now.
2. Navigate to **Utilities > Disruptive Mode Utility** and set Disruptive mode to **On**.
3. Navigate to **Utilities > Clean Userdefs**.
4. Select the data you want to set to the default settings.
5. Click **Clean Userdefs**.

For more information, see:	• Utilities on page 282
----------------------------	---

10.2.7.8 Third Party Software Licenses

Third Party Software Licenses displays the third-party software licenses used as part of the service platform.

The information on Third Party Software Licenses is available to all service class licenses.

To access Third Party Software Licenses, select **Utilities > Third Party Software Licenses**.

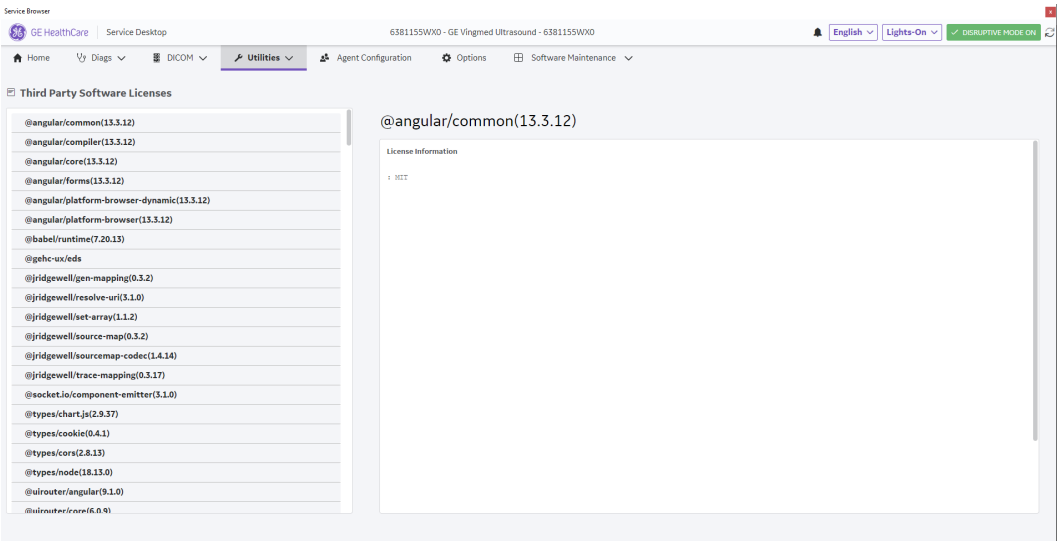


Figure 10-32 Third Party Software Licenses

For more information, see:	• Utilities on page 282
----------------------------	---

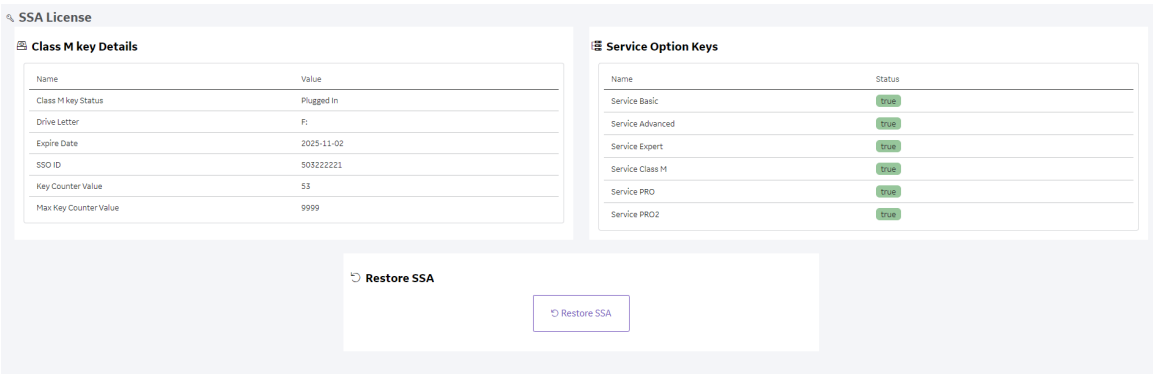
10.2.7.9 SSA License

SSA License provides a way to do the following:

- When inserted, view the details of an SSA key.
- View the status of the service class options.
- Restore an SSA license when the SSA key is not validating or when a remote log in shows as a Class A user.

The information on SSA License is available to all service class licenses.

To access SSA License, select **Utilities > SSA License**.



This table shows all the elements available on SSA License with descriptions.

Figure 10-33 SSA License

Table 10-21 SSA License

Element	Description
Class M Key Details	

Element	Description
Class M Key Status	Status of the SSA key. Valid values are: - Not Plugged In - Plugged In
Drive Letter	Drive where the SSA key is plugged into the Ultrasound system.
Expire Date	Date the SSA key is set to expire.
SSO ID	Identifier for the user assigned to the SSA key.
Key Counter Value	Number of times the SSA key has been used.
Max Key Counter Value	Number of remaining times the SSA key can be used.
Service Option Keys	
Name	Name of the service class option.
Status	Status of the access to the associated service class option. Valid values are: TrueFalse
Restore SSA	Restores the SSA license to the SSA key.

To view the Class M license information:

1. Insert the SSA key.
2. Navigate to **Utilities > SSA License**.
3. Under **Class M key Details**, view the values. For example, the SSO ID for the user assigned to the SSA key. If **Not Available** displays for all these values, the SSA key is not validating.

To restore an SSA key that is not validating:

1. Remove the SSA key from the Ultrasound system.
2. If open, close the Service desktop.
3. Restart the Ultrasound system.
4. Once the Ultrasound system has restarted, plug in the SSA key.
5. Navigate to **Utilities > SSA License**.
6. Click **Restore SSA**.
7. Check to see if the SSA key validates.

For more information, see:	Utilities on page 282
----------------------------	---

10.2.7.10 System Shutdown

With remote access, System Shutdown provides a way to shut down or restart the Ultrasound system. Disruptive mode must be On to shut down or restart the Ultrasound system. If Disruptive mode is Off, a warning message displays.

The information on System Shutdown is available to Class M licenses.

To access System Shutdown, select **Utilities > System Shutdown**.

This table shows all the elements available on System Shutdown with descriptions.

Figure 10-34 System Shutdown

Table 10-22 System Shutdown

Element	Description
Select Operation	
Shutdown System	When Disruptive mode is On, shuts down the Ultrasound system when Submit Request is pressed.
Restart System	When Disruptive mode is On, restarts the Ultrasound system when Submit Request is pressed.
Disruptive Mode	
Retain Disruptive Mode	Determines whether Disruptive mode is retained after an operation has been performed. Generally used by the remote engineer to regain access to disruptive functions after a reboot without having to request Disruptive mode be enabled again.
Submit Request	Performs the operation selected under Select Operation.

To remotely start, restart, or shut down the Ultrasound system:

1. Navigate to **Utilities > Disruptive Mode Utility** and set Disruptive mode to **On**.
2. Navigate to **Utilities > System Shutdown** and then select an operation:
 - To shut down the Ultrasound system, select **Shutdown System**.
 - To restart the Ultrasound system, select **Restart System**.
 - To shut down the application software, select **Shutdown Application**.
3. To retain the current setting for Disruptive mode, select **Retain Disruptive Mode**.
4. Click **Submit Request**.

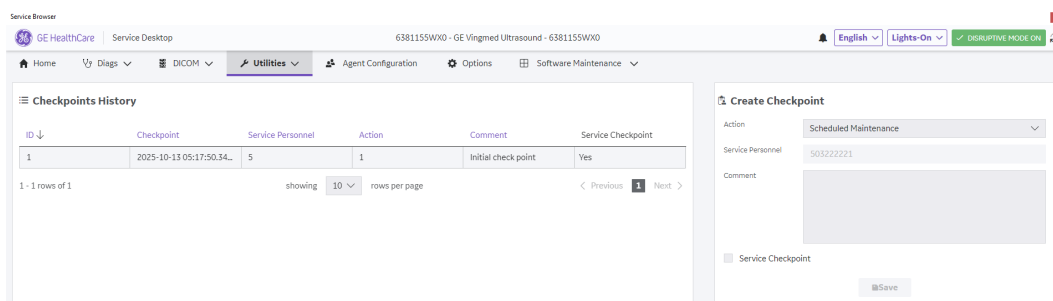
For more information, see:	• Utilities on page 282
----------------------------	---

10.2.7.11 Checkpoints

Checkpoints allows for the creation of checkpoints and displays the history of those checkpoints. Checkpoints are used to indicate when the Ultrasound system has been serviced, status has been reset, and information related to the service performed has been added.

The information on Checkpoints is available to Class C (Service Advanced and Service Expert) and Class M licenses.

To access Checkpoints, select **Utilities > Checkpoints**.



This table shows all the elements available on Checkpoints with descriptions.

Figure 10-35 Checkpoints

Table 10-23 Checkpoints

Element	Description
Checkpoints History	
Checkpoint	Displays the name of the checkpoint.
Service Personnel	Displays the ID for the logged in user.
Action	Displays the service action.
Comment	Displays comment text entered when the checkpoint was created.
Service Checkpoint	Displays whether the checkpoint is a service checkpoint or not.
Create Checkpoint	
Action	Selects the service action. Value values are: - Scheduled Maintenance - Unscheduled Maintenance - Upgrade - FMI
Service Personnel	ID for the logged in user. Value values are: - SSO ID: When a Class M key is plugged in, the SSO ID for the logged in user. - Remote user ID: If there is a remote connection, the remote user ID. - User ID: If the SSO ID and remote user ID are not available, the user ID for the logged in user. - Unknown: The logged in user ID is not available.
Comment	Enter comments for creating a checkpoint.
Service Checkpoint	Select whether to mark a service checkpoint or not.
Save	Saves the information.

To create a check point:

- Navigate to **Utilities > Checkpoints**.
- Under Create Checkpoints, from Action, select the action.
- In Comment, enter notes for the checkpoint.
- If this is a checkpoint for service, select **Service Checkpoint**.
- Click **Save**.

For more information, see:

- [Utilities](#) on page 282

10.2.7.12 Disruptive Mode

Disruptive Mode provides a way to enable and disable Disruptive mode. When enabled, Disruptive mode allows a remote user to access the Ultrasound system which can disrupt normal operation of the Ultrasound system. Disruptive mode can be remotely turned on when the logged-in user has Authorize Remote Service Access privilege. If the logged-in user does not have remote service privilege, displays this error message: "You do not have the required permission to perform this operation."

When initiating Disruptive mode from a remote location, a message displays on the Ultrasound system asking whether the user will allow Disruptive mode and notifying them that normal operation might be disturbed.

- If Yes is selected, then Disruptive mode is turned on.
- If No is selected, then Disruptive mode is not turned on and an error message display saying that permission to change the mode was denied.

When Disruptive mode is On, the Contact GE icon in the status bar changes to red and displays this message: "Due to Service testing restart needed."

When Disruptive mode is changed to Off, the Contact GE icon turns white and displays this message: "Due to Service testing restart needed." In addition, VCO and SSH stop, if running.

The information on Disruptive Mode is available to Class C (Service Advanced and Service Expert) and Class M licenses.

To access Disruptive Mode, select **Utilities > Disruptive Mode Utility**.

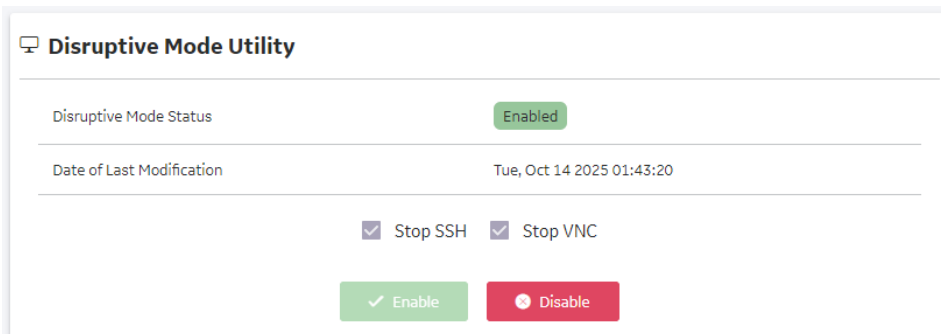


Figure 10-36 Disruptive Mode

For more information, see:

- [Utilities](#) on page 282

10.2.7.13 Data Transfer

Data Transfer provides a way to do the following:

- View information about past transfers of (APM) information.
- Set up automatic/scheduled transfer of allowed data files from the Ultrasound system to the server.
- Manually transfer data files allowed data files from the Ultrasound system to the server.

The information on Data Transfer is available to all service class licenses.

To access Data Transfer, open Service Desktop and select **Utilities > Data Transfer**.

Data Transfer

Type of Upload	Last Upload Status	Last Upload Attempt	Last Successful Attempt
System Logs	Success	Oct 14, 2025, 2:14:36 AM	Oct 14, 2025, 2:14:36 AM
CDF files	Success	Oct 13, 2025, 9:24:11 AM	Oct 13, 2025, 9:24:11 AM
VITA files	Success	Oct 13, 2025, 9:24:25 AM	Oct 13, 2025, 9:24:25 AM

Type of Upload

☒ All ☒ System Logs ☒ CDF files ☒ VITA files

☒ Scheduler

☒ All Days ☒ Mon ☒ Tue ☒ Wed ☒ Thu ☒ Fri ☒ Sat ☒ Sun

Save Settings

Send All

This table shows all the elements available on Data Transfer with descriptions.

Figure 10-37 Data Transfer

Table 10-24 Data Transfer

Element	Description
Type of Upload	Type of log file. For example, System Logs are incrementally transferred when automatic transfer is enabled. To enable automatic transfer, navigate to System Admin and, under Service, check Enable Automatic Request for Service. CDF files: VITA files:
Last Upload Status	Whether the last log file upload was successful or not.
Last Upload Attempt	Date and time the last log file upload was attempted.
Last Successful Attempt	Date and time the last log file was successfully uploaded.
Scheduler	When selected, enables the related day selections. For example, All Days, Monday, and Tuesday.
Save Settings	Saves the information.
Send All	Manually send the selected log files to the server.

10.2.7.14 SSH

SSH displays the status of the Secure Shell (SSH) server. SSH is a cryptographic network protocol for operating network services securely over an unsecured network. This protocol is used for remote login to the operational system. SSH allows remote login to the Ultrasound system and remote file transfer from the Ultrasound system. Remotely log into the Ultrasound system using the username “GEService” and its password.

The information on SSH is available to Class M licenses.

To access SSH, select **Utilities > SSH**.

SSH

Status SSH is Running.

Start

Stop

This table shows all the elements available on SSH with descriptions.

Figure 10-38 SSH

Table 10-25 SSH

Element	Description
Status	When Disruptive mode is on, displays the status of the SSH server. Valid values are: - SSH is stopped - SSH is running
Start	When Disruptive mode is ON, start the SSHD service.
Stop	When Disruptive mode is ON and the SSHD service is running, stops the SSHD service.

To start the SSH server:

- Navigate to **Utilities > Disruptive Mode Utility** and set Disruptive mode to On.
- Navigate to **Utilities > SSH**.
- Click **Start**.
- To stop the SSH server, select **Stop**.

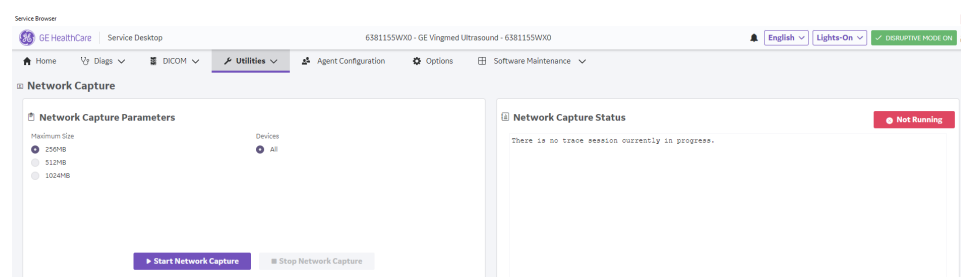
For more information, see:	Utilities on page 282
----------------------------	---

10.2.7.15 Network Capture

Network Capture displays network traffic between the Ultrasound system and configured devices. A network capture outputs two log files: one for main logging with no protected information and another including protected information. These log files are useful when debugging connectivity issues. Because these log files can be large, they are only kept for one week.

The information on Network Capture is available to all service class licenses.

To access Network Capture, select **Utilities > Network Capture**.



This table shows all the elements available on Network Capture with descriptions.

Figure 10-39 Network Capture

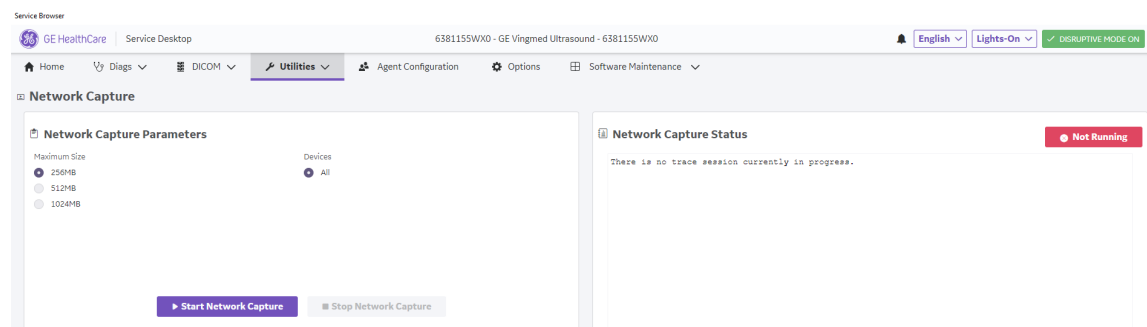
Table 10-26 Network Capture

Element	Description
Network Capture Parameters	

Element	Description
Maximum Size	Allowed size of the generated log file. Valid values are: 256MB 512MB 1024MB
Devices	DICOM-configured devices for which you want to capture information. If no additional devices are configured, only All will be available.
Start Network Capture	Select to start the process. This causes the network capture to start, enables the Stop button, and updates the Network Capture Status pane and changes the Status to Running.
Stop Network Capture	Select to stop the process.
Network Capture Status	
	Displays information about the status of the network capture. The language setting for this information is set in Windows and not through the Service desktop or Ultrasound system application software.
	Displays the status of the network capture. Valid values are: - Not Running - Running

To perform a network capture:

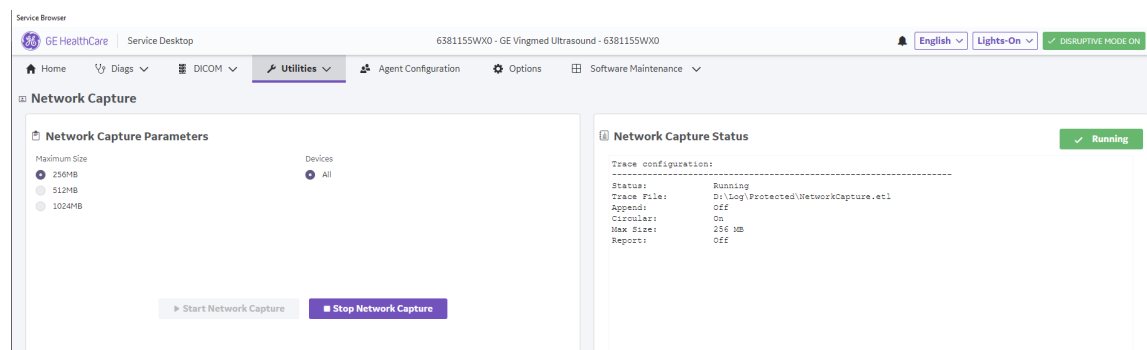
1. Navigate to **Utilities > Network Capture**.



2. From Network Capture, do the following:

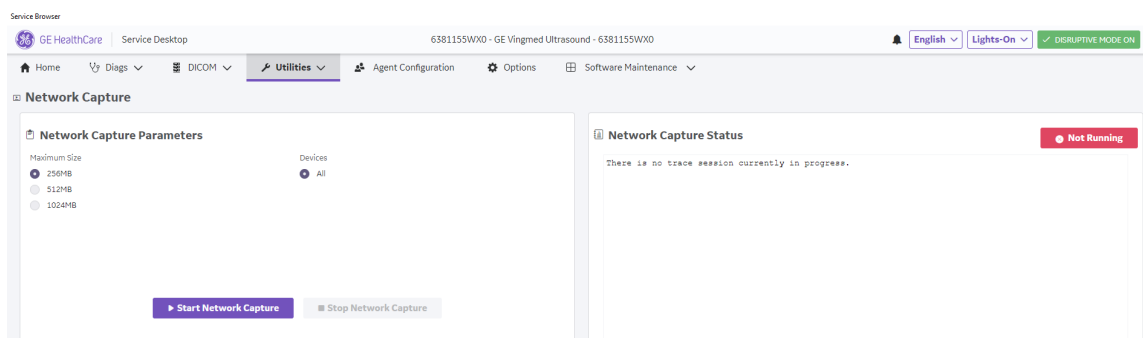
- Under **Maximum Size**, select the allowed size of the generated log file.
- Under **Devices**, select the DICOM-configured device for which you want to capture information. If no additional devices are configured, only All will be available.

3. Select **Start Network Capture** to start the process. This causes the network capture to start, enables the **Stop** button, and updates the **Network Capture Status** pane and changes the Status to **Running**.



4. Click the **Stop** button to end data collection. Stopping is a two-step process:

- Stops the data collection and immediately closes the .etl file.
- Collects additional diagnostic data that may help diagnose network issues. When the file is closed, you see “There is no trace session currently in progress”. When the remaining data is collected and the .cab file is closed, you are notified in the banner.



For more information, see:

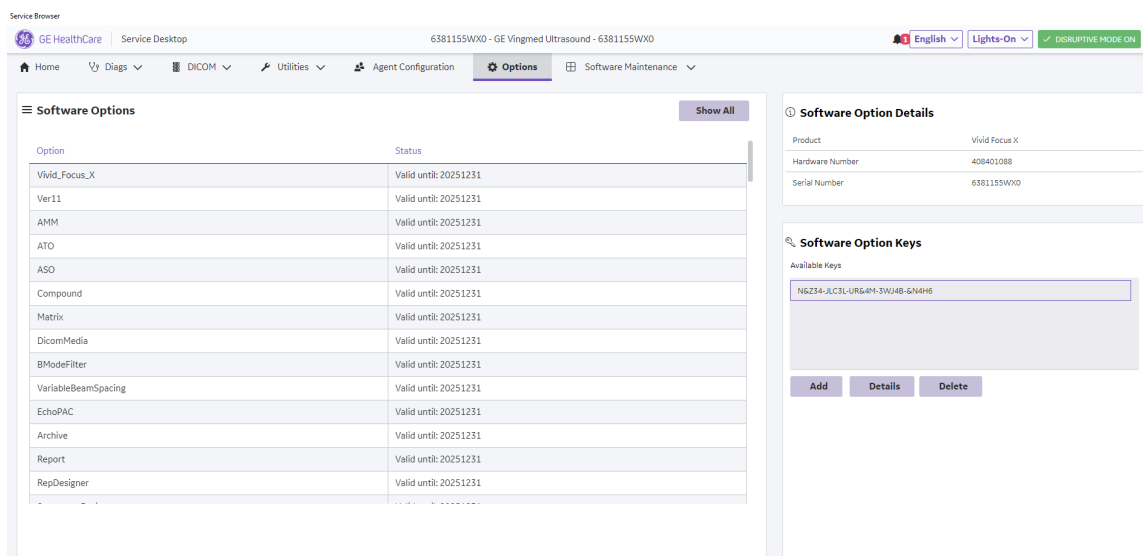
- [Utilities](#) on page 282

10.2.8 Options

Use Options to:

- View software options.
- View software option details.
- Add (or delete) a valid option key, add a duplicate option key, not add an invalid option key, and ask for confirmation before deleting an option key.
- View software option key details. Key details are a list of options that are enabled by a particular key. Under Available Keys, highlight the option string, select Details and then view the options on the left side of the screen. Press **Show All** to view all the activated options.

The information on Options is available to all service class licenses.



This table shows all the elements available on Options with descriptions.

Figure 10-40 Options

Table 10-27 Options

Element	Description
Software Options	
Option	Software options on the Ultrasound system.
Status	Status of the options on the Ultrasound system.
Software Option Details	
Product	Name of the product.
Hardware Number	Number for the hardware. The hardware number is the hash of the serial number that is used to generate the option key.
Serial Number	Serial number of the Ultrasound system.
Software Option Keys	
Available Keys	List of the option keys installed on the Ultrasound system.

10.2.9 Configuration Steps

Use Agent Configuration to:

- Edit and configure the following:
 - Enterprise host name in the agent.
 - Enterprise port number in the agent.
 - Proxy server in the agent.
 - Proxy port in the agent.
 - CRM number in the agent.
 - Display name in the agent.
- Set the serial number in the agent.
- Enter the username and password for the proxy.
- Reset the edited unsaved value.
- Update contact details
- Enable Automatic InSite Connection.
- If "Automatic InSite Connection" is enabled, AutoConnect Retry details can be checked

The information on Agent Configuration is available to all service class licenses.

Service Browser

GE HealthCare | Service Desktop | 6381155WX0 - GE Vingmed Ultrasound - 6381155WX0

English | Lights-On | DISRUPTIVE MODE ON

Home | Diags | DICOM | Utilities | **Agent Configuration** | Options | Software Maintenance

Agent Configuration ☐ Enable automatic InSite connection

Agent Status: Running Agent Registered: Yes Agent Quarantine: No Agent CRM Verified: Yes

Agent Model Number: CS_MLS_CVUS_VIVID_FOCUS_X Serial Number: 6381155WX0 CRM No: 6381155WX0 Display Name:

Agent Connected: Yes Last Connected Time: Mon, Oct 13 2025, 05:10:16 Agent Version: 4.7.2

▼ Contact Details

▼ Advanced Configuration

▼ Proxy Configuration

[Submit Changes](#) [Reset Form](#)

This table shows all the elements available on Agent Configuration with descriptions.

Figure 10-41 Agent Configuration

Table 10-28 Agent Configuration

Element	Description
Agent Configuration	
Enable Automatic InSite Connection	This device includes capability to automatically populate and activate a GEHC InSite connection for remote service features. This feature automatically populates the following fields: CRM No, Enterprise Server, Enterprise Host, Enterprise Port and Proxy Configuration fields (if detected). Editing any field in this screen requires de-selecting this option. This will fully disable automatic connection features and will then requires manual entry and submission to activate GEHC InSite connection.
Agent Status	Status for the agent. Valid values are: - Running - Not Running
Agent connected	
Agent Version	Version of the agent.
Agent Registered	Registered status of the agent. Valid values are: - Yes – The agent is registered in the back office. - No – The agent is not registered in the back office. - Not Available – The agent is not configured or running.
Agent Quarantine	Quarantine status of the agent. Valid values are: - Yes – The agent has more than one device registered with the same CRM Number in the back office. This scanner cannot send data back to GE or be remotely accessed. - No – The agent has one device registered with the listed CRM Number in the back office. - Not Available – The agent is not configured or running.
Agent CRM Verified	CRM verified status of the agent. Valid values are: - Yes – The agent is verified in the back office. - No – The agent is not verified in the back office. - Not Available – The agent is not configured or running.

Element	Description
Agent Model Number	GE Healthcare part number for the ultrasound system. The same number as listed on the rating plate.
Serial Number	Serial number of the agent (read-only). If the agent is not registered with a serial number, this field is populated with the serial number of the Ultrasound system. The serial number of the agent is tied to the serial number of the Ultrasound system.
CRM No	Customer Relationship Management (CRM) number. System identifier assigned to the customer unit by the service region. CRM is pre-populated by adding Serial number to the CRM number. The CRM number is editable.
Display Name	Displayed name of the agent.
Contact Details	Phone number for the person at the customer site a GE remote service engineer would contact. The phone number is entered during installation and reviewed at every service call to make sure the information is correct.
Advanced Configuration	
Enterprise Server	Name of the enterprise server.
Enterprise Host	Number of the enterprise host.
Enterprise Port	Number of the enterprise port.
Proxy Configuration	
Enable Proxy	Enables the proxy server.
Proxy Server	When Enable Proxy is selected, name of the proxy server IP.
Proxy Port	When Enable Proxy is selected, number of the proxy server port.
Credentials	
Enable Proxy Credentials	Enables the proxy credentials.
Username	When Enable Proxy Credentials is selected, name of the user.
Password	When Enable Proxy Credentials is selected, password for the user.

10.2.9.1 Configuration Steps

Please choose the corresponding server from Table 7-30: 'Enterprise server list' on page 7-83, and ensure that 'Enable automatic Insite Connection' is selected before complete Agent Configuration setting.

Service Browser

Agent Model Number CS,ULS,CVUS,VIVID, FOCUS, S Serial Number TC001005 CRM No TC001005 Display Name

Agent Connected No Agent Version 4.7.2

▼ Contact Details

▲ Advanced Configuration

Enterprise Configuration Attention: According to legal regulation, please choose the correct Enterprise server.

Server Production Enterprise Host inSITE.gehealthcare.com Enterprise Port 443

Flexera Configuration

Software Download URL

Add Enterprise Configuration

Production

Production-EU

Production-CN

Others

Proxy Configuration

☐ Enable Proxy

Proxy Server

Proxy Port

☐ Enable Proxy Credentials

Username

Password

Submit Changes Reset Form

© 2025 Common Service Desktop 3.2.1.87

Figure 10-42 Select proper server

Service Browser

GE HealthCare Service Desktop TC001005 - GE Vingmed Ultrasound - TC001005

English Lights-On DISRUPTIVE MODE OFF

Home Diags DICOM Utilities **Agent Configuration** Options Software Maintenance

▲ Agent Configuration Enable automatic InSite connection

Agent Status Running Agent Registered No Agent Quarantine No Agent CRM Verified No

Agent Model Number CS,ULS,CVUS,VIVID, FOCUS, S Serial Number TC001005 CRM No TC001005 Display Name

Agent Connected No Agent Version 4.7.2

▼ Contact Details

▼ AutoConnect Retry Details

▲ Advanced Configuration

Enterprise Configuration Attention: According to legal regulation, please choose the correct Enterprise server.

Server Production Enterprise Host inSITE.gehealthcare.com Enterprise Port 443

Flexera Configuration

Software Download URL <https://gehealthcare-us.flexera.com/>

Add Enterprise Configuration

Proxy Configuration

☐ Enable Proxy

☐ Enable Proxy Credentials

Figure 10-43 Check 'Enable automatic Insite Connection'

Table 10-29 Enterprise server list

Ship to country	Enterprise Server
(AD) Andorra, (AL) Albania, (AT) Austria, (AX) Åland Islands, (BA) Bosnia and Herzegovina, (BE) Belgium, (BG) Bulgaria, (CH) Switzerland, (CY) Cyprus, (CZ) Czech Republic, (DE) Germany, (DK) Denmark, (EE) Estonia, (ES) Spain, (FI) Finland, (FO) Faroe Islands, (FR) France, (GB) United Kingdom, (GG) Guernsey, (GI) Gibraltar, (GR) Greece, (HR) Croatia, (HU) Hungary, (IE) Ireland, (IM) Isle of Man, (IS) Iceland, (IT) Italy, (JE) Jersey, (LI) Liechtenstein, (LT) Lithuania, (LU) Luxembourg, (LV) Latvia, (MC) Monaco, (MD) Moldova, (ME) Montenegro, (MK) North Macedonia, (MT) Malta, (NL) Netherlands, (NO) Norway, (PL) Poland, (PT) Portugal, (RO) Romania, (RS) Serbia, (SE) Sweden, (SI) Slovenia, (SJ) Svalbard and Jan Mayen, (SK) Slovakia, (SM) San Marino, (VA) Vatican City, (UK) United Kingdom	Production-EU
(CN) China, (HK) Hong Kong, (MO) Macau, (TW) Taiwan	Production-CN
All other countries	Production

10.2.10 Configuration Steps

Please choose the corresponding server from Table 7-30: 'Enterprise server list' on page 7-83, and ensure that 'Enable automatic Insite Connection' is selected before complete Agent Configuration setting.

Service Browser

Agent Model Number CS,ULS,CVUS,VIVID, FOCUS, S Serial Number TC001005 CRM No TC001005 Display Name

Agent Connected No Agent Version 4.7.2

▼ Contact Details

▲ Advanced Configuration

Enterprise Configuration Attention: According to legal regulation, please choose the correct Enterprise server.

Server Production Enterprise Host inSITE.gehealthcare.com Enterprise Port 443

Flexera Configuration

Software Download URL

Add Enterprise Configuration

Production

Production-EU

Production-CN

Others

Proxy Configuration

☐ Enable Proxy

Proxy Server

Proxy Port

☐ Enable Proxy Credentials

Username

Password

Submit Changes Reset Form

© 2025 Common Service Desktop 3.2.1.87

Figure 10-44 Select proper server

Service Browser

GE HealthCare Service Desktop TC001005 - GE Vingmed Ultrasound - TC001005

English Lights-On DISRUPTIVE MODE OFF

Home Diags DICOM Utilities **Agent Configuration** Options Software Maintenance

▲ Agent Configuration Enable automatic InSite connection

Agent Status Running Agent Registered No Agent Quarantine No Agent CRM Verified No

Agent Model Number CS,ULS,CVUS,VIVID, FOCUS, S Serial Number TC001005 CRM No TC001005 Display Name

Agent Connected No Agent Version 4.7.2

▼ Contact Details

▼ AutoConnect Retry Details

▲ Advanced Configuration

Enterprise Configuration Attention: According to legal regulation, please choose the correct Enterprise server.

Server Production Enterprise Host inSITE.gehealthcare.com Enterprise Port 443

Flexera Configuration

Software Download URL <https://gehealthcare-us.flexera.com/>

Add Enterprise Configuration

Proxy Configuration

☐ Enable Proxy

☐ Enable Proxy Credentials

Figure 10-45 Check 'Enable automatic Insite Connection'

Table 10-30 Enterprise server list

Ship to country	Enterprise Server
(AD) Andorra, (AL) Albania, (AT) Austria, (AX) Åland Islands, (BA) Bosnia and Herzegovina, (BE) Belgium, (BG) Bulgaria, (CH) Switzerland, (CY) Cyprus, (CZ) Czech Republic, (DE) Germany, (DK) Denmark, (EE) Estonia, (ES) Spain, (FI) Finland, (FO) Faroe Islands, (FR) France, (GB) United Kingdom, (GG) Guernsey, (GI) Gibraltar, (GR) Greece, (HR) Croatia, (HU) Hungary, (IE) Ireland, (IM) Isle of Man, (IS) Iceland, (IT) Italy, (JE) Jersey, (LI) Liechtenstein, (LT) Lithuania, (LU) Luxembourg, (LV) Latvia, (MC) Monaco, (MD) Moldova, (ME) Montenegro, (MK) North Macedonia, (MT) Malta, (NL) Netherlands, (NO) Norway, (PL) Poland, (PT) Portugal, (RO) Romania, (RS) Serbia, (SE) Sweden, (SI) Slovenia, (SJ) Svalbard and Jan Mayen, (SK) Slovakia, (SM) San Marino, (VA) Vatican City, (UK) United Kingdom	Production-EU
(CN) China, (HK) Hong Kong, (MO) Macau, (TW) Taiwan	Production-CN
All other countries	Production

10.3 Request for Service (RFS)

Create a Request

Users can create and submit an error report to GE Healthcare digital service.

NOTE

Submission will only be successful if the agent is already connected.

1. Select the Contact GE Icon on the status bar at the bottom of the screen.



2. The InSite service menu is displayed. You will see these options:

- **Service Desktop**
- **Request for Service**
- **Connect to GE Healthcare**
- **Connect to Clinical Lifeline**
- **Cancel**

3. Select **Request for Service**

4. The **Contact GE** screen is displayed

5. Select **Create RFS** from the top right of the screen.

6. Fill out the required information and press **Submit**

Table 10-31 Request for Service - Required Information

Element	Description
System ID	
Problem Type	What type of issue are you experiencing? Valid values are: • Service • Application This is a required field

Element	Description
Problem Area	In which area are you experiencing an issue? Valid values are: • Hardware Electrical • Hardware Mechanical • Network • Image Quality • Peripherals This is a required field
Problem Description	Description of the issue in more detail. This is a required field
First Name	First name of submitter. This is a required field
Last Name	Last name of submitter. This is a required field
Phone Number	Phone number of submitter. This is a required field
Email	Email of submitter.

Selecting **RFS History** on the **Contact GE** screen will display all previously submitted requests and include the information described in table [Request for Service \(RFS\)](#) on page 307.

10.4 Check Point Voltages

Output voltage (usually 12V DC) of the TPS Power Supply can be checked via a LED on the TBE Backend Base Board. This green-light status LED (= ON) is visible from outside.

During boot-up further status LEDs are enabled.

10.5 Battery Check

10.5.1 Battery safety



DANGER

Failure to read and follow the battery pack safety instructions and warnings below may result in personal injury and property damages.

- **Do not** disassemble or modify the battery pack.
- Excessive vibration outside the final product, dropping, puncture, contact with metals, or tampering with the battery can cause it to fail.
- **Do not** short circuit the battery pack.
- **Do not** keep a battery pack in your pocket, purse, or anywhere together with other metal (conductive) objects.
- Never use force to install (insert) the battery pack
- If the battery pack is damaged in any way stop using immediately
- Immediately discontinue use of the battery if, while using, charging, or storing the battery, the battery emits an unusual smell, feels hot, changes color or shape, or appears abnormal in any other way.
- After the battery is fully discharged, don't leave it fully discharged, charge battery as soon as possible.
- Only charge battery packs in authorized GE HealthCare devices or chargers, never use a third-party charger.
- Only use GE HealthCare designated battery packs inside a GE HealthCare system and/or GE HealthCare Charger, never use an aftermarket or non-GE HealthCare authorized battery pack, if you are unsure, contact GE HealthCare for confirmation. **Failure to follow this may lead to smoke, fire or explosion due to incompatibility with non-GE HealthCare battery packs or chargers.**
- Store in a cool, dry place. Never store or charge batteries in extreme temperatures, such as a hot car; near heat sources or fire; in pressurized or very cold environments.
- The battery contains safety and protection devices, which, if damaged, may cause the battery pack to fail.
- Exposure to liquids can cause internal corrosion or damage to the cells or to the Battery Management System (BMS). The BMS protects the battery from overcharging, high self-discharge, or imbalanced charging of the cells, any of which can present the possibility of hazards during charging or use.
- Avoid unattended charging of battery packs, or where objects such as carpet, furniture, wood, vinyl floors, curtains or other flammable objects are nearby.
- If the battery pack is not meeting expected capacity (e.g., scanning time is significantly reduced compared to a new product) or the pack and/or software is indicating end of service life, discontinue using the battery pack immediately and replace with an authorized GE HealthCare battery pack for this product.
- The battery pack shall be replaced every two years for optimal safety and functionality of the system.
- Once end of service life notification is received, all rechargeable batteries shall be removed within 3 years and no new battery shall be installed in the device. Since there is no longer availability for GEHC authorized batteries, all batteries placed in the system are not in compliance with this system after this time frame.

10.5.2 Power state indications

When the system is not equipped with a battery, a status bar icon indicates that it runs on AC power only.



Figure 10-46 Status bar icon: no battery is present on the system

The **Power** button on the Operator Panel is lit in green.

When the system is equipped with a battery but is running on AC, the battery status bar icon shows the battery charge level range, as well as a lightning symbol as an indicator for AC connection.

Note that the AC indicator is displayed whether the battery is charging or not.



Figure 10-47 Status bar icons: the system has a battery with full / medium / empty charge level, but is running on AC

Hover the trackball pointer over the battery icon to display a tooltip with the exact charge level and the remaining time until full charge.

The **Power** button on the Operator Panel is lit in green.

When the system runs on battery, with AC disconnected, the battery status bar icon shows the battery charge level range without the indicator for AC connection.



Figure 10-48 Status bar icon: the system runs on battery with full / medium / empty charge level

The **Power** button on the Operator Panel is lit in blue.

Hover the trackball pointer over the battery icon to display a tooltip with the exact charge level and the remaining time where scanning is still possible.

The remaining run time is indicated next to the icon. It is an estimation of the remaining time where scanning is still possible. It is displayed in red font if the remaining time is 1 minute or less, orange font if up to 5 minutes, and white font if more than 5 minutes.

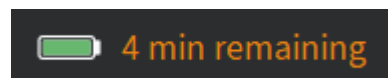


Figure 10-49 When running on battery, the remaining time is displayed next to the battery icon

10.5.3 Battery information

For details about the battery installed on the system, navigate to **Config > Utility > Service**.

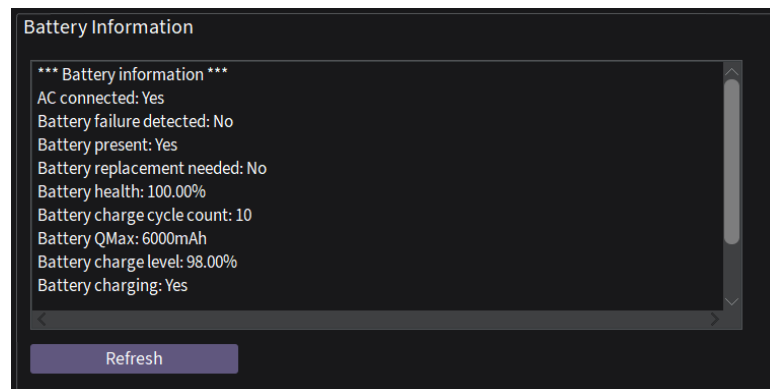


Figure 10-50 Battery information

10.5.4 Battery Scan Mode

On a system equipped with a fully charged battery, in case of voluntary or accidental AC power loss, the system will remain in the same state and continue working, until the battery level drops to *Empty* range.

NOTE

This feature requires 'Battery scan mode' option enabled.

The estimated remaining time to use the scanner on battery is displayed in the status bar.

Without 'Battery scan mode' option enabled, the system can only support maximum 5 minutes scanning. After 5 minutes, A dialog offers the user to switch to **Transportation** mode or to shut the system down, with no possibility to go back to scan mode until AC power is restored.

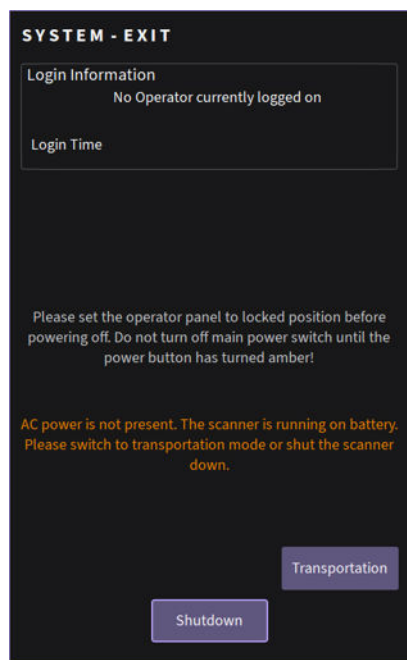


Figure 10-51 System Exit dialog

When the battery charge level reaches 0%, a graceful shutdown is initiated.



WARNING

The battery shall be removed from the system when in storage for more than one year.

10.5.5 Shipping and Long-term Storage

Prior to shipping, the absolute charge level of the scanner battery must be less than 30%.

Prior to long-term storage, the absolute charge level of the scanner battery must be between 25% and 30%.

To verify if the scanner is currently ready for shipping or long-term storage, navigate to **Config > System > Battery** and select the **Shipping / storage** charging mode (see the figure below).

When the **Shipping / storage** charging mode is selected, information boxes indicate if the system is ready for shipping or long-term storage.

The **Battery** window also displays the absolute battery charge level detailed per battery pack.

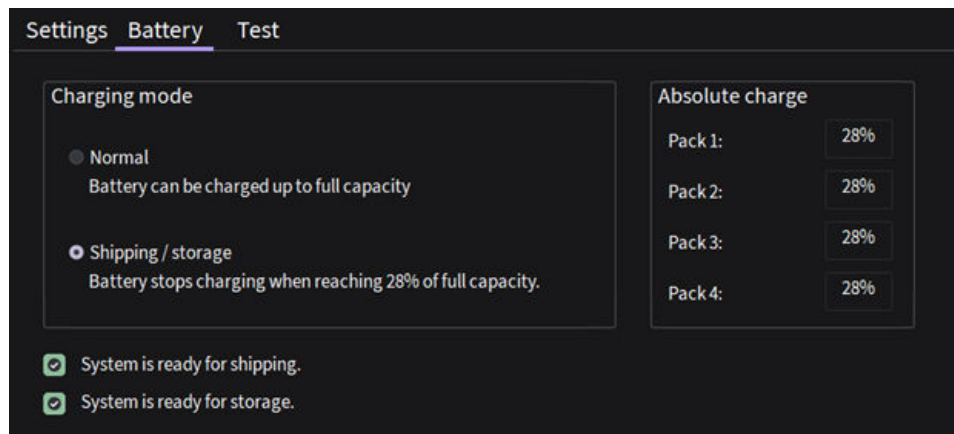


Figure 10-52 Config > System > Battery page with Shipping / storage charging mode

With the **Shipping / storage** charging mode selected, the battery will not charge above 28%. To prepare the system for shipping or long-term storage, perform the following sequence:

- Make sure the system is powered with AC
- Navigate to **Config > System > Battery**
- Select the **Shipping / storage** charging mode
- Disconnect the AC cable and let the battery discharge until shutdown
- Reconnect the AC cable and turn the system on
- Let the system charge above 25% if it is below that level.

Charging stops when the charge level reaches 28%, which is appropriate for shipping or long-term storage.

When the **Shipping / storage** charging mode is active, the battery status bar icon includes a warning indicator and its tooltip states that the **Shipping / storage** mode is on.

NOTE

Remove the battery package and transport or store it under the following conditions after proper packaging.

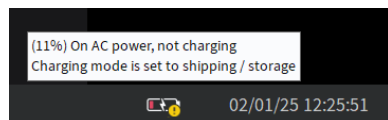


Figure 10-53 Battery icon and tooltip with Shipping / storage mode selected

On receipt of the shipped system or after long-term storage, unselect the **Shipping / storage** charging mode:

- Make sure the system is powered with AC
- Navigate to **Config > System > Battery**
- Select the **Normal** charging mode
- Let the battery charge fully.

10.5.6 Battery replacement needed

The battery service life depends on the usage pattern of the batteries. Frequent recharge will reduce battery service life. The battery is designed for at least 300 charge cycles.

The battery service life is checked at every system startup. If the battery is reaching its end of life and needs to be replaced, a warning dialog will be displayed at startup to request the user to organize the replacement.

When the system is on, the battery status bar icon will include a replacement warning indicator.



Figure 10-54 Status bar icon: the battery needs to be replaced



DANGER

The batteries must be replaced every two years, and also if fully charged batteries are not capable of maintaining system power for more than half of the expected time for a new battery. Please organize a replacement.

10.5.7 Battery failure is detected

If a failure is detected with the battery hardware, it is marked as out of order. Any AC power loss, voluntary or accidental, will cause the system to shutdown immediately.

If the system already runs on battery when the failure is detected, the system will shutdown immediately.

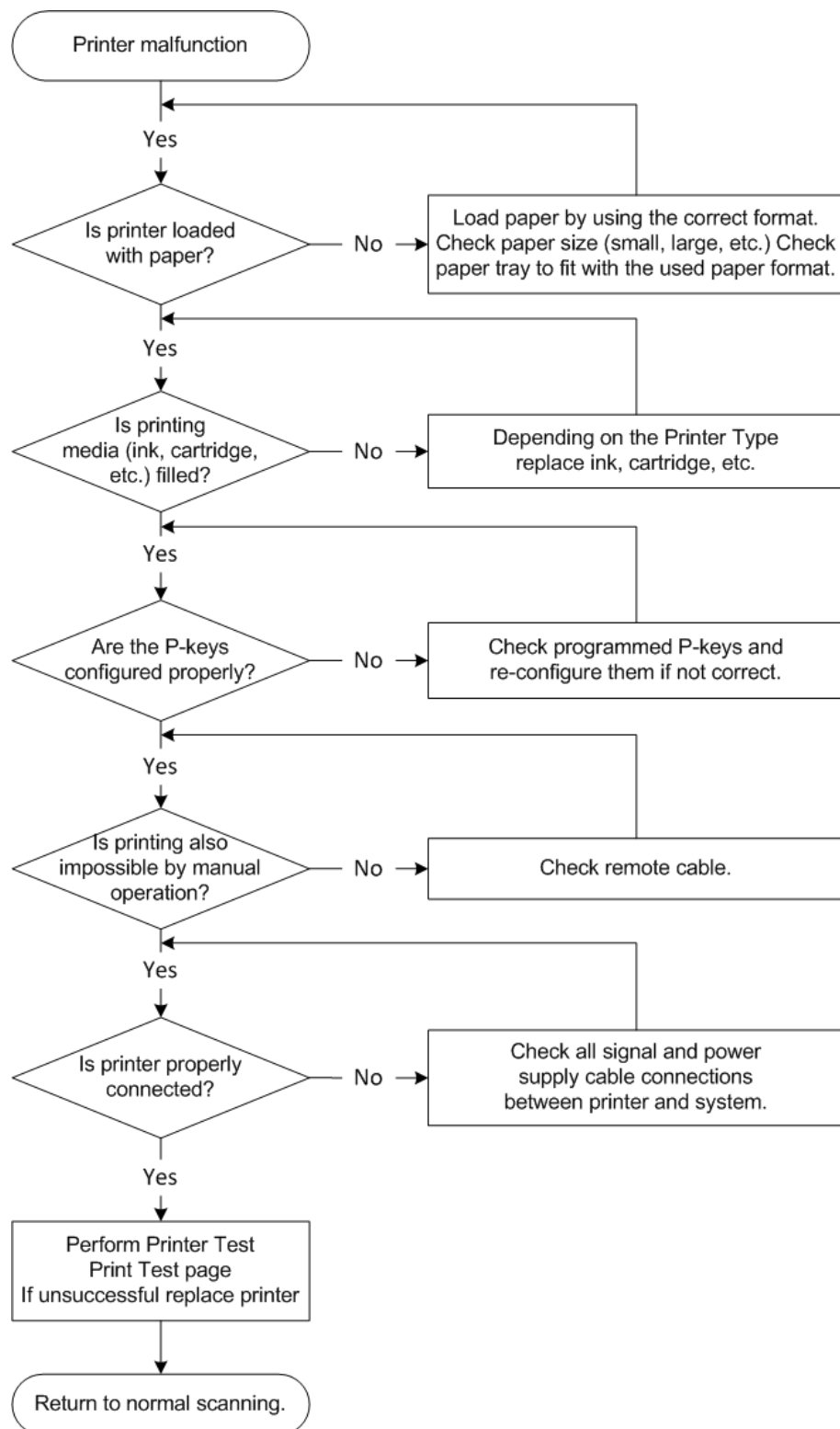


Figure 10-55 Status bar icon: the battery is out of order

10.6 Monitor Troubleshooting

Fault symptom	Check these items
No image	Check the power cord is properly connected.
	Check the video cable is properly connected.
	Check no pins of the video cable are bent.
	Check if video is present on backplane.
Picture is fuzzy	Adjust the picture contrast and picture brightness. Some SVGA cards having an excessive video output level will cause a fuzzy picture at the maximum contrast level.
Video test patterns are not clear, bright, parallel or square	Replace the monitor.

10.7 Printer Malfunction



Chapter 11

Parts

In this chapter

Parts List Groups.	318
Power Cords.	320
CRU parts.	323
Optional Peripherals and Accessories.	327
System Manuals.	332
Probes.	333

11.1 Parts List Groups

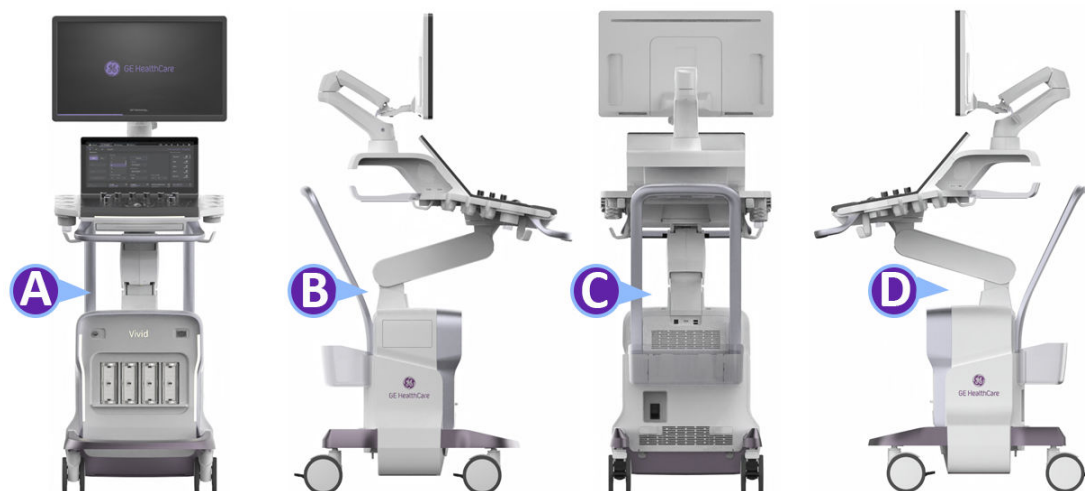


Figure 11-1 Vivid Explorer Console Views of LEFT Side (A) - BACK/REAR (B) - FRONT (C) - RIGHT Side (D)

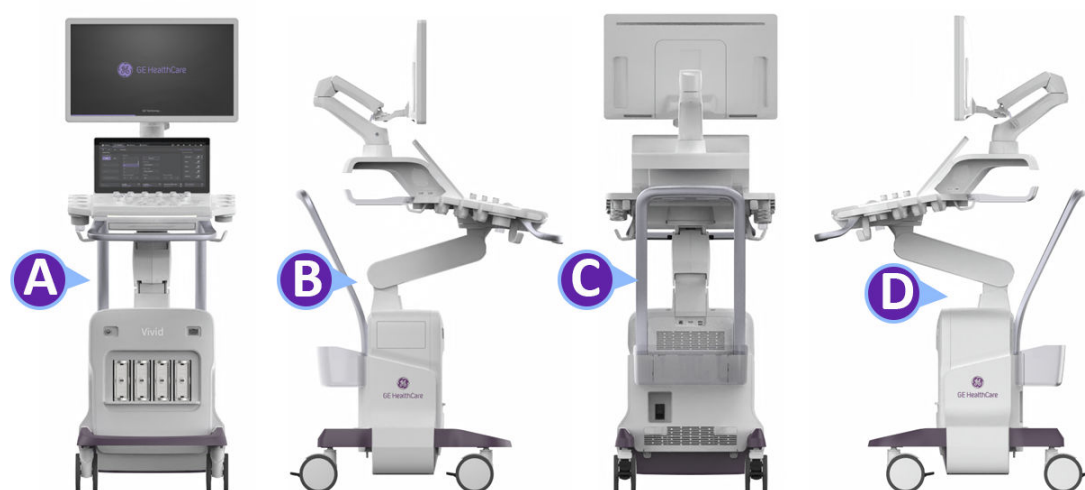


Figure 11-2 Vivid Advanced Console Views of FRONT Side (A) - LEFT (B) - BACK/REAR (C) - RIGHT Side (D)



Figure 11-3 Vivid Focus Console Views of FRONT Side (A) - LEFT (B) - BACK/REAR (C) - RIGHT Side (D)

Table 11-1 Mechanical and user accessible parts

Part Group Name	Description
CRU parts on page 323	All CRU parts
Power Cords on page 320	Ultrasound system power cords
System Manuals on page 332	Manuals
Probes on page 333	Probes

11.2 Power Cords

Table 11-2 Power Cords

Item	Part Name	HCAT#	Spare Part Number	Description	Qty	CRU
1	Power Cord - Europe 230V	H46342LZ	6736105-3	Power Cord Europe 230V/240V used at Mains Input Connector C14 (10A type) 	1	Y
2	Power Cord - USA C13 (Hops. grade)	H46712LW	6736114-3	Power Cord - USA C13 (Hops. grade) 	1	Y
3	Power Cord - Japan C13 (Hops. grade)	H44512LY	6736111-3	Power Cord - Japan C13 (Hops. grade) 	1	Y
4	Power Cord - Australia	H46712LZ	6736102-3	Power Cord - Australia 	1	Y
5	Power Cord - United Kingdom	H46712LM	6736107-3	Power Cord - United Kingdom 	1	Y

Item	Part Name	HCAT#	Spare Part Number	Description	Qty	CRU
6	Power Cord - South Africa	H46712LN	6736112-3	Power Cord - South Africa 	1	Y
7	Power Cord - Argentina	H46712LP	6736101-3	Power Cord - Argentina 	1	Y
8	Power Cord - Israel	H46712LR	6736109-3	Power Cord - Israel 	1	Y
9	Power Cord - Switzerland	H46712LS	6736113-3	Power Cord - Switzerland 	1	Y
10	Power Cord - Denmark, Red Shielded	H46712LT	6736115-3	Power Cord - Denmark, Red Shielded 	1	Y

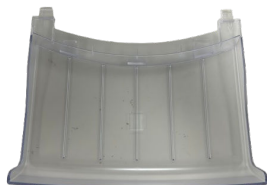
Item	Part Name	HCAT#	Spare Part Number	Description	Qty	CRU
11	Power Cord - Denmark, Gray	H46692LK	6736106-12	Power Cord Denmark used at Mains Input Connector C14 (10A type) 2 pcs 	1	Y
12	Power Cord - Italy	H46722LD	6736110-3	Power Cord - Italy 	1	Y
13	Power Cord - Brazil	H46752LW	6736103-3	Power Cord - Brazil 	1	Y

11.3 CRU parts

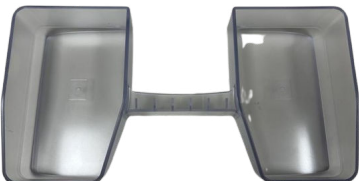
NOTE

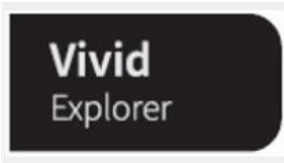
CRU (Customer Replacement Unit) parts are defined to cover components for both BIOMED and end users.

Table 11-3 CRU parts

Item	CRU Part No.	Description	Qty	VE	VA	VF
1	TC207522-S	OPIO Top Cover rear CRU 	1	Y	Y	Y
2	TC207304-S	OPIO Upper Deck Storage Tray CRU 	1	Y	Y	N
3	5933625-S	ANkey film 	1	Y	Y	Y
4	TC207301-S	Trolley Lower Front Tray CRU 	1	Y	Y	Y
5	TC207101-S	Trolley Left Cover CRU 	1	Y	Y	Y

Item	CRU Part No.	Description	Qty	VE	VA	VF
7	TC207100-S	Trolley Right Cover CRU 	1	Y	Y	Y
8	TC307076-S	Air filter CRU 	1	Y	Y	Y
9	TC307041-S	Gel Cup Small CRU 	1	Y	Y	Y
10	TC207517-S	Encoder Knob kits for VE CRU 	1	Y	N	N
11	TC207518-S	Encoder Knob kits for VA CRU 	1	N	Y	N
12	TC207519-S	Encoder Knob kits for VF CRU 	1	N	N	Y

Item	CRU Part No.	Description	Qty	VE	VA	VF
13	TC207302-S	Trolley Rear Basket CRU 	1	Y	Y	N
14	TC207117-S	Charge Box 	1	Y	Y	Y
15	TC207118-S	4S2P Battery Packs, 4pcs (14.4V Li-ion 5900 mAh Rechargeable) with bracket 	4	Y	Y	Y
16	TC207303-S	Trolley Mounted dual Storage BIN CRU 	1	Y	Y	Y
17	TC207541-S	Retractable cable hook CRU 	1	Y	Y	Y
18	TC314070-S	Monitor Sticker for Dimension 	1	Y	N	N

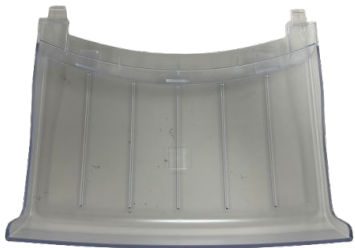
Item	CRU Part No.	Description	Qty	VE	VA	VF
19	TC307157-S	Nameplate kit for VA 	1	N	Y	N
20	TC307158-S	Nameplate kit for VF 	1	N	N	Y
21	TC307207-S	Name plate kit for VE 	1	Y	N	N
22	5974261-S	Ergonomic P2D Probe Holder CRU	1	Y	Y	Y

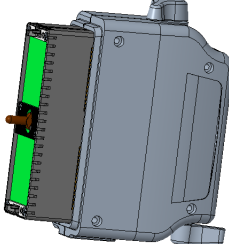
11.4 Optional Peripherals and Accessories

Table 11-4 Optional Peripherals and Accessories

Item	HCAT#	Spare Part Number	Description	Qty	CRU	FRU
1	H41402LS		UPP-110HD BW PAPER 1 ROLL	1	N	Y
2	H48392DT	5962058-S	Transend DVD Writer kit 	1	N	Y
3	H48702AR	5718360	Suns USB foot switch 	1	N	Y
4	H48962CG	5933000-S	NETGEAR A8000 USB Adapter 	1	Y	N
5	H48802DD	SGN200199	Network breakaway adapter 	1	Y	N
6	H48982AP	5810766	2 Ferrite cores and instruction	1	N	Y
7	H48782DH	TC207118-S	Spare Battery Pack 	4	Y	N

Item	HCAT#	Spare Part Number	Description	Qty	CRU	FRU
8	H48612DJ	TC207309-S	Gel Warmer kit for VA, VF 	1	N	Y
9	H48792DQ	TC207117-S/ TC207118-S	Charge Box with 4 batteries  	1	Y	N
10	H48782DK	TC207117-S	Charge Box 	1	Y	N

Item	HCAT#	Spare Part Number	Description	Qty	CRU	FRU
11	H48782DL	TC207301-S	Lower Front Tray 	1	Y	N
12	H48782DM	TC207302-S	Rear Storage Bin 	1	Y	N
13	H48782DN	TC207303-S	Trolley Mounted Dual Storage Bin 	1	Y	N
14	H48782DP	TC207304-S	Upper Deck Storage Tray 	1	Y	N
15	H48782DQ	TC207541-S	Retractable Cable Hook 	1	Y	N
16	H48782DJ	5990042-S	Printer Support Kit 	1	N	Y

Item	HCAT#	Spare Part Number	Description	Qty	CRU	FRU
17	H48792DA	TC308027-S	Rear Handle 	1	N	Y
18	H48792DB	5981788-S	Optional 2TB SSD 	1	N	Y
19	H48802DE	5981786-S	Optional 1TB SSD 	1	N	Y
20	H48772DB	TC309040-S	ICE RS-DLP Adaptor 	1	Y	N
21	H48232DT	5877427-S	Crystal ICECord-RS	1	Y	N
22	H48072BT	5834536	EP Lab Accessory Kit 	1	Y	N
23	H48762DW	TC207501-S	Vivid Advanced/Focus Alphanumeric Keyboard – English	1	N	Y
24	H48762DY	TC207508-S	Vivid Alphanumeric Keyboard – French	1	N	Y
25	H48762DZ	TC207510-S	Vivid Alphanumeric Keyboard – German	1	N	Y

Item	HCAT#	Spare Part Number	Description	Qty	CRU	FRU
26	H48772DA	TC207511-S	Vivid Alphanumeric Keyboard – Italian	1	N	Y
27	H48772DM	TC207529-S	Vivid Alphanumeric Keyboard – Russian	1	N	Y
28	H48772DN	TC207531-S	Vivid Alphanumeric Keyboard – Danish	1	N	Y
29	H48772DP	TC207532-S	Vivid Alphanumeric Keyboard – Norwegian	1	N	Y
30	H48772DQ	TC207533-S	Vivid Alphanumeric Keyboard – Swedish	1	N	Y
31	H48772DR	TC207534-S	Vivid Alphanumeric Keyboard – Finnish	1	N	Y
32	H48962DN	TC207543-S	Vivid Alphanumeric Keyboard – Spanish	1	N	Y
33	H48962DP	TC207544-S	Vivid Alphanumeric Keyboard – Portuguese	1	N	Y
34	H48772DC	TC5933625-S	Vivid Alphanumeric Keyboard Protective Film	1	Y	N
35	H45531HS	KZ200800	Bite Hole Indicator	1	N	Y
36	H45551NM	KX200272	TEE Storage rack	1	N	Y
37	H48842DD	6007672	Vivid V211 Physical Software USB	1	Y	N

11.4.1 Optional Peripherals and Accessories

ECG Cables

Table 11-5 ECG Cables

Item	Category	HCAT#	FRU Part Number	Description
1	AHA(USA)-Adult	H45601SB	2106305-001	ecg trunk cable, aha
2	AHA(USA)-Adult	H45601SC	2106390-001	ecg leadwire set, aha
3	IEC(Europe)-Adult	H45601SD	2106305-003	ecg trunk cable, iec
4	IEC(Europe)-Adult	H45601SE	2106390-003	ecg leadwire set, iec
5	IEC(Europe)-Neo	H45601SG	2106306-003	ecg trunk cable, neonatal din 3-lead, iec
6	AHA(USA)-Neo	H45601SF	2106306-001	ecg trunk cable, neonatal din 3-lead, aha
7		H45601SK	2106403-003	DIN Converter, 3-Lead
8		H45021LL	S2420410	ECG EXT.CABLE VS-SERVICE KIT

11.5 System Manuals

All documentation related information is included in the 'OEM and Other Manuals' chapter.

For a list of product manuals, see: [Product Manuals - Software v211](#) on page 385

11.6 Probes

NOTE

Please observe that some probes are not applicable on all systems (depending on system type). Check with your local distributor or GE HealthCare sales representative for availability.

Table 11-6 Probe list

Item	Service Part Number	HCAT#	Description	VE	VA	VF
1	5446030	H44901AE	M5Sc-D 	Y	Y	N
2	5692037	H40482LS	4Vc-D 	Y	N	N
3	5499322	H45021RT	12S-D 	Y	Y	N
4	5499316	H45021RP	6S-RS 	N	N	Y
5	5499321	H44901AB	12S-RS 	N	N	Y

Item	Service Part Number	HCAT#	Description	VE	VA	VF
6	5920393	H46462LS	6Sc-D 	Y	Y	N
7	5978046	H48062AU	4Sc-RS 	N	N	Y
8	5499510	H40442LM	9L-D 	Y	Y	N
9	5499499	H40432LN	11L-D 	Y	Y	N
10	5499511	H40442LL	9L-RS 	N	N	Y

Item	Service Part Number	HCAT#	Description	VE	VA	VF
11	5499501	H40402LY	12L-RS 	N	N	Y
12	5843289	H48062AJ	L4-20t-RS 	N	N	Y
13	5499606	H40472LT	C1-6-D 	Y	Y	N
14	5890488	H40462LN	C2-9-D 	Y	Y	N
15	5493012	H40482LB	C3-10-D 	Y	Y	N
16	5488477	H4000SR	4C-RS 	N	N	Y
17	5499508	H40402LS	8C-RS 	N	N	Y

Item	Service Part Number	HCAT#	Description	VE	VA	VF
18	5499608	H40462LA	C1-5-RS 	N	N	Y
19	5499592	H40442LK	IC5-9-D 	Y	Y	N
20	5722023	H48691PJ	IC9-RS 	N	N	Y
21	5867041	H45581CS	9VT-D 	Y	Y	N
22	5729428	H45581BJ	6Vt-D 	Y	Y	N

Item	Service Part Number	HCAT#	Description	VE	VA	VF
23	5729431	H45551ZE	6Tc-RS 	N	N	Y
24	5729436	H45551CA	P2D-RS 	N	N	Y
25	5729434	H4830JE	P2D-Lemo 	Y	Y	N

Item	Service Part Number	HCAT#	Description	VE	VA	VF
26	5729437	H4830JG	P6D-Lemo	Y	Y	N
27	5499594	H40452LL	L8-18i-D 	Y	Y	N
28	5499609	H40462LF	L8-18i-RS 	N	N	Y
29	NA	NA	AcuNav Crystal 	Y	Y	N
30	NA	NA	SoundStar Crystal 	Y	Y	N

Item	Service Part Number	HCAT#	Description	VE	VA	VF
31	NA	NA	NUVISION 	Y	N	N
32	NA	NA	NUVISION NAV 	Y	N	N

Chapter 12

Parts Replacement

In this chapter

Service Safety Considerations.	342
Returning/Shipping System, Probes and Repair Parts.	343
Operator Panel Top Cover Rear Replacement.	344
Trolley Left Cover Replacement.	345
Trolley Right Cover Replacement	346
Air Filter Replacement.	347
Battery Replacement	348
Charge Box Module Replacement	352
Trolley Mounted Dual Storage Bin Replacement.	355
Trolley Rear Basket Replacement.	356
OPIO Upper Deck Storage Tray.	357
Gel Cup Replacement.	358
Trolley Lower Front Tray Replacement.	359
AN Keyboard Protective Film Replacement.	361
Encoder collection Replacement.	363
Monitor Sticker Replacement.	367
Nameplate kit Replacement.	368

12.1 Service Safety Considerations



ELECTRICAL HAZARD

Dangerous voltages, capable of causing death, are present in this equipment.

Use extreme caution when handling, testing and adjusting.



WARNING

Before performing any service or removing covers, turn off the system, disconnect the AC power and wait until the power button light turns off (approximately 40 seconds if the system has batteries). Also ensure not to press the power button during the service. Remove the batteries before performing service.



WARNING

If the covers are removed from an operating Vivid Explorer/Vivid Advanced/Vivid Focus, some metal surfaces may be warm enough to pose a potential heat hazard if touched, even while in shutdown mode.



WARNING

Use all Personal Protection Equipment (PPE) such as gloves, safety shoes, safety glasses, and kneeling pads, to reduce the risk of injury.



CAUTION

Do not disassemble or alter the Batteries.

Only use Batteries approved by GE HealthCare as suitable for use with the ultrasound system.

Replace all batteries at the same time to avoid mix-up with used and new battery packs.

Remove the batteries before performing service.

12.2 Returning/Shipping System, Probes and Repair Parts

When returning or shipping the Vivid Explorer/Vivid Advanced/Vivid Focus system in the original packaging:

- the control console must be centered and locked in “unextended” position
- system must be lowered to its minimum height with the monitor flipped down

Equipment being returned must be clean and free of blood and other infectious substances.

GE HealthCare policy states that body fluids must be properly removed from any part or equipment prior to shipment. GE HealthCare employees, as well as customers, are responsible for ensuring that parts/equipment have been properly decontaminated prior to shipment. Under no circumstance should a part or equipment with visible body fluids be removed or shipped from a clinic or site (for example, body coils or ultrasound probes).

The purpose of the regulation is to protect employees in the transportation industry, as well as the people who will receive or open this package.

NOTE

The US Department of Transportation (DOT) has defined items that have been “saturated and/or dripping with human blood that are now caked with dried blood; or which were used or intended for use in patient care” as Regulated Medical Waste for transportation purposes and must therefore be transported as hazardous materials.

NOTE

The user/service staff should dispose of all waste properly in accordance with federal, state, and local waste disposal regulations.

The Vivid Explorer/Vivid Advanced/Vivid Focus system is not meant to be used for long-term storage of patient data or images. The user is responsible for the data on the system and a regular backup is highly recommended.

If the system is sent for repair, please ensure that any patient information is backed up and erased from the system before shipping. It is always possible during system failure and repair to lose patient data. GE HealthCare is not responsible for the loss of this data.

If PHI (Patient Healthcare Information) data needs to be sent to GE HealthCare employees for service purposes, GE HealthCare will ascertain agreement from the customer. Patient information shall only be transferred by approved service processes, tools and devices restricting access, protecting or encrypting data where required, and providing traceability in the form of paper or electronic documents at each stage of the procedure while maintaining compliance with cross-border restrictions of patient information transfers.

12.2.1 Repacking the Ultrasound System

NOTE

Shipping the Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound system in its original packaging is only possible when the control console is centered and locked in position, the system is lowered to its minimum height and the monitor is flipped down and monitor arm locked.

Follow the unpacking procedure in opposite order to pack the ultrasound system.

12.3 Operator Panel Top Cover Rear Replacement

CRU Number

TC207522-S

Manpower

One person, 5 minutes

Tools

None

Preparations

Power down the ultrasound system as described in [Complete Power Down](#) on page 92

12.3.1 Operator Panel Top Cover Rear - Removal Procedure

1. Remove the Operator Panel Top Cover Rear by hands.



Figure 12-1 Remove Operator Panel Top Cover Rear

12.3.2 Operator Panel Top Cover Rear - Installation Procedure

1. Install the Operator Panel Top Cover Rear in the reverse order of the removal.

12.4 Trolley Left Cover Replacement

CRU Number

TC207101-S

Manpower

One person, 5 minutes

Tools

None

Preparations

Power down the ultrasound system as described in [Complete Power Down](#) on page 92

12.4.1 Trolley Left Cover - Removal Procedure

1. Remove the Trolley Left Cover by hands.



Figure 12-2 Remove Trolley Left Cover

12.4.2 Trolley Left Cover - Installation Procedure

1. Install the Trolley Left Cover in the reverse order of the removal.

12.5 Trolley Right Cover Replacement

CRU Number

TC207100-S

Manpower

One person, 5 minutes

Tools

None

Preparations

Power down the ultrasound system as described in [Complete Power Down](#) on page 92

12.5.1 Trolley Right Cover - Removal Procedure

1. Remove the Trolley Right Cover by hands.



Figure 12-3 Remove Trolley Right Cover

12.5.2 Trolley Right Cover - Installation Procedure

1. Install the Trolley Right Cover in the reverse order of the removal.

12.6 Air Filter Replacement

CRU Number

TC307076-S

Manpower

One person, 5 minutes

Tools

None

Preparations

Power down the ultrasound system as described in [Complete Power Down](#) on page 92

12.6.1 Air Filter - Removal Procedure

1. Use both hands to securely push and pull downside the two outward-extending places at the bottom of the body to pull out the air filter.

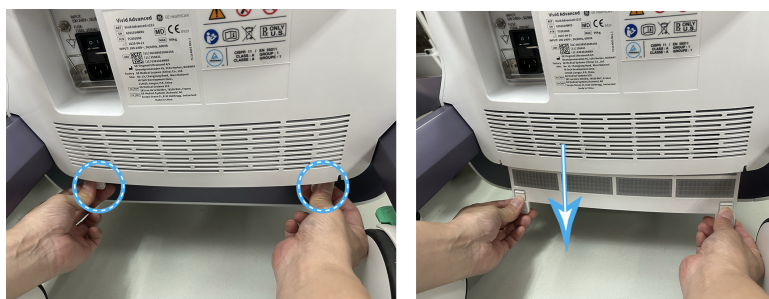


Figure 12-4 Remove Air Filter

12.6.2 Air Filter - Installation Procedure

1. Install the Air Filter in the reverse order of the removal.

12.7 Battery Replacement

FRU Number

TC207118-S

Manpower

One person, 5 minutes

Tools

None

Preparations

1. Power down the ultrasound system as described in [Complete Power Down](#) on page 92.
2. Disconnect the power supply.

12.7.1 Battery - Removal Procedure

1. Remove Trolley Left Cover by hands, refer to [Trolley Left Cover Replacement](#) on page 345.
2. Use one hand to press down the lock lever of the rotary push-to-close latch while at the same time use the other hand to pull out the charge box.



Figure 12-5 Remove the Charge Box

3. Unscrew the screw to open the battery box cover



Figure 12-6 Open Battery Box Cover

4. Remove the Battery.



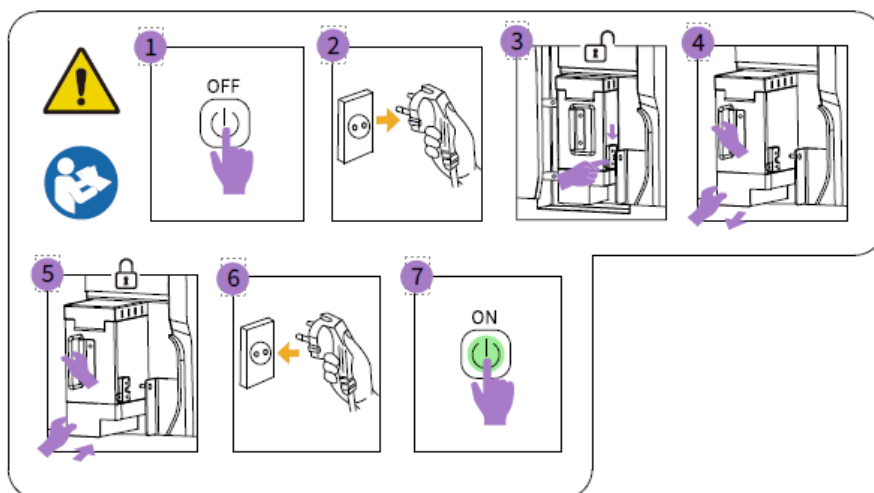
Figure 12-7 Remove Battery

12.7.2 Battery - Installation Procedure

1. Install the Battery in the reverse order of the removal.

NOTE

There is a label on the charge box indicating the installation sequence. Please strictly follow the order shown on the label to install the charge box to ensure it functions properly.



1. (1) Press on/off button on the ultrasound system to turn off the system.
2. (2) Disconnect the power source to the ultrasound system.
3. (3)~(4) Use one hand to press down the lock lever of the rotary push-to-close latch while at the same time use the other hand to pull out the charge box.
4. (5) Carefully push battery box into main frame with both hands. When the Charge Box is fully installed on the system, the latch will automatically lock.
5. (6) Connect the power source to the ultrasound system.
6. (7) Press on/off button on the ultrasound system to turn on the system.

Figure 12-8 Charge Box Installation Label

NOTE

If the system fails to start normally due to battery replacement not being performed in the correct order specified on the label, please disconnect the 220V power supply for no less than 60 seconds and then reconnect it. If the system still cannot start properly, please contact service personnel.

2. Perform Functional Checks.

12.7.3 Functional Checks for Battery Replacement

After the Battery is installed, please follow the steps below to check whether the Battery is functioning properly.

1. Check if the Charge Box is fixed securely.
2. Check the battery icon on the status bar of the screen. Ensure that the battery level is displayed correctly.
 - Refer to [Figure 12-9 Battery icon location on the system](#) on page 351 for the battery icon location on the system.



Figure 12-9 Battery icon location on the system

3. Confirm if the value for battery health status is displayed on Utility page.
 - Go to **Utility > Config > About > Battery** for checking the batttery health status.

Imaging

Meas/Text

Report

Connectivity

Security

System

About

System Version

SW Version

Firmware Version

HW Version

Probes

Battery

Pack no.	Manufacturer	Model	Manufacture date	Serial no.	Battery health	Cycle count	Absolute charge
1	Texas Instruments	5808900-4	28/04/2025	434	100%	8	100%
2	Texas Instruments	5808900-4	28/04/2025	422	100%	8	100%
3	Texas Instruments	5808900-4	22/03/2025	1265	100%	8	80%
4	Texas Instruments	5808900-4	28/04/2025	536	100%	8	100%

i

Battery charging options can be set on the System tab, under Battery.

Figure 12-10 Check Battery Health Status

4. For those who have purchased the options for 'Transportation Mode' and 'Scanner on Battery', verify that these functions operate correctly.

12.8 Charge Box Module Replacement

FRU Number

TC207117-S

Manpower

One person, 5 minutes

Tools

None

Preparations

1. Power down the ultrasound system as described in [Complete Power Down](#) on page 92.
2. Disconnect the power supply.

12.8.1 Charge Box Module - Removal Procedure

1. Remove Trolley Left Cover by hands, refer to [Trolley Left Cover Replacement](#) on page 345.
2. Use one hand to press down the lock lever of the rotary push-to-close latch while at the same time use the other hand to pull out the charge box.



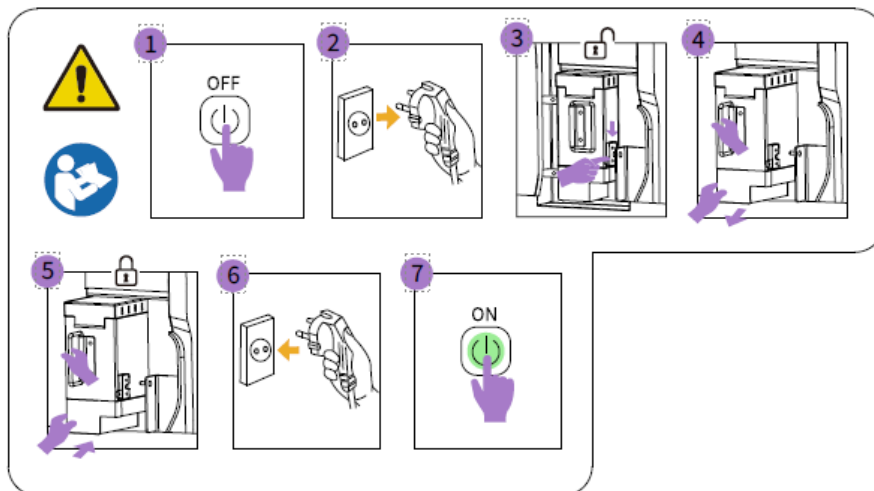
Figure 12-11 Remove the Charge Box

12.8.2 Charge Box Module - Installation Procedure

1. Install the Charge Box Module in the reverse order of the removal.

NOTE

There is a label on the charge box indicating the installation sequence. Please strictly follow the order shown on the label to install the charge box to ensure it functions properly.



1. (1) Press on/off button on the ultrasound system to turn off the system.
2. (2) Disconnect the power source to the ultrasound system.
3. (3)~(4) Use one hand to press down the lock lever of the rotary push-to-close latch while at the same time use the other hand to pull out the charge box.
4. (5) Carefully push battery box into main frame with both hands. When the Charge Box is fully installed on the system, the latch will automatically lock.
5. (6) Connect the power source to the ultrasound system.
6. (7) Press on/off button on the ultrasound system to turn on the system.

Figure 12-12 Charge Box Installation Label

NOTE

If the system fails to start normally due to battery replacement not being performed in the correct order specified on the label, please disconnect the 220V power supply for no less than 60 seconds and then reconnect it. If the system still cannot start properly, please contact service personnel.

2. Perform Functional Checks.

12.8.3 Functional Checks for Charge Box Module Replacement

After the Charge Box is installed, please follow the steps below to check whether the Charge Box is functioning properly.

1. Check if the Charge Box is fixed securely.
2. Check the battery icon on the status bar of the screen. Ensure that the battery level is displayed correctly.
 - Refer to [Figure 12-9 Battery icon location on the system](#) on page 351 for the battery icon location on the system.



Figure 12-13 Battery icon location on the system

3. Confirm if the value for battery health status is displayed on Utility page.
 - Go to **Utility > Config > About > Battery** for checking the batttery health status.

Imaging

Meas/Text

Report

Connectivity

Security

System

About

System Version

SW Version

Firmware Version

HW Version

Probes

Battery

Pack no.	Manufacturer	Model	Manufacture date	Serial no.	Battery health	Cycle count	Absolute charge
1	Texas Instruments	5808900-4	28/04/2025	434	100%	8	100%
2	Texas Instruments	5808900-4	28/04/2025	422	100%	8	100%
3	Texas Instruments	5808900-4	22/03/2025	1265	100%	8	80%
4	Texas Instruments	5808900-4	28/04/2025	536	100%	8	100%

i

Battery charging options can be set on the System tab, under Battery.

Figure 12-14 Check Battery Health Status

4. For those who have purchased the options for 'Transportation Mode' and 'Scanner on Battery', verify that these functions operate correctly.

12.9 Trolley Mounted Dual Storage Bin Replacement

CRU Number

TC207303-S

Manpower

One person, 5 minutes

Tools

None

Preparations

Power down the ultrasound system as described in [Complete Power Down](#) on page 92

12.9.1 Trolley Mounted Dual Storage Bin - Removal Procedure

1. Release the two clips of the Trolley Mounted dual Storage Bin from the system, then remove the Trolley Mounted Dual Storage Bin by hands.

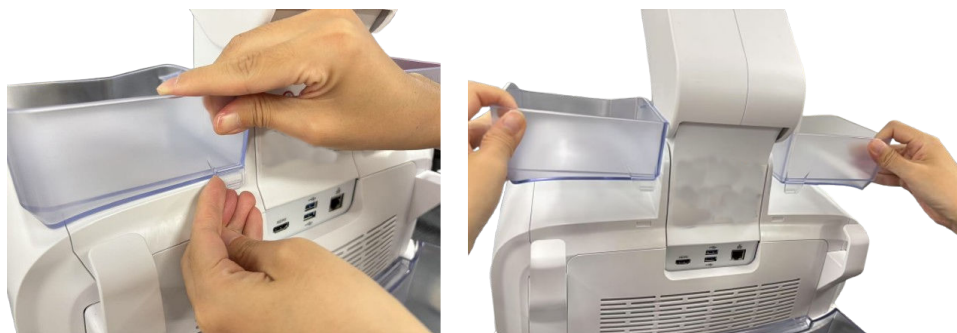


Figure 12-15 Remove Trolley Mounted Dual Storage Bin

12.9.2 Trolley Mounted Dual Storage Bin - Installation Procedure

1. Install the Trolley Mounted Dual Storage Bin in the reverse order of the removal.

12.10 Trolley Rear Basket Replacement

CRU Number

TC207302-S

Manpower

One person, 5 minutes

Tools

None

Preparations

None

12.10.1 Trolley Rear Basket - Removal Procedure

1. Lift the Trolley Rear Basket out of the Rear Cover slot by hands.



Figure 12-16 Remove Trolley Rear Basket

12.10.2 Trolley Rear Basket - Installation Procedure

1. Install the Trolley Rear Basket in the reverse order of the removal.

12.11 OPIO Upper Deck Storage Tray

CRU Number

TC207304-S

Manpower

One person, 5 minutes

Tools

None

Preparations

Power down the ultrasound system as described in [Complete Power Down](#) on page 92

12.11.1 OPIO Upper Deck Storage Tray - Installation Procedure

1. Slide off the small cover behind the Operator Panel.
2. Install the OPIO Upper Deck Storage Tray in place of the removed small cover.



Figure 12-17 Install OPIO Upper Deck Storage Tray

12.11.2 Operator Rear Tray - Removal Procedure

1. Remove the Operator Rear Tray in the reverse order of the installation.

12.12 Gel Cup Replacement

CRU Number

TC307041-S

Manpower

One person, 5 minutes

Tools

None

Preparations

None

12.12.1 Gel Cup - Removal Procedure

1. Gently lift the gel cup and take it out of the gel holder.



Figure 12-18 Remove Gel Cup

12.12.2 Gel Cup - Installation Procedure

1. Install the Gel Cup in the reverse order of the removal.

12.13 Trolley Lower Front Tray Replacement

CRU Number

TC207301-S

Manpower

One person, 5 minutes

Tools

None

Preparations

None

12.13.1 Trolley Lower Front Tray - Removal Procedure

1. Take out the Trolley Lower Front Tray from Trolley Base by hands.



Figure 12-19 Remove Trolley Lower Front Tray

12.13.2 Trolley Lower Front Tray - Installation Procedure

1. Install the Trolley Lower Front Tray in the reverse order of the removal.

NOTE

To ensure the installation easier, align the two pads on the Trolley Lower Front Tray with the front housing to install it.



Figure 12-20 Install Trolley Lower Front Tray

12.14 AN Keyboard Protective Film Replacement

CRU Number

5933625-S

Manpower

One person, 5 minutes

Tools

None

Preparations

Power down the ultrasound system as described in [Complete Power Down](#) on page 92

12.14.1 AN Keyboard Protective Film - Installation Procedure

1. Peel off the orange backing adhesive from the AN Keyboard Film.

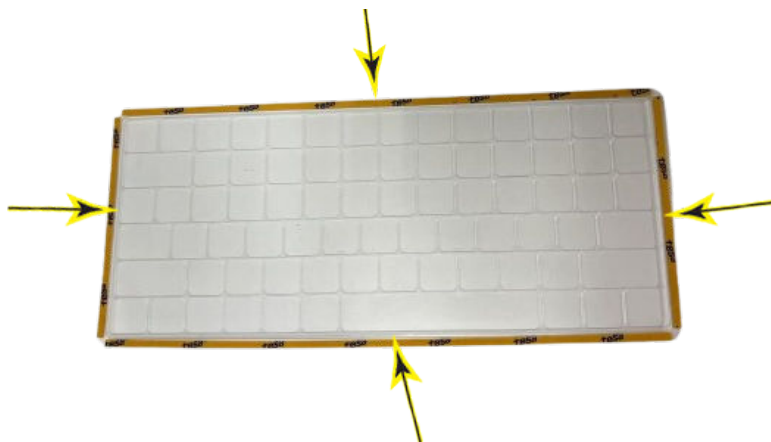


Figure 12-21 Peel off Adhesive

2. Align the Film and apply it over the AN Keyboard and secure it.

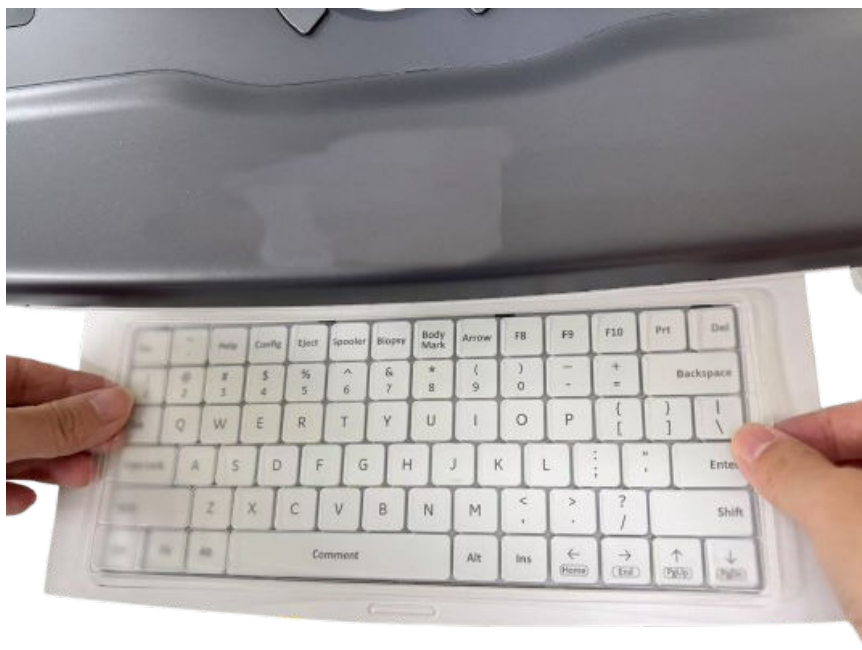


Figure 12-22 Apply the Film

NOTE

After applying the film, please press keys around the edges to make it stick more securely.

12.14.2 AN Keyboard Protective Film - Removal Procedure

1. Remove the AN Keyboard Protective Film in the reverse order of installation.

12.15 Encoder collection Replacement

CRU Number

TC207517-S, TC207518-S and TC207519-S

Manpower

One person, 10 minutes

Tools

None

Preparations

Power down the ultrasound system as described in [Complete Power Down](#) on page 92

12.15.1 Encoder collection - Removal Procedure

1. Remove the encoder key caps directly by hand.



Figure 12-23 Remove the Encoder Key Cap

12.15.2 Encoder collection - Installation Procedure

Encoder collection includes two types of encoders: (1) Encoder key cap without silk print; (2) Encoder key cap with silk print. These two types of encoder key caps use two different installation procedures.



Figure 12-24 Encoder Collection

For encoder key cap without silk print

1. Align the keycap with the encoder notch to install it.



Figure 12-25 Install the Encoder Collection

NOTE

To facilitate installation, please ensure that the notch on the keycap is aligned with the notch on the encoder shaft.

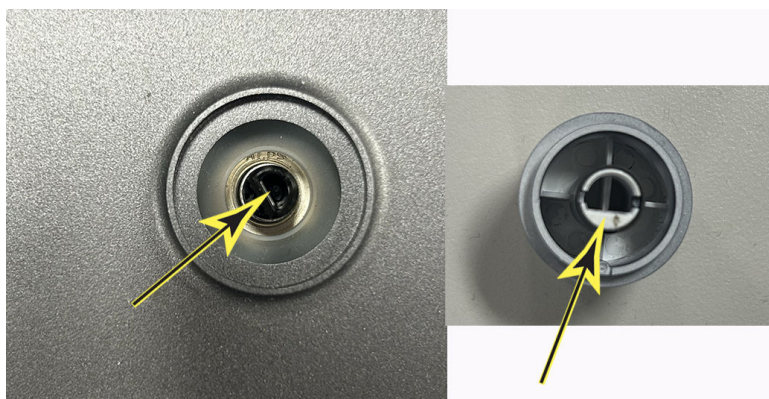


Figure 12-26 Align the notch of keycap with notch of encoder shaft

For encoder key cap with silk print

The encoder key cap with silk print has two parts: (1) the big rotary; (2) the big cap. When installing encoder key cap with silk print, you should respectively install these two parts.

1. Align the big rotary with the encoder notch to install it.

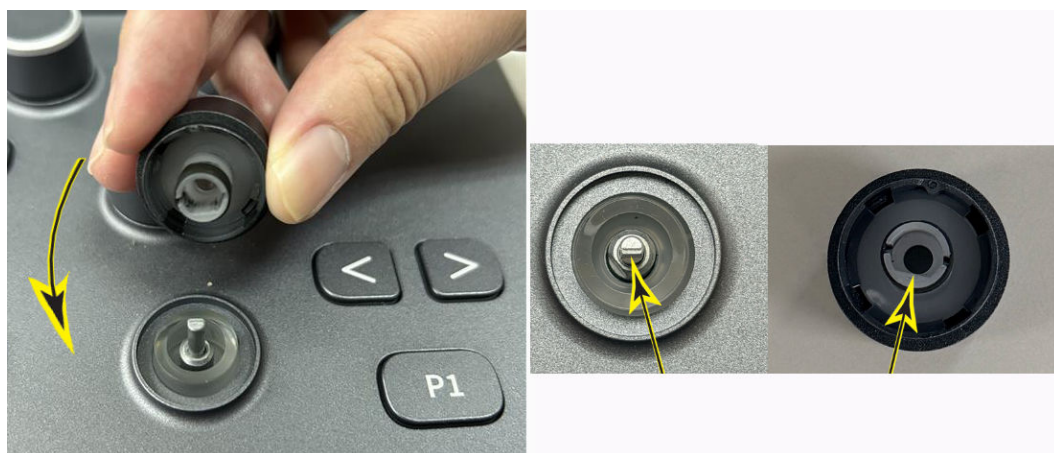


Figure 12-27 Install big rotary with the encoder notch

2. Align the big cap with the encoder notch to install it.



Figure 12-28 Install big cap with the encoder notch

12.15.3 Functional Checks after Encoder Collection Replacement

TBD

12.16 Monitor Sticker Replacement

CRU Number

TC314070-S

Manpower

One person, 5 minutes

Tools

None

Preparations

None.

12.16.1 Monitor Sticker - Installation Procedure

1. Peel off the adhesive backing of the Monitor Sticker and attach it to the bottom-right corner of the Monitor frame.



Figure 12-29 Attach the Monitor Sticker

12.17 Nameplate kit Replacement

CRU Number

TC307157-S, TC307158-S, TC307207-S, TC307208-S and TC307209-S

Manpower

One person, 5 minutes

Tools

None

Preparations

1. Power down the ultrasound system as described in [Complete Power Down](#) on page 92
2. Check the model name information on the rating plate and ensure the nameplate to be replaced is correct.

12.17.1 Nameplate Kit - Installation Procedure

1. Peel off the adhesive backing of the nameplate kit and attach it to the reserved area on the left side of the Operator Panel.



Figure 12-30 Location of Nameplate Kit

Chapter 13

Care and Maintenance

In this chapter

Introduction to Care and Maintenance. 370

Warnings. 371

Why do Maintenance. 372

Maintenance Task Schedule. 373

Tools Required. 375

System Maintenance. 376

13.1 Introduction to Care and Maintenance

This chapter describes Care and Maintenance on the ultrasound system and peripherals. These procedures are intended to maintain the quality of the Ultrasound system's performance. Read this chapter completely and familiarize yourself with the procedures before performing a task.

13.2 Warnings



CAUTION - AVOID STATIC ELECTRICITY

Practice good ESD prevention. Wear an anti-static strap when handling electronic parts and even when disconnecting/ connecting cables.

Notice

Do not pull out or insert circuit boards while power is on.

Notice

Do not operate this ultrasound system unless all board covers and frame panels are securely in place. System performance and cooling require this.

Notice

To ensure the mutual protection and safety of GE HealthCare service personnel and our customers, all equipment and work areas must be clean and free of any hazardous contaminants before a Service Engineer starts a repair. This includes, but is not limited to, decontamination and/or sterilization, depending on the application or use of the medical device.

13.3 Why do Maintenance

13.3.1 Preventive Maintenance Inspections

It has been determined by engineering that your Vivid Explorer/Vivid Advanced/Vivid Focus system does not have any high wear components that fail with use, therefore no Preventive Maintenance Inspections are mandatory. However, some Customers Quality Assurance Programs may require additional tasks and/or inspections at a different frequency than listed in this manual.

13.3.2 Quality Assurance

In order to gain accreditation from organizations such as the American College of Radiology (USA), it is the customer's responsibility to have a quality assurance program in place for each system. The program must be directed by a medical physicist, the supervising radiologist/physician or an appropriate designee.

Routine quality control testing must occur regularly. The same tests are performed during each period so that changes can be monitored over time and effective corrective action can be taken.

Testing results, corrective action and the effects of corrective action must be documented and maintained on the site.

Your GE service representative can help you with establishing, performing and maintaining records for a quality assurance program. Please contact us for coverage information and/or price for service.

13.4 Maintenance Task Schedule

13.4.1 How often should Care and Maintenance Tasks be performed?

The Customer Care Schedule (see: [Table 13-1 Customer care schedule](#) on page 373) specifies how often your Vivid Explorer/Vivid Advanced/Vivid Focus should be serviced and outlines items requiring special attention.

NOTE

It is the customer's responsibility to ensure that the Vivid Explorer/Vivid Advanced/Vivid Focus care and maintenance is performed as scheduled in order to retain its high level of safety, dependability and performance.

Your GE Service Representative has an in-depth knowledge of your Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound system and can best provide competent, efficient service. Contact GE HealthCare for coverage information and/or price for service.

The service procedures and recommended intervals shown in the Customer Care Schedule assumes that you use your Vivid Explorer/Vivid Advanced/Vivid Focus for an average patient load (10 per day) and not use it as a primary mobile system which is transported between diagnostic facilities.

NOTE

If conditions exist which exceed typical usage and patient load, then it is recommended to increase the care and maintenance frequencies.

Abbreviations used in the [Table 13-1 Customer care schedule](#) on page 373:

- D = Daily
- W = Weekly
- M = Monthly
- A = Annually

Table 13-1 Customer care schedule

Item	Service at indicated Time	D	W	M	A	Notes
Air Filter Grid	Remove the filter grid and clean the air filter grid with vacuum cleaner from outside.			•		more frequently depending on your environment
Air Filter Grid	Remove filter grid, back top cover and back cover and clean the housing from inside. (vacuum cleaner and soft brush)				•	more frequently depending on your environment
AC Mains Cable	Inspect AC mains cable			•		mobile systems check weekly
Cables and Connectors	Check if all cables are fixed and well seated at the correct position and if there is no mechanical damage visible.				•	also after corrective maintenance
User Interface	Clean alphanumerical keyboard, functional keys, digital potentiometers. (vacuum cleaner, lukewarm soap water on a soft, damp cloth)		•			Be careful not to get the cloth too wet so that moisture does not enter the loudspeakers or other keys!
Monitor, Touch Panel and Probe holder	Clean monitor surface, Touch Panel and probe holder with a fluid detergent in warm water on a soft, damp cloth.		•			Be careful not to get the cloth too wet so that moisture does not enter the entire system.
Mechanical parts	Clean and inspect the mechanical function of wheels, casters, brakes and swivel locks as well as side door, foot rest, front and rear handle, and monitor holder. Remove dust and coupling gel.			•		mobile systems check daily
Control Console movement	Check translation/rotation and height adjustment (elevation)				•	more frequently at mobile systems

Item	Service at indicated Time	D	W	M	A	Notes
Trackball Check	Check proper operation (cursor movement X, Y direction)	•				If failure occurs go to trackball cleaning.
Trackball Cleaning	Remove trackball ring and take out the trackball. Clean the trackball with soft tissue. (See Chapter 18 in the User Manual)				•	
Disk Drives (Data Backup)	Test image filing (Archive) import and export data capability (DVD/CD Drive)		•	• *		* save the image filing data weekly or at least monthly on DVD/CD depending on the number of examinations
Safe Probe Operation	Clean probes and probe cables and check acoustic lens housing (cracks) and probe cables. In case of mechanical damage, don't use them! Danger: Safety risk for operator and patient.	• *				* or before each use
Probe Connectors	Remove dust/dirt of all probe connectors. Clean with vacuum cleaner if dust is visible.			•		
Ultrasound system Leakage Current Checks					•	also after corrective maintenance or as required by your facilities QA program
Peripheral Leakage Current Checks					•	also after corrective maintenance or as required by your facilities QA program
Surface Probe Leakage Current Checks					•	also after corrective maintenance or as required by your facilities QA program
Endocavity Probe Leakage Current Checks					•	also after corrective maintenance or as required by your facilities QA program
Measurement Accuracy Checks					•	also after corrective maintenance or as required by your facilities QA program
Probe/Phantom Checks	Check gain changes, vary the focus and check reaction on screen. <i>NOTE! The use of a Phantom is not required during Care and Maintenance. Customer may use it as part of their Quality Assurance Program tests.</i>				•	also after corrective maintenance or as required by your facilities QA program
Functional Checks of all probes Functional Checks on page 216					•	also after corrective maintenance or as required by your facilities QA program

13.5 Tools Required

NOTE

For a list of required tools for servicing the ultrasound system, refer to Chapter 3 - Tools.

13.6 System Maintenance

13.6.1 Preliminary Checks

The preliminary checks take about 15 minutes to perform. Refer to the Vivid Explorer/Vivid Advanced/Vivid Focus user documentation whenever necessary.

Table 13-2 system checks

Step	Item	Description
1	Ask and Listen	Ask the customer if they have any problems or questions about the equipment.
2	Power up	- Turn the system power on and verify that all fans and peripherals turn on. - Watch the displays during power up to verify that no warning or error messages are displayed.
3	Probes	Verify that the ultrasound system properly recognizes all probes.
4	Displays	Verify proper display on the Monitor and Touch Panel (when present).
5	Presets	Backup all customer presets onto appropriate media).
6	Image Archive	Backup the Image Archive onto appropriate media

13.6.2 Functional Checks

The functional checks take about 60 minutes to perform. Refer to the Vivid Explorer/Vivid Advanced/Vivid Focus user documentation whenever necessary.

13.6.2.1 Ultrasound System Checks

Table 13-3 ultrasound system functional checks

Step	Item	Description
1	B Mode	Verify basic B Mode (2D) operation. Check the basic ultrasound system controls that affect this mode of operation.
2	M Mode	Verify basic M Mode operation. Check the basic ultrasound system controls that affect this mode of operation.
3	C Mode	Verify basic CFM Mode (Color Flow Mode) operation. Check the basic ultrasound system controls that affect this mode of operation.
4	PD Mode	Verify basic PD Mode (Power Doppler Mode) operation. Check the basic ultrasound system controls that affect this mode of operation.
5	Doppler Modes	Verify basic Doppler Mode operation (PW and CW if available). Check the basic ultrasound system controls that affect this mode of operation.
6	3D Mode	Verify basic 3D Mode operation. Check the basic ultrasound system controls that affect this mode of operation.
7	RealTime 4D Mode (optional)	Verify basic RealTime 4D Mode operation. Check the basic ultrasound system controls that affect this mode of operation.
8	*Applicable Software Options	Verify the basic operation of all optional modes. Check the basic system controls that affect each option's operation.

Step	Item	Description
9	Console Test	Perform the Test Procedure to verify that all controls are OK.
10	Monitor	Verify basic Monitor display functions.
11	Touch Panel	Verify basic Touch Panel display functions.
12	Peripherals	see: Peripheral/Option Checks on page 377

NOTE

* Some software may be considered standard depending upon system model configuration.

13.6.2.2 Peripheral/Option Checks

If any peripherals or options are not part of the ultrasound system configuration, the check can be omitted.

Table 13-4 Approved peripheral/hardware option functional checks

Step	Item	Description
1	Media	Verify media drive(s) read/write properly. Clean if necessary.
2	B/W Printer	Verify hardcopy output of the B/W video page printer. Clean heads and covers if necessary.
3	DICOM	Verify that DICOM is functioning properly. Send an image to a DICOM device.
4	Foot Switch	Verify that the foot switch is functioning as programmed. Clean as necessary.
5	ECG	Verify basic operation with customer.

13.6.2.3 Mains Cable Inspection

Table 13-5 mains cable inspection, as appropriate

Step	Item	Description
1	Unplug Cord	Disconnect the mains cable from the wall and ultrasound system.
2	Inspect	Inspect it and its connectors for damage of any kind.
3	Verify	Verify that the LINE, NEUTRAL and GROUND wires are properly attached to the terminals, and that no strands may cause a short circuit.
4	Verify	Inlet connector retainer is functional.

13.6.2.4 Optional Diagnostic Checks

To complete the Vivid Explorer/Vivid Advanced/Vivid Focus checks, view the error logs and run desired diagnostics.

13.6.2.5 View the Logs

1. Review the ultrasound system error log for any problems.
2. Check the temperature log to see if there are any trends that could cause problems in the future.

13.6.3 Physical Inspection

Table 13-6 Physical checks

Step	Item	Description
1	Labeling	Verify that all ultrasound system labeling is present and in readable condition.
2	Covers	Where applicable, verify all covers are secured in place and are properly aligned with other covers. Replace any covers that are damaged.
3	Control Console	Inspect the control console. Record any damaged or missing items. (Replace faulty components, as required).
4	Control Console Movement	• Where applicable, verify ease of control console movement in all acceptable directions. • Where applicable, ensure that it latches in position as required.
5	Control Console Lighting	Check proper operation of all control console.
6	Monitor	Inspect the Display for scratches and bad pixels. Verify proper operation of Contrast and Brightness controls. Where applicable, confirm that the monitor arm allows: • swiveling the screen to the left and to the right • folding the screen to the locked position • release and adjustment backwards and forwards • can be adjusted in the up/down positions <i>Note: Monitor arm movement may vary and is not applicable to all ultrasound systems.</i>
7	Cables and Connectors	Check all internal cable harnesses and connectors for wear and secure connector seating. Pay special attention to foot switch assembly and probe strain or bend reliefs.
8	Shielding & Covers	Check to ensure that all EMI shielding, internal covers, air flow panels and screws are in place. Missing covers and hardware could cause EMI/RFI problems during scanning.
9	Probe Holders	Where applicable, inspect the Probe Holders for cracks or damage.
10	Power and System Status Indicators	Check for proper operation of all Power and System Status Indicators.
11	External I/O	Check all connectors for damage and verify that the labeling is good.
12	Wheels and Brakes	• Check all wheels and casters for wear and verify operation of foot brake, to stop the system from moving, and release mechanism. • Check all wheel locks and wheel swivel locks for proper operation.
13	Battery	Where applicable, check that the batteries are not damaged, do not leak, do not emit an odor, and are not deformed or discolored. Observe all warnings and cautions for transportation battery handling, recharging, storing, and/or disposal.

13.6.4 Cleaning

The User Manual of the Vivid Explorer/Vivid Advanced/Vivid Focus system provides a complete description of system/probe care, maintenance, cleaning and disinfection.

Notice

- Before cleaning the Vivid Explorer/Vivid Advanced/Vivid Focus system switch it off and disconnect from AC mains.
- Electric parts must be protected from liquids that drip or seep into the system.
- Electrical contacts and connectors must not be cleaned.

- Do not use high-proof alcohol for cleaning. Only use cleaning/disinfecting agents listed in the User Manual of the Vivid Explorer/Vivid Advanced/Vivid Focus system.

NOTE

It is the responsibility of the user to decide which cleaning procedures are necessary to ensure a safe working environment.

13.6.4.1 General Cleaning

Frequent and diligent cleaning of the Vivid Explorer/Vivid Advanced/Vivid Focus ultrasound system reduces the risk of spreading infection from person to person and helps to maintain a clean work environment.

Table 13-7 general cleaning

Step	Item	Description
1	Console	Use cleaning agent on a soft, non-abrasive, damp cloth to carefully wipe the entire system. Be careful not to get the cloth too wet so that moisture does not enter the console. Caution: DO NOT allow any liquid to drip or seep into the system.
2	Monitor + Touch Panel	Clean Monitor surface and Touch Panel with cleaning agent on a soft, non-abrasive, damp cloth. Caution: DO NOT spray any liquid directly onto Vivid Explorer/Vivid Advanced/Vivid Focus covers, Monitor, keyboard, etc.

13.6.5 Probe Maintenance

13.6.5.1 Probe Related Checks

Table 13-8 probe related checks

Step	Item	Description
1	Probes	Thoroughly check the ultrasound system probe connectors and remove dust from inside the connector sockets if necessary. Visually check for damage.
2	Probes	Verify that the ultrasound system properly recognizes all probes.
3	Probe Holder	Clean probe holders (they may need to be soaked to remove excess gel).

13.6.5.2 Basic Probe Care

The Vivid Explorer/Vivid Advanced/Vivid Focus system user manuals and various probe handling cards provide a complete description of probe care, maintenance, cleaning and disinfection. Ensure that you are completely familiar with the proper care of GE Healthcare probes.

Ultrasound probes can be easily damaged by improper handling. Review the User Manual of Vivid Explorer/Vivid Advanced/Vivid Focus for more details. Failure to follow these precautions can result in serious injury and equipment damage. Failure to properly handle or maintain a probe may also void its warranty.

Any evidence of wear indicates the probe cannot be used.

Do a visual check of the probe and ultrasound system sockets before plugging in a probe.

TEE and Interoperative probes often have special considerations and individual probe user manuals. For TEE and Interoperative probes also refer to their separate user manuals.

13.6.5.3 Basic Probe Cleaning and/or Disinfection

Refer to the Vivid Explorer/Vivid Advanced/Vivid Focus User Manual for details on cleaning and disinfection.

NOTE

Biological Hazard: To help protect yourself from blood borne diseases, wear approved disposable gloves. These are made of nitrile derived from vegetable starch to prevent allergic latex reactions.



CAUTION

Disinfect a defective probe before you return it. Be sure to tag the probe as being disinfected.



CAUTION

Follow the Care Card instructions supplied with each probe (inside the probe boxes) for disinfectants and gels that are compatible with the surface material of the probes.

NOTE

For the latest list of compatible cleaning solutions and disinfectants refer to: <http://www.gehealthcare.com/transducers>.

13.6.6 Using a Phantom

The use of a Phantom is not required during Care and Maintenance. Customer may use it as part of their Quality Assurance Program tests.

Chapter 14

Optional Features

In this chapter

Optional Features for Vivid Explorer/Vivid Advanced/Vivid Focus.	382
--	-----

14.1 Optional Features for Vivid Explorer/Vivid Advanced/Vivid Focus

There is currently no information applicable to this product.

Chapter 15

OEM and Other Manuals

In this chapter

OEM Manuals.	384
Product Manuals for Vivid Explorer/Vivid Advanced/Vivid Focus.	385

15.1 OEM Manuals

None.

15.2 Product Manuals for Vivid Explorer/Vivid Advanced/Vivid Focus

The User Manuals in the supported languages and the General Service Manual (English) are all included on removal media (User Manual UFD).

The on-board user manuals must be installed from the User Manual UFD.

15.2.1 Product Manuals - Software v211

Table 15-1 Product Manuals for Vivid Explorer/Vivid Advanced/Vivid Focus(v211)

ITEM	PART NUMBER	DESCRIPTION	QTY	CRU	FRU
1	6007611-1xx	Vivid Explorer/Vivid Advanced/Vivid Focus v211 User Manual xx: see Product Manual Language codes on page 385	1	N	N
2	6007613-1xx	Vivid Explorer/Vivid Advanced/Vivid Focus v211 Release Notes xx: see Product Manual Language codes on page 385	1	N	N
3	6007617-1xx	Vivid Explorer/Vivid Advanced/Vivid Focus v211 Advanced Reference Manual xx: see Product Manual Language codes on page 385. Only available in English and French	1	N	N
4	6007618-1EN	Vivid Explorer/Vivid Advanced/Vivid Focus v211 General Service Manual (Available for the customer)	1	N	N
5	5982035-1EN	Vivid Explorer/Vivid Advanced/Vivid Focus/Vivid Explorer Vet/Vivid Focus Vet v211 Technical Service Manual	1	N	N
6	5982037-1EN	Vivid Explorer/Vivid Advanced/Vivid Focus/Vivid Explorer Vet/Vivid Focus Vet v211 Highly Confidential Service Manual	1	N	N
7	6007622-199	Vivid Explorer/Vivid Advanced/Vivid Focus v211 eIFU leaflet	1	N	N

Table 15-2 Related manuals

ITEM	PART NUMBER	DESCRIPTION	QTY	CRU	FRU
1	GD294142-1EN	View-X with Epiphan Pearl Nano Broadcaster Installation Manual	1	N	N

15.2.2 Product Manual Language codes

The following language codes are used in product manual part numbers:

Table 15-3 Language Codes for Manuals

Code	Language
EN	English
BG	Bulgarian

Code	Language
CS	Czech
DA	Danish
DE	German
EL	Greek
ES	Spanish
ET	Estonian
FI	Finnish
FR	French
HR	Croatian
HU	Hungarian
ID	Indonesian
IT	Italian
JA	Japanese
KK	Kazakh
KO	Korean
LT	Lithuanian
LV	Latvian
NL	Dutch
NO	Norwegian
PL	Polish
PB-BR	Portuguese Brazil
PT-PT	Portuguese Europe
RO	Romanian
RU	Russian
SK	Slovak
SL	Slovenian
SR	Serbian
SV	Swedish
TR	Turkish
UK	Ukrainian
VI	Vietnamese
ZH-CN	Chinese

Chapter 16

Favorite Links

In this chapter

Documentation Web Site Links. 388

Other links. 389

16.1 Documentation Web Site Links

Support Documentation Library Web Site

<http://www.gehealthcare.com/DocumentationLibrary>

Customer Documentation Portal

<https://www.gehealthcare.com/documentation>

Product Security Portal

<https://www.gehealthcare.com/services/lifecycle-management/product-security-portal/security>

Sustainability reports

<http://www.gehealthcare.com/about/sustainability/reports>

16.2 Other links

GE HealthCare

<https://www.gehealthcare.com>

Vivid Club

<https://www.vividechoclub.net>

Ultrasound Probe Web Site

<https://www.gehealthcare.com/products/ultrasound/ultrasound-transducers>

GEHC Chemicals Compatibility

<https://cleaning.gehealthcare.com/>

System Cleaning Site

<https://cleaning.gehealthcare.com/>

Chapter 17

Glossary

In this chapter

Glossary.	392
-------------------	-----

17.1 Glossary

Term	Description
3D	Three Dimensional
4D	Four Dimensional. 4D is the same as Three Dimensional + Realtime
AC	Alternating Current
APM	Asset Performance Tool
ASSY	Assembly
BEP	Back End Processor
CPU	Central Processing Unit
CRM	Customer Relationship Management (number)
CRU	Customer Replaceable Unit
CTRL	Control
CW	Continuous Wave Doppler
DC	Direct Current
DVI	Digital Visual Interface
ECG	Electro Cardio Graphy
eIfU	electronic Instructions for Use
EXT.	External
FE	1: Field Engineer 2: Front End
FPGA	Field Programmable Gate Arrays
FRU	Field Replacable Unit
FRU Y	Replacement Part
FRU N	Non Stock Part
GB	Gigabyte
HDU	High Definition Ultrasound
IFU	Instructions for Use
I/O	Input/output
INT	Internal
LCD	Liquid Crystal Display
OP	Operating Panel
PS	Power Supply
PWR	Power
PW	Pulsed Wave Doppler
QTY	Quantity used per ultrasound system

Term	Description
RAM	Random Access Memory
SSA	Secure Service Access (by means of an "USB Class M" SSA dongle + PIN)
SSD	Solid-State Drive
TB	Terabyte
TBE	Tandem Back End
UI	User interface
VCO	Virtual Console Observation
WPA	WI-Fi Protected Access

A

- about (system information) [196](#)
- accessories [112](#)
- Accessories
 - Ordering [52](#)
 - Requesting a catalog [52](#)
- Accessory List [112](#)
- acoustic indexes display [204](#)
- adjustable components description [131](#)
- admin tab [197](#)
- advanced security features [248](#)
- Air Filter
 - replacement [347](#)
- alignment
 - monitor and touch screen alignment [209](#)
- Alphanumeric Keyboard Configuration [205](#)
- AN Keyboard Protective Film
 - replacement [361](#)
- appendix
 - pre-installation [80](#)
- archive presets to removable media [197](#)
- auxiliary devices safety [33](#)
- average setup time [84](#)

B

- back up patient archive and system configurations [221](#)
- back up system configurations and patient archive [221](#)
- Backup [197](#)
- backup and restore [241](#)
- Battery
 - replacement [348](#)
- Battery check [310](#)
- battery disposal [79](#)
- battery safety [31](#), [310](#)
- battery settings [195](#)
- Battery support
 - Power state indications [310](#)
- before unpacking the equipment [97](#)
- block diagrams
 - when used [49](#)
- brake pedal operation [85](#)
- buttons sheet [171](#)

C

- cables
 - power cords [320](#)
- calibration [209](#)
- casters, brakes, direction lock checks [216](#)
- Castors [129](#)
- CE compliance [73](#)
- central listing [130](#)
- certified electrical contractor statement - for USA only [xvii](#)
- Charge Box Module
 - replacement [352](#)
- circuit breaker [68](#)
- complete power down [92](#)

- completing the setup, See
- compliance [73](#)
- configuration [133](#)
 - wireless network [139](#), [140](#)
- connect external USB hard disk [114](#)
- connect UFD (USB flash drive) [114](#)
- connect USB memory stick [114](#)
- connecting black and white printer [112](#)
- connecting the DVD Writer Kit [113](#)
- connecting the ECG [112](#)
- connecting the ultrasound system to a power source [87](#)
- connecting the wireless network adapter [115](#)
- connection of accessories [112](#)
- Connectivity [135](#)
- connectivity setup [136](#)
- connectivity setup worksheet [70](#)
- connectivity tab [135](#)
- contact information [52](#)
- Contacts
 - Clinical questions [52](#)
 - Internet [52](#)
 - Service questions [52](#)
- control console positioning [84](#)
- conventions for this publication [48](#)
- conventions used in manual [48](#)
- cooling, See heat output (cooling)
- CRU parts
 - spare parts [323](#)
- customer assistance [52](#)
- Customize configuration
 - Users and security [181](#)
- cybersecurity precautions [33](#)

D

- damage in transportation [xvi](#), [99](#)
- data and time settings [194](#)
- dataflow sheet [164](#)
- definition of safety messages [28](#)
- desirable features [61](#)
- DICOM option pre-installation requirements [69](#)
- DICOM sheet [145](#)
- DICOM spooler [159](#)
- disable usb external media [204](#)
- disconnect power cable [94](#)
- disk encryption [248](#)
- Disk encryption [186](#)
- disk management sheet [174](#)
- Disk space management [174](#)

E

- eDelivery [222](#)
- electrical requirements for Vivid [67](#)
- electrical safety [31](#)
- electromagnetic compatibility [73](#)
- electromagnetic interference survey [73](#)
- electrostatic discharge prevention [73](#)
- EMC compliance [80](#)

EMI limitations [75](#)
EMI prevention/abatement [75](#)
EMI protection [102](#)
Encoder collection
 replacement [363](#)
environmental dangers [64](#)
environmental protection [79](#)
environmental requirements [64](#)
environmental requirements)
 if the ultrasound system is very cold or hot [66](#)
environmental specifications [66](#)
EQC and IQC [218](#)
examine all packages [98](#)
external connection of other peripheral devices [64](#)
External monitor
 Connection [124](#)

F

floor plan with ultrasound system and workstation in same room [63](#)
floor plan, minimal, suggestion [61](#)
floor plan, recommended suggestion [62](#)
Footswitch [113](#)
Functional Check [120](#)
functional checks [216](#)
 Battery [351](#)
 casters, brakes, direction lock checks [216](#)
 Charge Box [354](#)
 Encoder Collection [366](#)
 mechanical function checks [216](#)
functional hazards [33](#)

G

Gel Cup
 replacement [358](#)
general electrical requirements [67](#)
general information [44](#), [220](#)
glossary [392](#)

H

hazard definitions [28](#)
 standard hazard icons [48](#)
heat output (cooling) [66](#)
how to access supplier manuals for product options [80](#)
human safety [28](#)

I

illustration of the ultrasound system (front/rear/right/left) [46](#)
imaging, See
important conventions: see hazard definitions [28](#)
Information
 Requesting [52](#)
inrush current [67](#)
insite setup [181](#)
InSite setup [213](#)
Install Battery Pack [120](#)
Install Charge Box [120](#)
Install Crystal ICECord-RS [120](#)
Install Ergonomic P2D Probe Holder [120](#)
Install Ferriter Core [121](#)
Install Gel Warmer [119](#)

Install the Baskets and Trays [120](#)
Install the Elastic Cable Hook [117](#)
Install the Network Breakway Adapter [118](#)
Installation [106](#)
installation instructions
 environmental dangers [64](#)
 patient environment [33](#)
installation warnings [84](#)
instructions for use [126](#)

K

keyboard setup [204](#)

L

label
 transportation box [99](#)
Label Location [42](#)
label, icon conventions [48](#)
language and units settings [194](#)
language disclaimers [iv](#)
LED descriptions [130](#)
legal notes [xv](#)
lighting [68](#)
list of chemicals associated with servicing and installing [33](#)
location settings [194](#)
Lock main display [128](#), [217](#)
lockout/tagout (LOTO) requirements [33](#)
long-term storage mode [195](#)
LOTO requirements [33](#)

M

main monitor position controls [128](#)
main monitor positions and lock functional checks [216](#)
Manpower [106](#)
manufacturer name and address [43](#)
measurments and text, See
mechanical alignment procedures [128](#)
mechanical function checks [34](#), [216](#)
mechanical installation [97](#)
mechanical safety [29](#)
minimal floor plan suggestion [61](#)
monitor and touch screen alignment [209](#)
Monitor Sticker
 replacement [367](#)

N

Nameplate kit
 replacement [368](#)
network proxy settings [138](#)
network sheet [135](#)
networking requirements [69](#)
 stand-alone system (without network connection) [69](#)
 system Connected to hospital's network [69](#)

O

OEM manuals [384](#)
omission and errors [xviii](#)
open service screen [203](#)
Operator Panel Top Cover Rear

Operator Panel Top Cover Rear ()
 replacement [344](#)
 OPIO Upper Deck Storage Tray
 replacement [357](#)
 option catalog numbers [80](#)
 optional features [382](#)
 optional peripherals and accessories [327](#), [331](#)
 options [110](#)
 other sheet [180](#)

P

packaging details [103](#)
 packing instructions [102](#)
 packing ultrasound system [343](#)
 password for ADM [220](#)
 patient environment [33](#)
 peripheral devices [64](#)
 external connection of other peripheral devices [64](#)
 peripherals, *See* optional peripherals and accessories
 photos [49](#)
 physical inspection [102](#)
 position controls for main monitor [128](#)
 position of the tilt indicators [98](#)
 power cable [68](#)
 power cords [320](#)
 power on and boot up [88](#)
 power on installation (completing the setup) [86](#)
 power on warnings [86](#)
 power outlets [68](#)
 power requirements [68](#)
 power shut down [91](#)
 Preparations [106](#)
 preparing for setup [102](#)
 preparing for transportation [96](#)
 preparing the site [60](#)
 prerequisite
 customer provided prerequisite [220](#)
 password for ADM [220](#)
 prescription device statement [35](#)
 presets [197](#)
 printer [179](#)
 Printing
 Printer configuration [179](#)
 Probes [333](#)
 Connect the probe [89](#)
 DLP Probe ports [90](#)
 RS Probe Ports [91](#)
 product and packaging
 product labels [36](#)
 product and packaging information [35](#)
 product labels [36](#)
 product locator installation card [126](#)
 product manuals [385](#)
 publication conventions [48](#)
 purchaser responsibilities [60](#)
 purpose of the DICOM network function [69](#)
 purpose of the operator manual(s)
 purpose of the user manual(s) [45](#)

R

receiving and unpacking the equipment [97](#)
 receiving the ultrasound system [98](#)
 recommended floor plan suggestion [62](#)

related documents, *See*
 remote path sheet [173](#)
 Removable media
 Ejecting [115](#), [173](#)
 Formatting [172](#)
 repacking ultrasound system [343](#)
 report, *See*
 request for service [213](#), [307](#)
 required facility needs [60](#)
 restore, *See* backup and restore
 Restore data [197](#)
 returning system, probes, parts [343](#)
 revision history [iii](#)
 RFS) [213](#), [307](#)

S

safety [28](#)
 auxiliary devices [33](#)
 battery safety [31](#), [310](#)
 electrical safety [31](#)
 human safety [28](#)
 mechanical safety [29](#)
 safety and regulatory procedures [28](#)
 safety messages
 definition of safety messages [28](#)
 safety reminders [86](#)
 scope [45](#)
 screenshots [49](#)
 Security [181](#)
 security tab [181](#)
 seismic considerations [63](#)
 select show test image [204](#)
 Service
 Requesting [52](#)
 service desktop [259](#)
 service form [97](#)
 service safety considerations [342](#)
 service screen, open [203](#)
 service tab [203](#)
 set monitor resolution [203](#)
 setup after software loading
 procedure [240](#)
 Shipping and long-term Storage [312](#)
 shipping mode [195](#)
 shipping system, probes, parts [343](#)
 site log [51](#)
 software configuration checks [34](#)
 software installation [220](#)
 Software Loading [231](#)
 software options configuration [110](#)
 Software Reload [234](#)
 standard hazard icons [48](#)
 Streaming video, *see* View-X [110](#), [125](#)
 suggested floor plan with ultrasound system and workstation
 in same room [63](#)
 System
 Software update [222](#)
 system acclimation time [66](#)
 system configuration [133](#)
 system disposal [79](#)
 system manufacturer name and address [43](#)
 system naming [45](#)
 system presets [197](#)
 system setup [84](#)
 system tab [194](#)

T

- TEE Probe Storage Rack [122](#)
- tilt indicator has triggered or is missing [98](#)
- Tilt Watch indicators [35](#), [97](#)
- time and manpower requirements
 - preinstallation [78](#)
- time synchronization settings [194](#)
- tools available via the service desktop, InSite etc. [82](#)
- tools required for care and maintenance [375](#)
- tools sheet [172](#)
- touch screen calibration [210](#)
- transfer sheet [171](#)
- transport preparation [96](#)
- transportation
 - damage [xvi](#)
- transportation box label [99](#)
- transportation damage [99](#)
- Trolley Left Cover
 - replacement [345](#)
- Trolley Lower Front Tray
 - replacement [359](#)
- Trolley Mounted Dual Storage Bin
 - replacement [355](#)
- Trolley Rear Basket
 - replacement [356](#)
- Trolley Right Cover
 - replacement [346](#)
- typical users of the documentation [45](#)

U

- UDI [43](#)
- ultrasound equipment quality check (EQC and IQC) [218](#)
- ultrasound IQC and EQC form [218](#)
- Unlock main display [128](#), [216](#)
- unpacking the equipment [99](#)
- updates [106](#)
- upgrades [105](#)
- USB
 - remarks for using external USB devices [115](#)
 - USB devices, remarks for use [115](#)
- user manuals, *See* instructions for use
- Users [181](#)

V

- verify customer order [102](#)
- View-X [110](#), [125](#)

W

- warnings for care and maintenance [371](#)
- warnings for receiving and unpacking [97](#)
- where to find the general service manual [50](#)
- where to find the privacy and security manuals [50](#)
- where to find the service notes [50](#)
- where to find the user manuals [50](#)
- wireless network configuration [139](#), [140](#)
- wireless network settings [71](#)

